Exploring new physics with proton interactions at the LHC

LABORATÓRIO DE INSTRUMENTAÇÃO
E FÍSICA EXPERIMENTAL DE PARTÍCULAS

Work done so far

Central exclusive production processes in proton-proton collisions.

Recent research has focused on lepton pair, WW/ZZ boson and top quark (tt) production, leading to:

- Precision Tests of the Standard Model
- Enhanced Sensitivity to New Physics: the ability to isolate exclusive events makes CEPs valuable for searching BSM;
- Forward Proton Tagging: the use of forward proton detectors - like PPS - improves event selection and provides more information about scattered protons, leading to more accurate measurements;

Why proton-proton collisions?

- Proton collisions at the LHC offer a unique environment for probing the Standard Model (SM)
- These collisions allow exploration of rare processes and potential new physics that are not accessible in lower-energy experiments.

Future work?

- Exploring the Anomalous Magnetic Moment of the Tau Lepton: unlike electrons and muons, leptons have higher mass and shorter lifetime, making it challenging to study the tau's g-2 - requires high-energy experiments;
- Since taus are 3rd generation of leptons and heavier than electrons and muons, they may be even more sensitive to Beyond SM physics;

Thank you
for your attention

Exploring new physics with proton interactions at the LHC

LABORATÓRIO DE INSTRUMENTAÇÃO
E FÍSICA EXPERIMENTAL DE PARTÍCULAS

Physics of anomalous tau couplings

Recent research has focused on lepton pair, WW/ZZ boson and top quark (tt) production, leading to:

- Precision Tests of the Standard Model
- Enhanced Sensitivity to New Physics: the ability to isolate exclusive events makes CEPs valuable for searching BSM;
- Forward Proton Tagging: the use of forward proton detectors - like PPS - improves event selection and provides more information about scattered protons, leading to more accurate measurements;

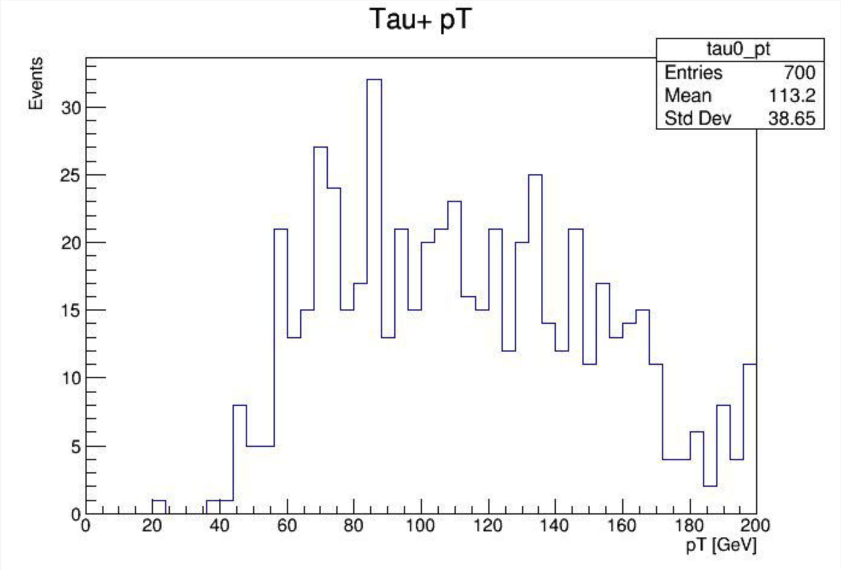
Analysis

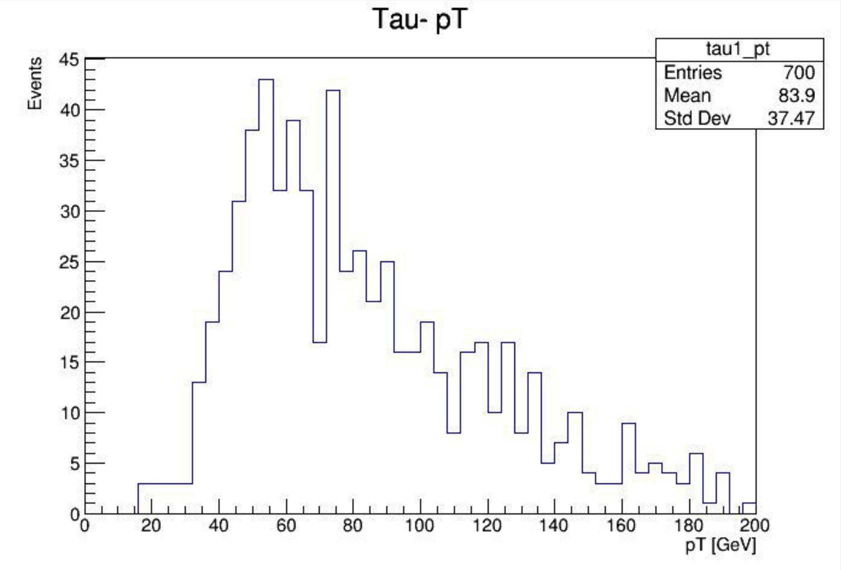
- Proton collisions at the LHC offer a unique environment for probing the Standard Model (SM)
- These collisions allow exploration of rare processes and potential new physics that are not accessible in lower-energy experiments.

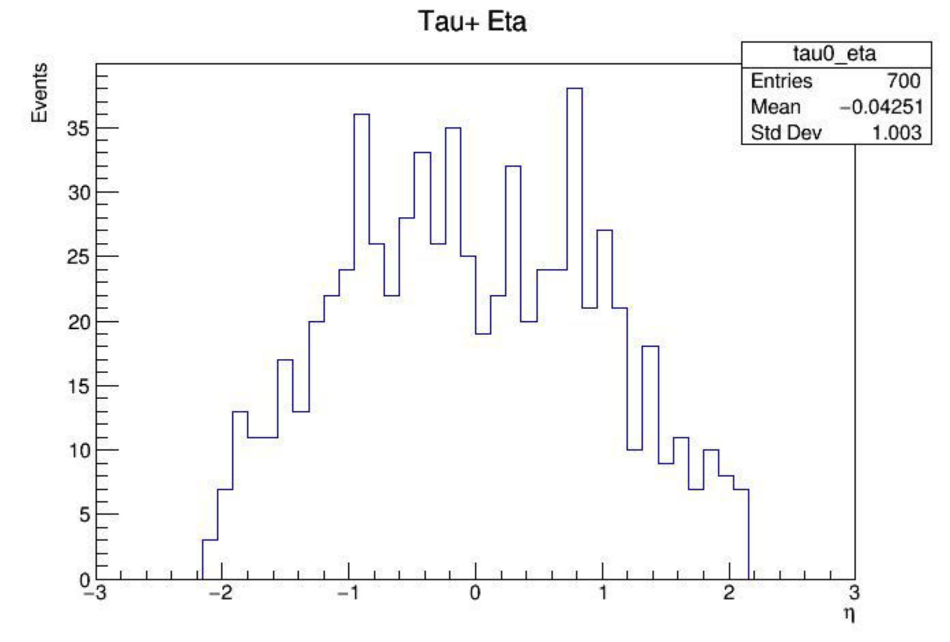
Analysis Improvements

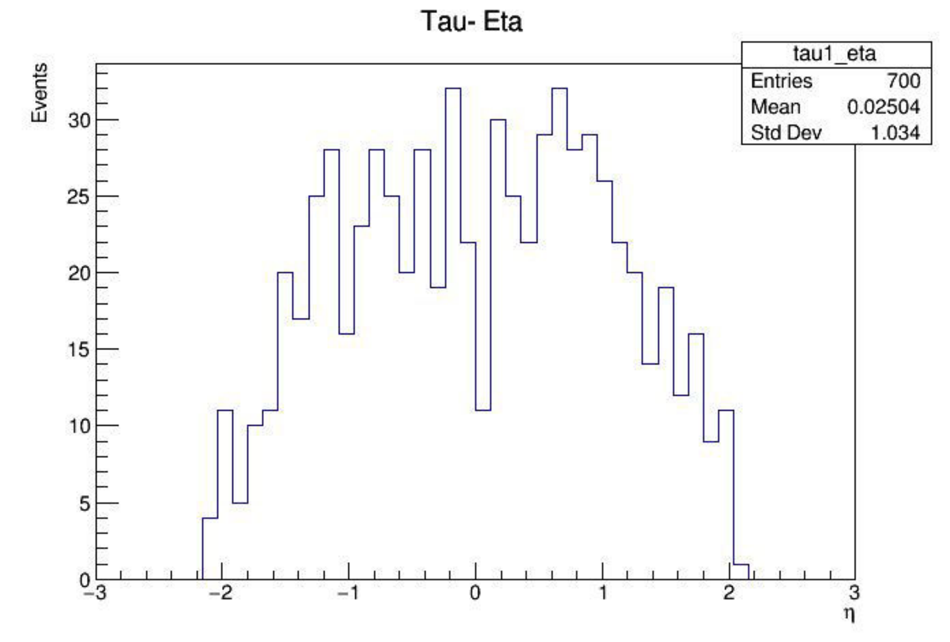
- Exploring the Anomalous Magnetic Moment of the Tau Lepton: unlike electrons and muons, leptons have higher mass and shorter lifetime, making it challenging to study the tau’s g-2 - requires high-energy experiments;
- Since taus are 3rd generation of leptons and heavier than electrons and muons, they may be even more sensitive to Beyond SM physics;

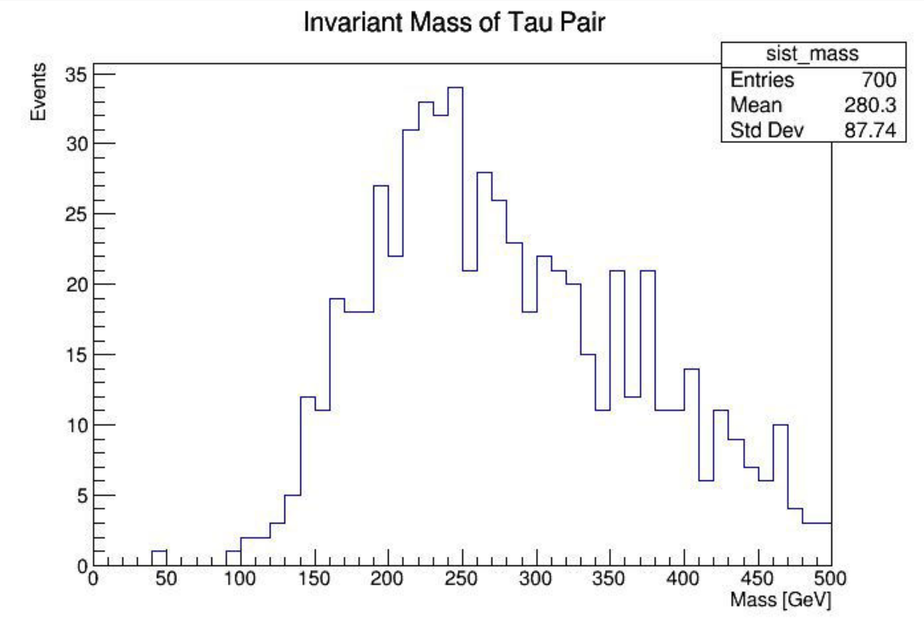
few plots - show properties of expected signal

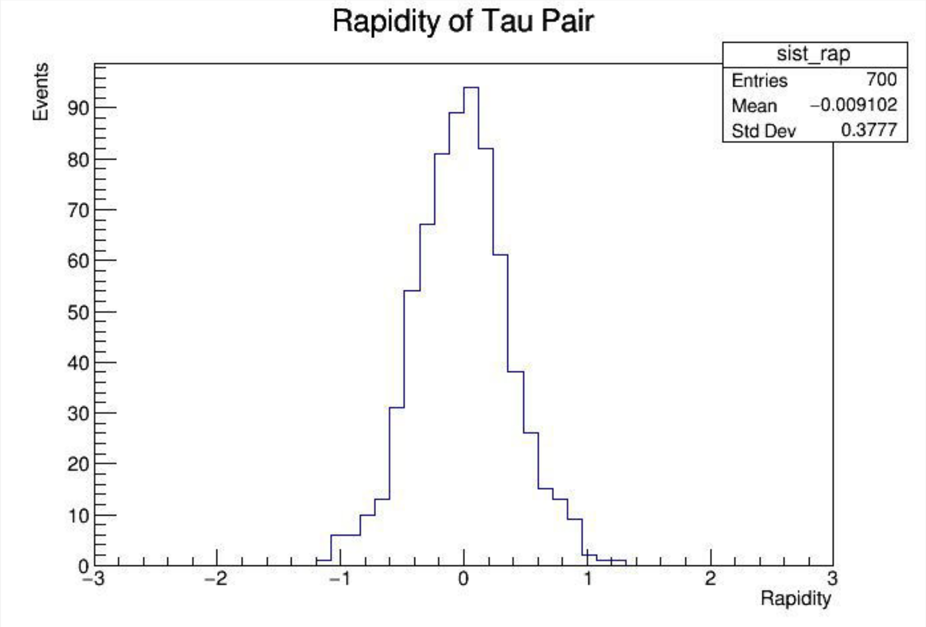


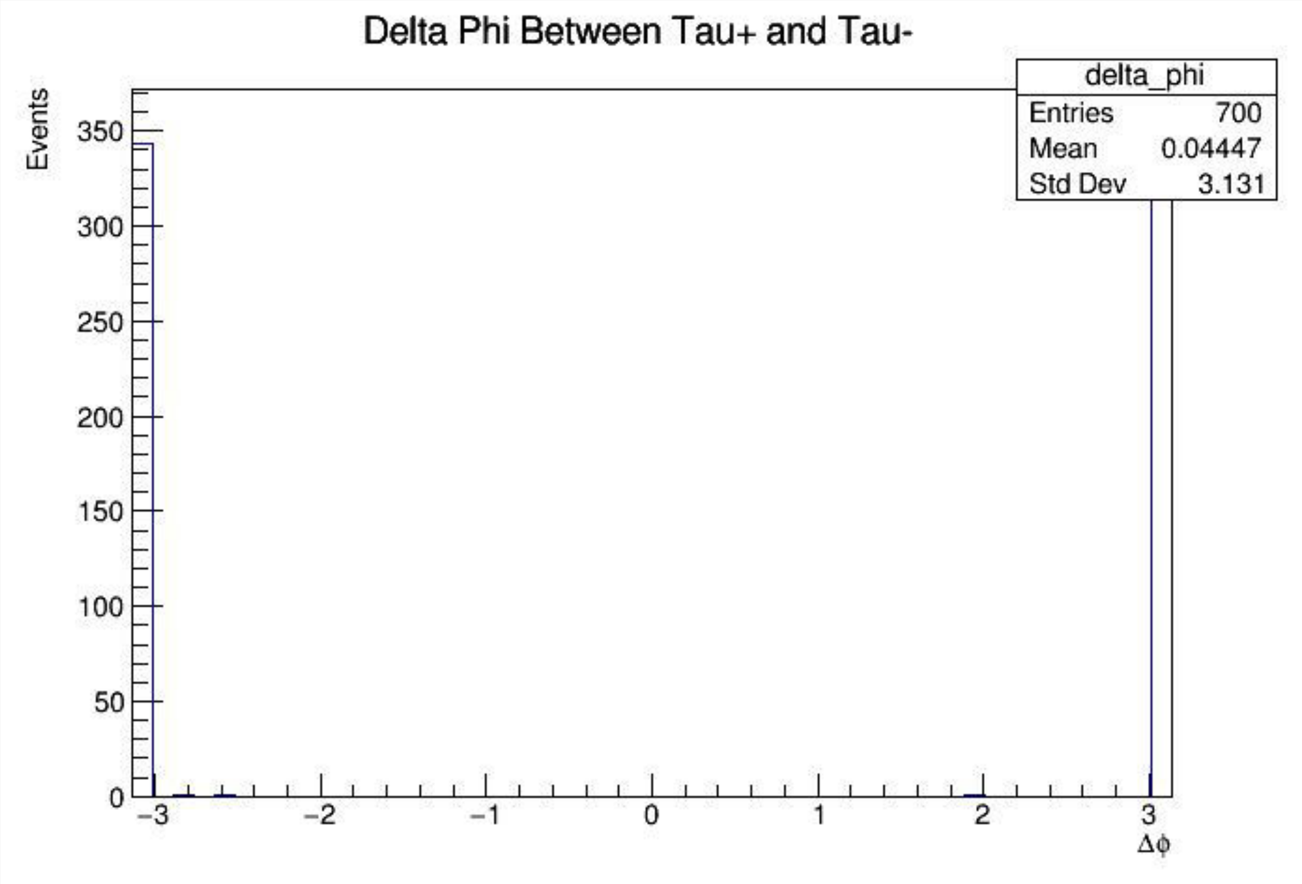



Tau+ Eta
tau0_eta
Entries 700
Mean −0.04251
Std Dev 1.003
Events
η

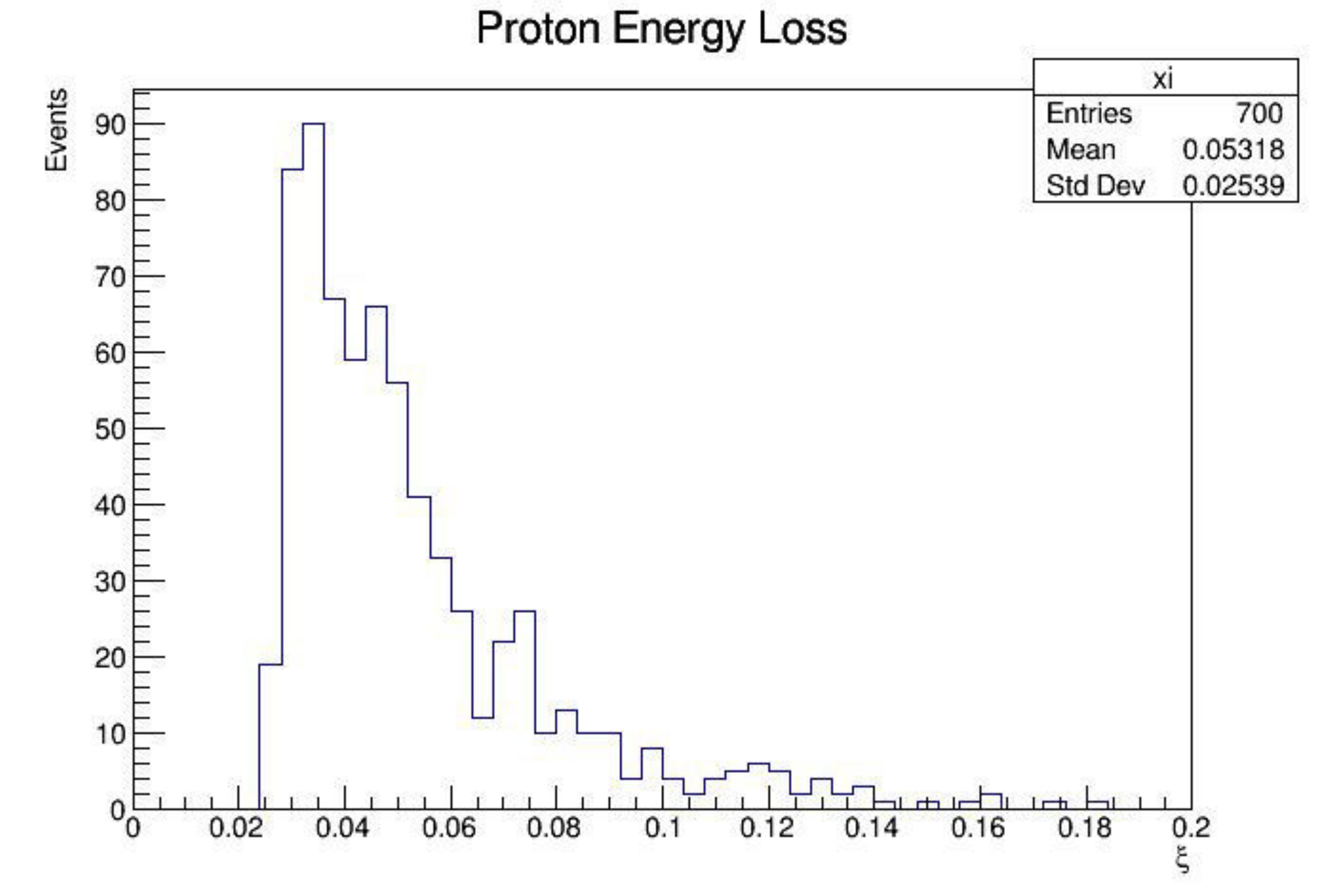

Tau- Eta
tau1_eta
Entries 700
Mean 0.02504
Std Dev 1.034
Events
η


Invariant Mass of Tau Pair
sist_mass
Entries 700
Mean 280.3
Std Dev 87.74
Events
Mass [GeV]


Rapidity of Tau Pair
sist_rap
Entries 700
Mean −0.009102
Std Dev 0.3777
Events
Rapidity




```
Successfully loaded TTree: tree
tau0_phi: 6.93062e-310, tau1_phi: 0
```



Proton Energy Loss
xi
Entries 700
Mean 0.05318
Std Dev 0.02539
Events
0
10
20
30
40
50
60
70
80
90
0
0.02
0.04
0.06
0.08
0.1
0.12
0.14
0.16
0.18
0.2
ξ

Thank you
for your attention

LIP-CMS weekly meeting

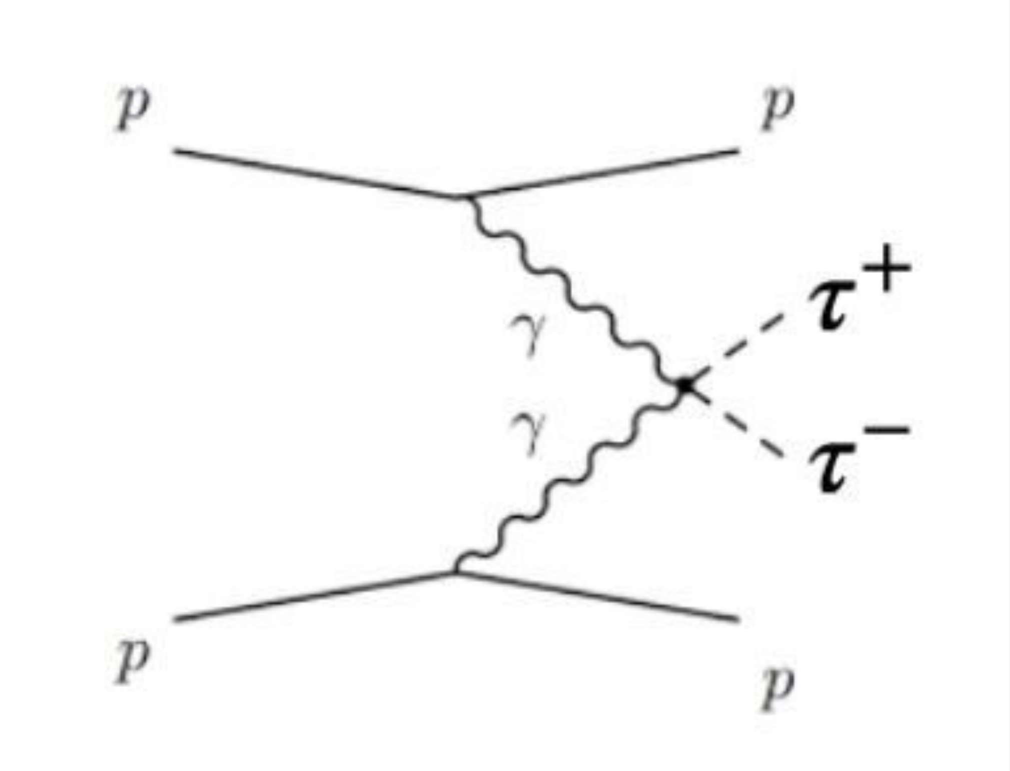
05/02/2025

Exploring new physics with photon interactions at the LHC

Madalena Ferreira

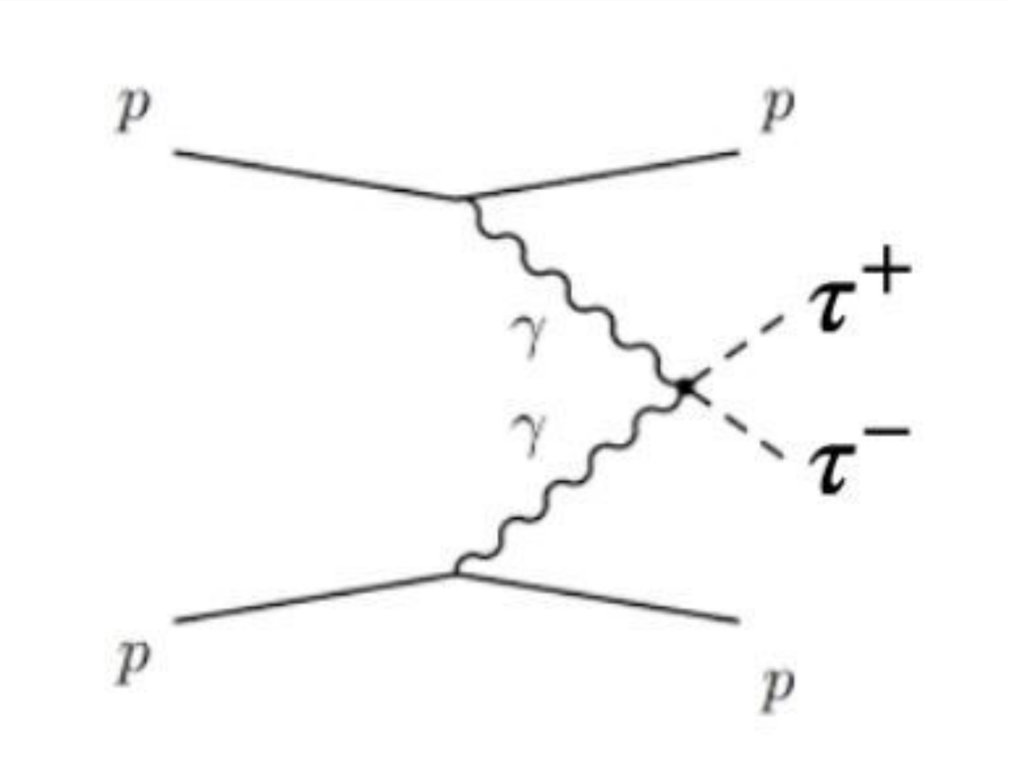
Tau anomalous magnetic moment

- Tau anomalous coupling refer to the deviations in the interactions involving tau leptons compared to the predictions made by the standard model;
- The tau anomalous magnetic moment, a_τ measures the deviation of the tau's magnetic dipole moment from the value predicted by the Dirac equation (g=2).
- Such deviations could point to physics beyond the standard model;
- Central exclusive production of tau pairs ($\tau^+\tau^-$) has direct sensitivity to a_τ through photon-photon interactions ($\gamma\gamma\rightarrow\tau^+\tau^-$) , and can be used to study the gyromagnetic factor (g) of the τ lepton;



$$pp \rightarrow pX(\rightarrow \tau\tau)p,$$

Central exclusive production of tau pairs



- In this interaction protons are tagged by the by PPS, while the decay products of the X system are detected by the CMS central detector.
- Focus on the fully hadronic decay mode (both taus decay hadronically)
- Other decay modes
  - e + τh decay mode: one τ decays hadronically and the other into an electron + two neutrinos;
  - μ + τh decay mode: one τ decays hadronically and the other into a muon + two neutrinos;
  - e + μ decay mode: one τ decays in an electron + two neutrinos, and the other τ decays in a muon + two neutrinos, $\tau_e\tau_\mu$.

Existing analysis/results

- Recent studies of the muon magnetic moment show that it does not match theoretical predictions. If the tau magnetic moment also deviates from the standard model predictions, it could confirm new physics.
- Matteo’s analysis studies the four mentioned decay modes using data from 2018 ($\sqrt{s}$=13 TeV, 54.9^{-1}).
  - Uses BDT to separate dominant backgrounds such as Drell-Yan, QCD and tt jets.
  - Sets an upper limit on the production cross section;
  - Sets constraints on the tau gyromagnetic factor.

Existing analysis/results

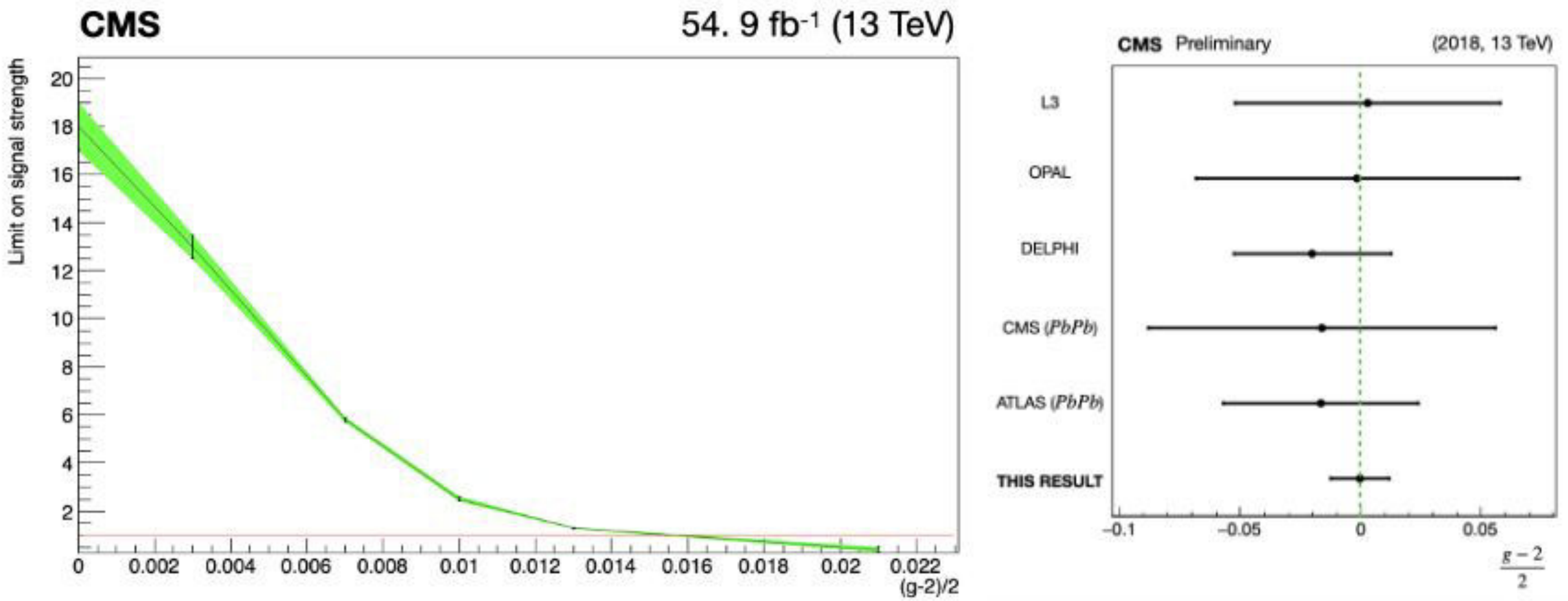


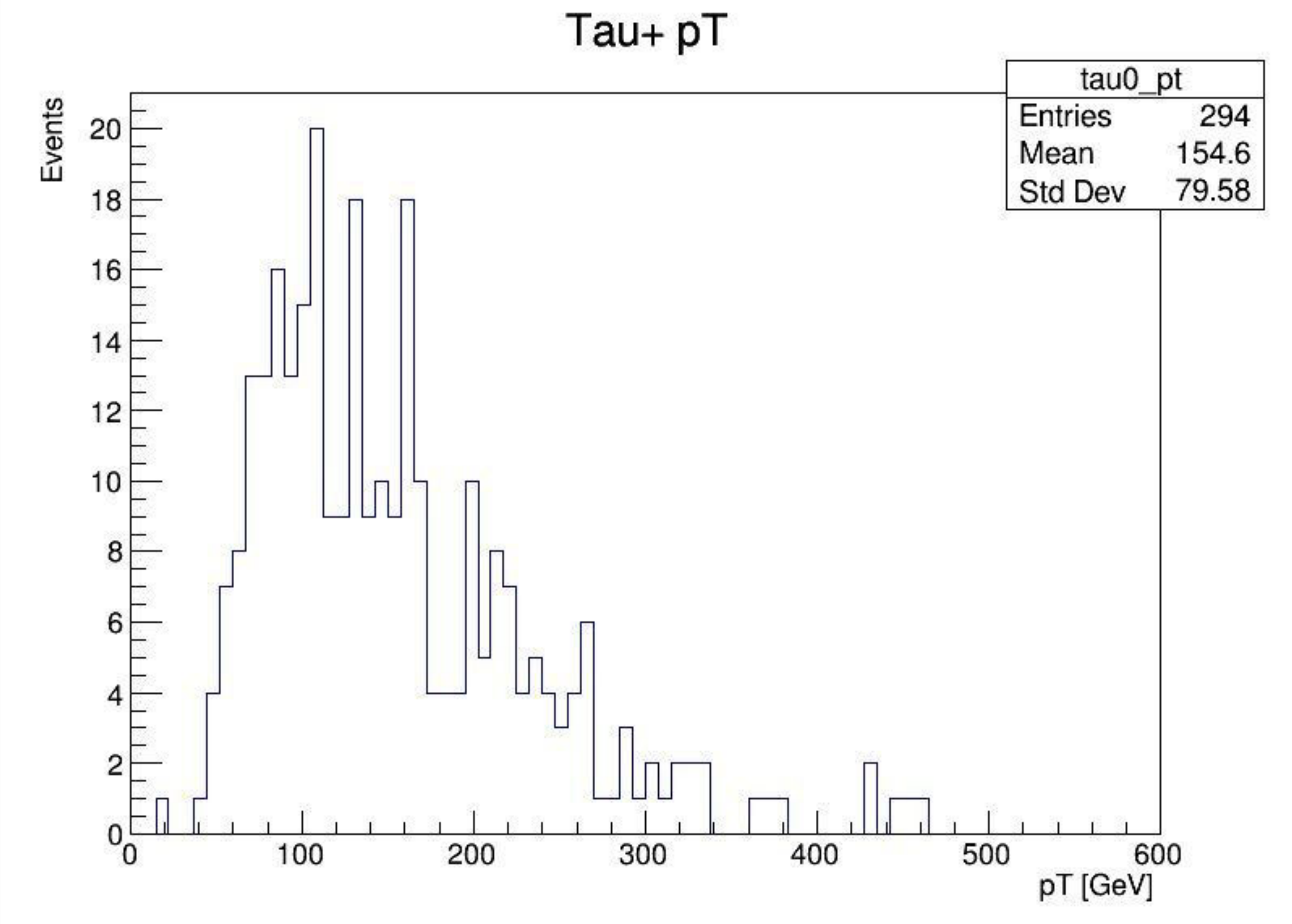
Figure 6: Limits on the signal strength as a function of a_τ (left); limits on a_τ for the different results (right).

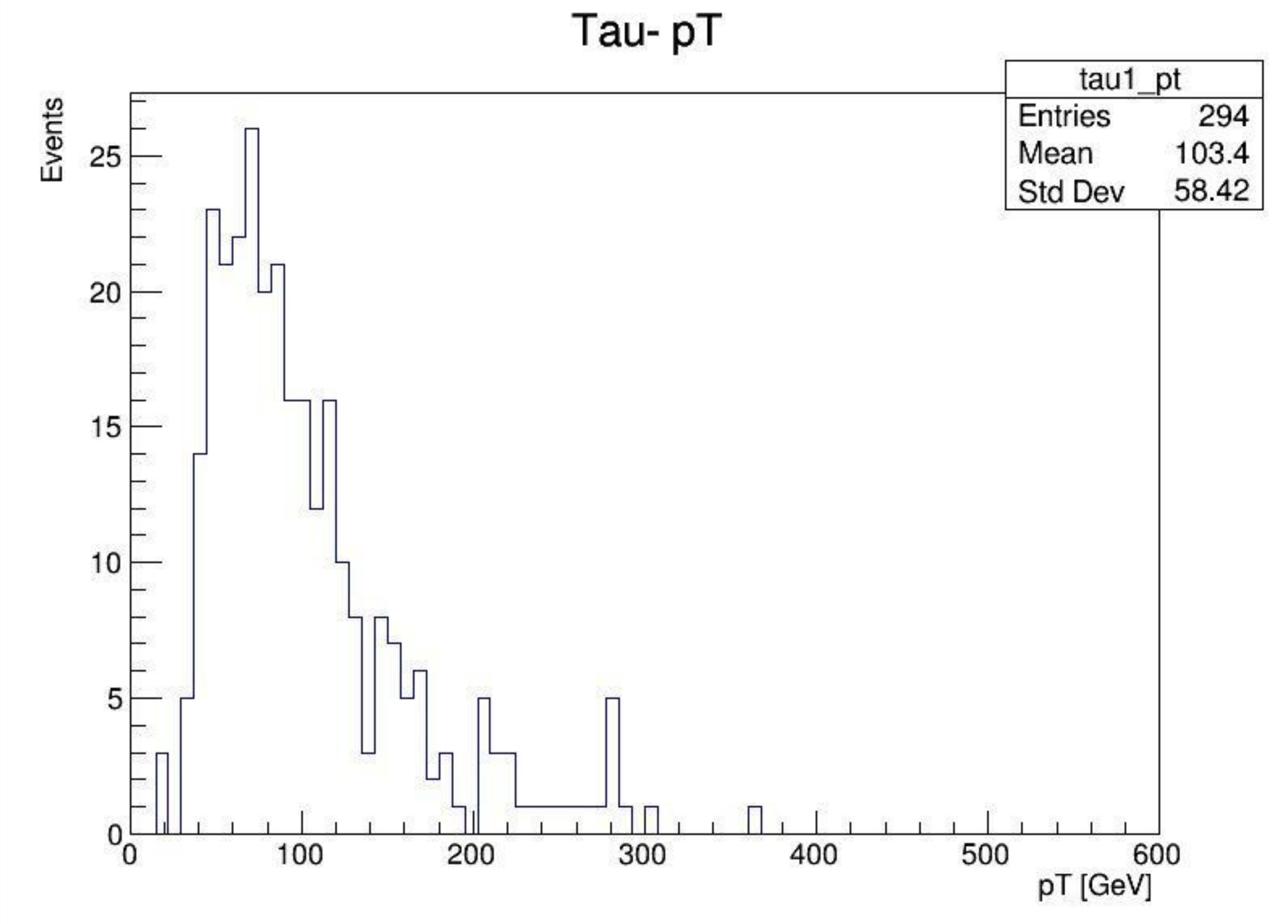
Variables of interest

- Transverse moment (pT) of the tau system
- System Rapidity: vectorial sum of the rapidity (Y) of the $\tau^+\tau^-$ pair decay products
- Acoplanarity: characterised by the presence of back-to-back τs
- $M_{CMS} - M_{PPS}$
- $Y_{CMS} - Y_{PPS}$
- Proton energy loss, ξ
- Rapidity distributions of the tau system

Plots

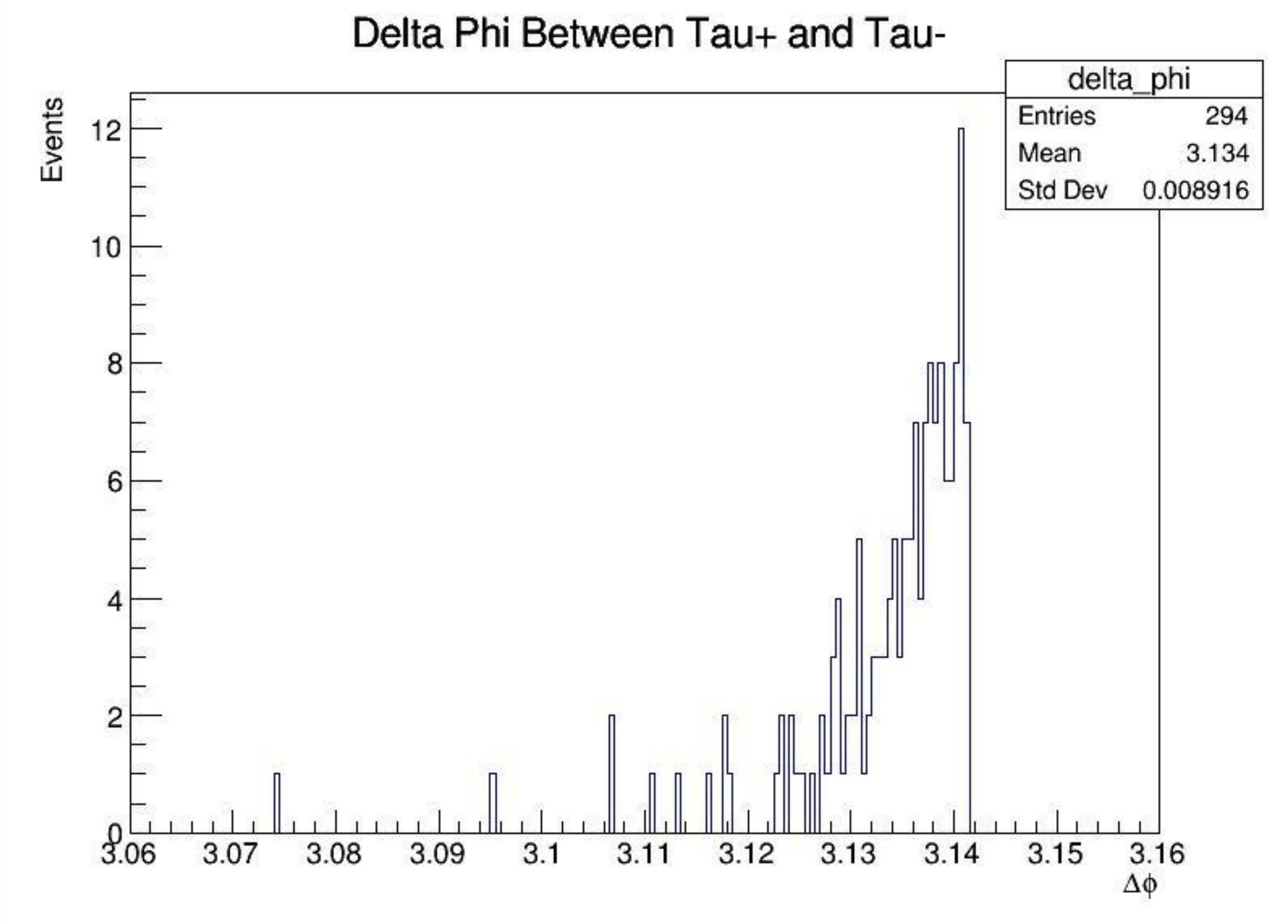
Monte carlo generated sample, 2018 run, 23k events





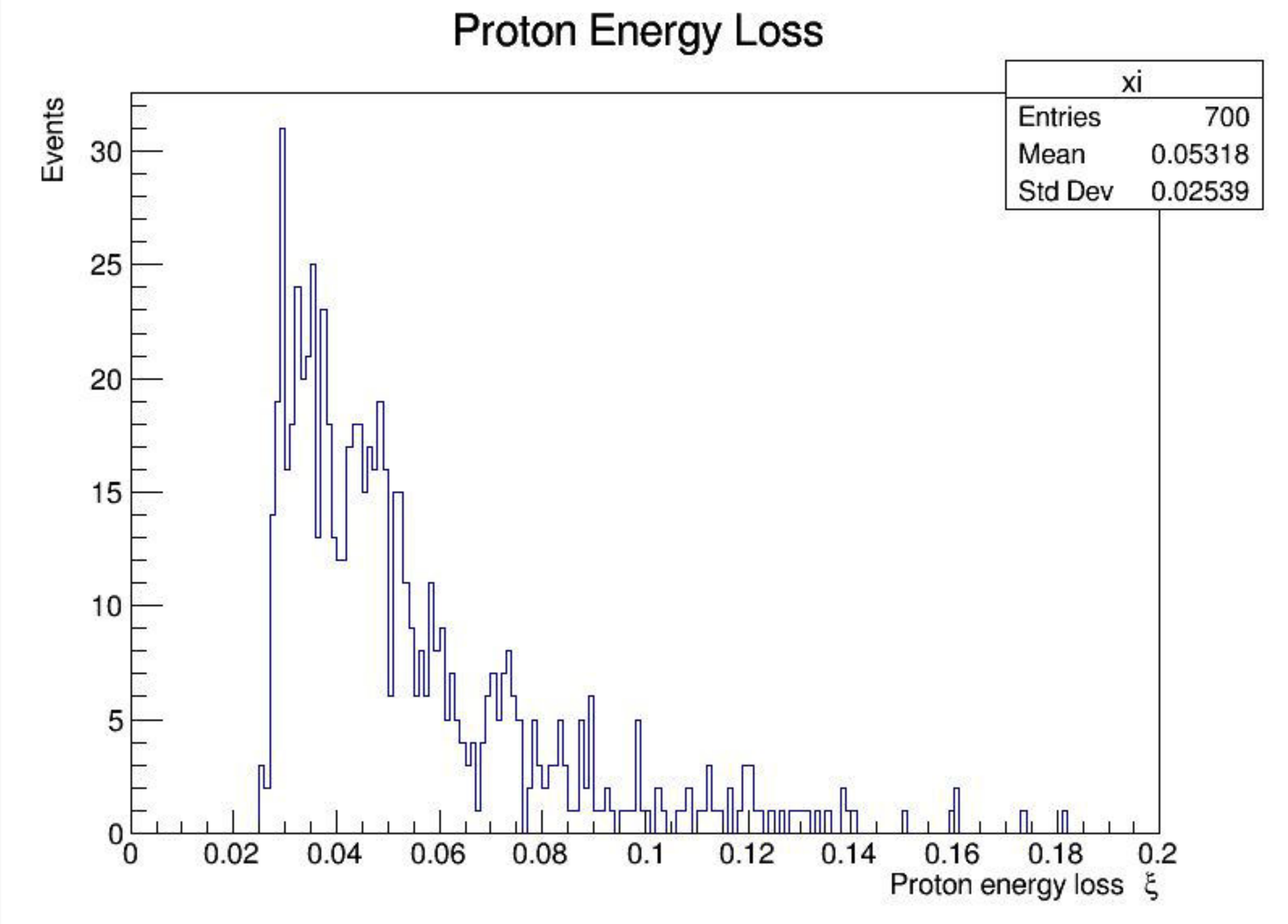
Transverse momentum: momentum of the tau-pair system projected onto the plane perpendicular to the beamline (z-axis)

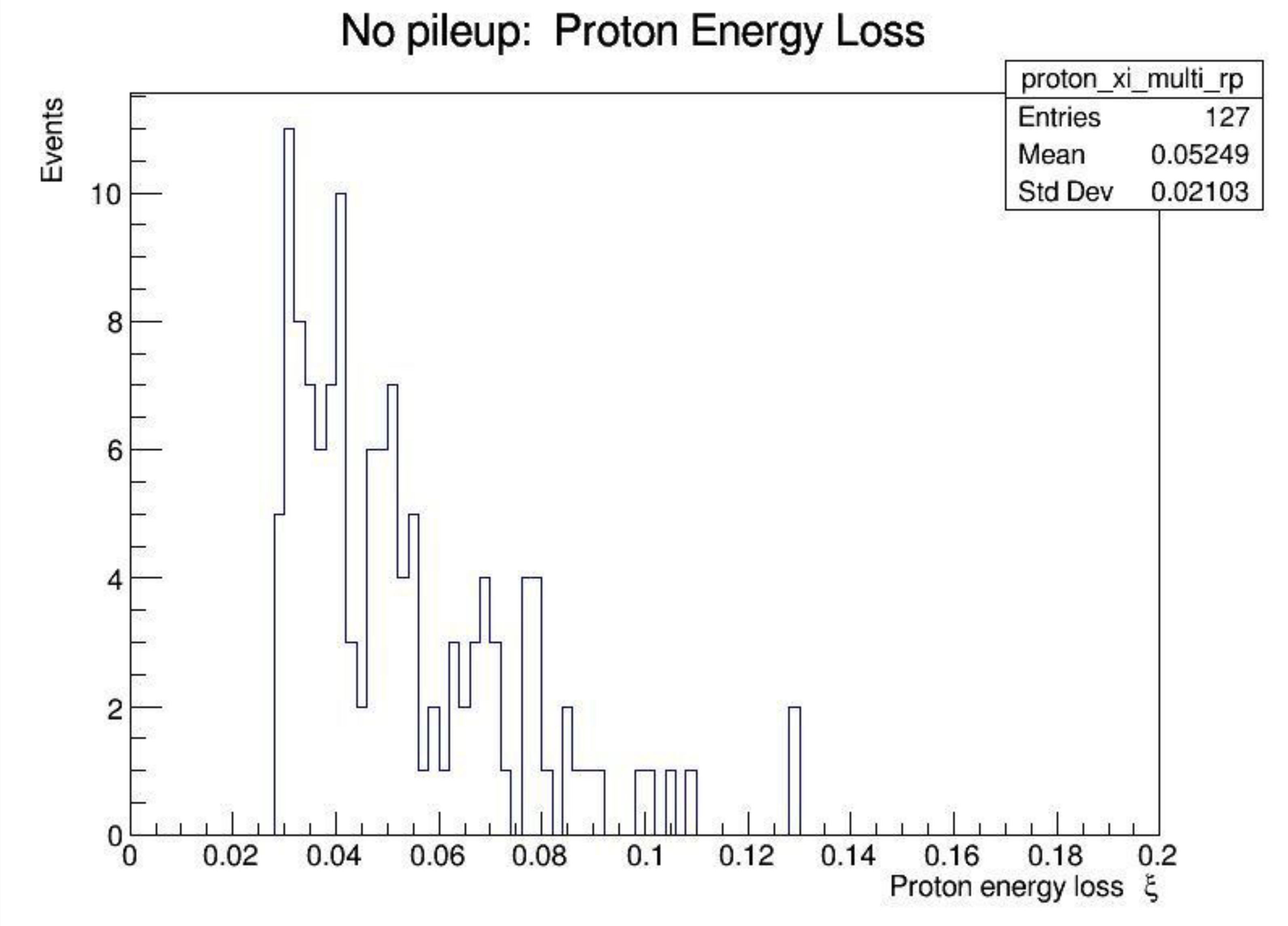
Plots



$$\Delta\phi = \phi(\tau_1) - \phi(\tau_0)$$

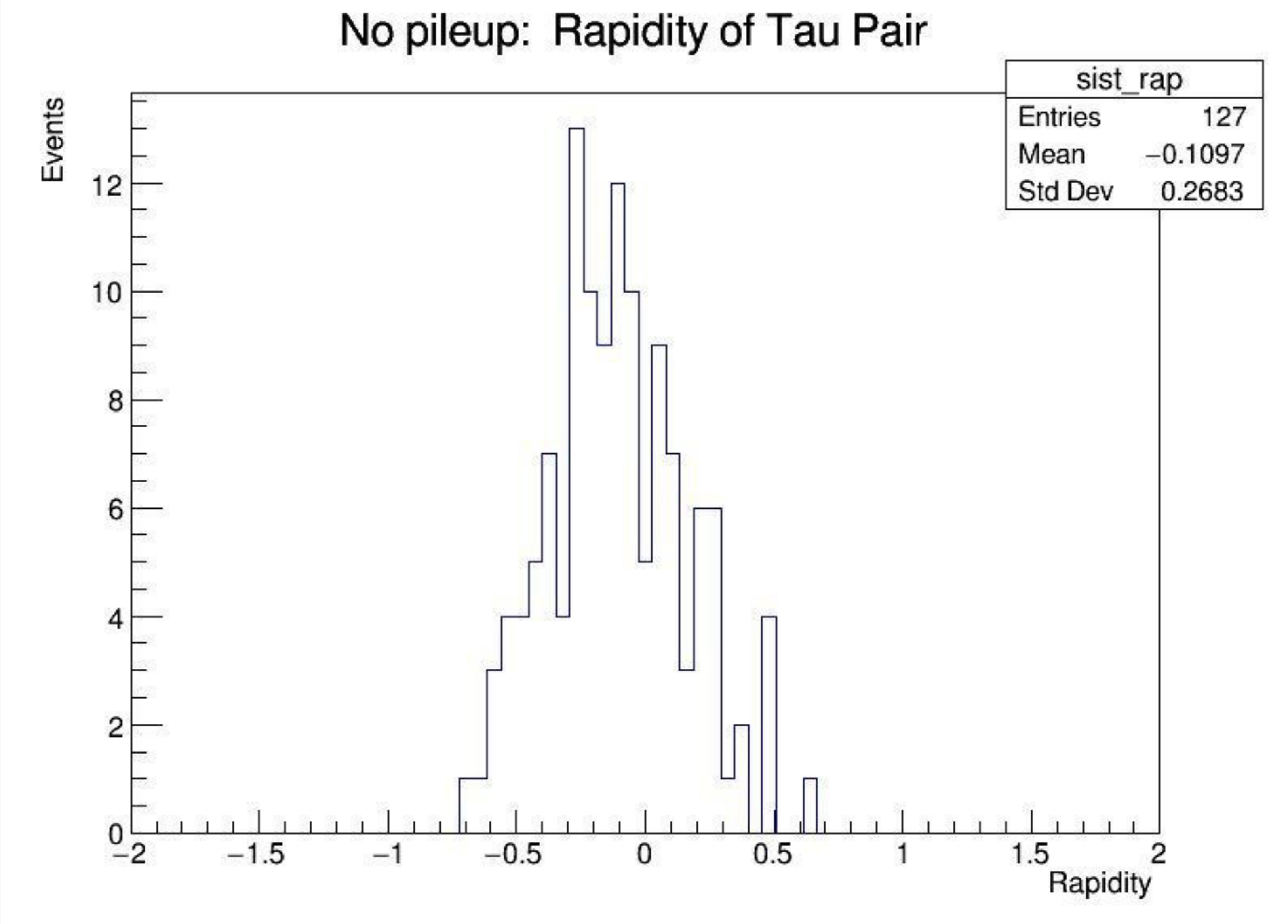
Plots

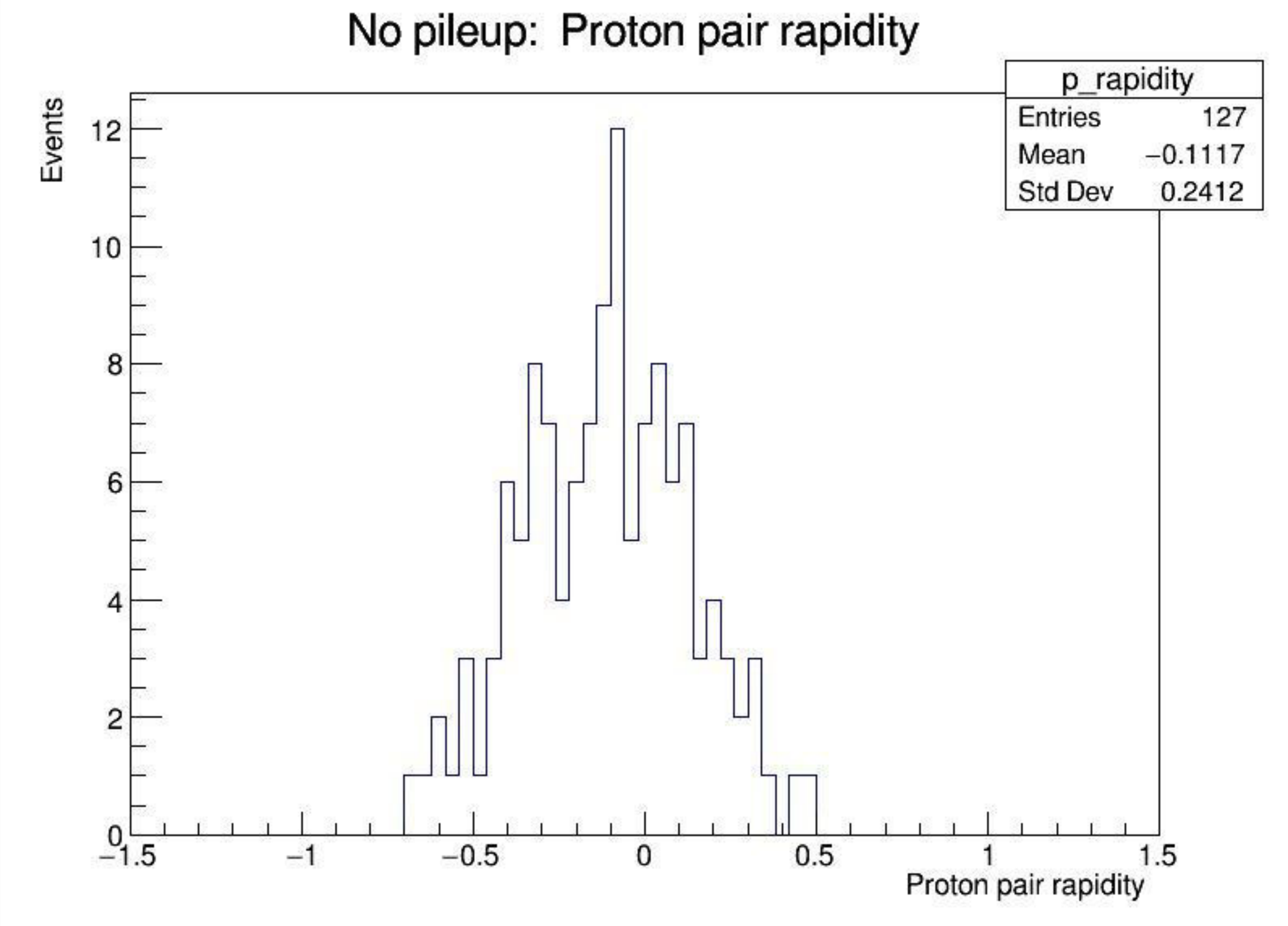




Fractional energy loss of protons tagged by PPS

Plots



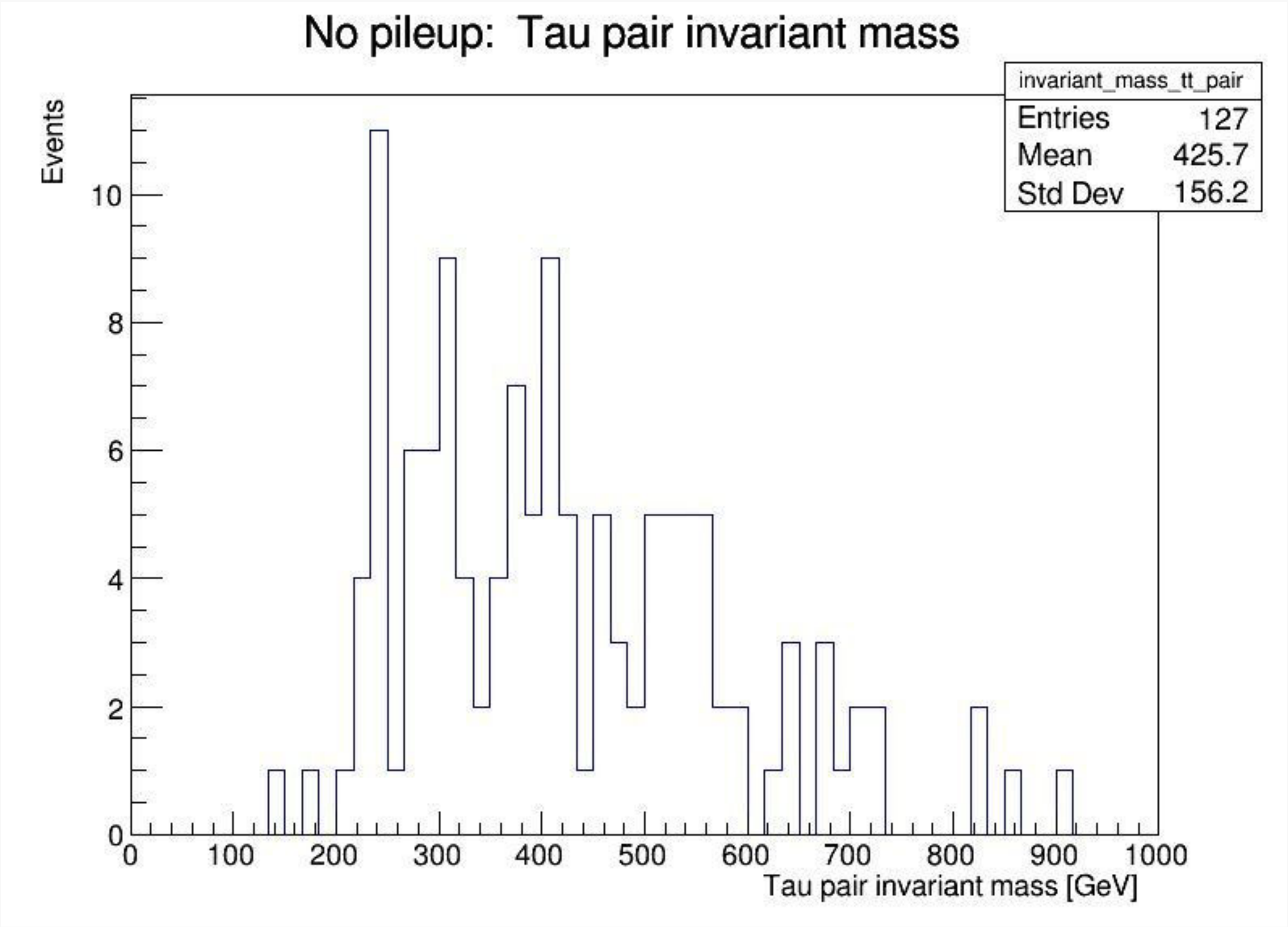


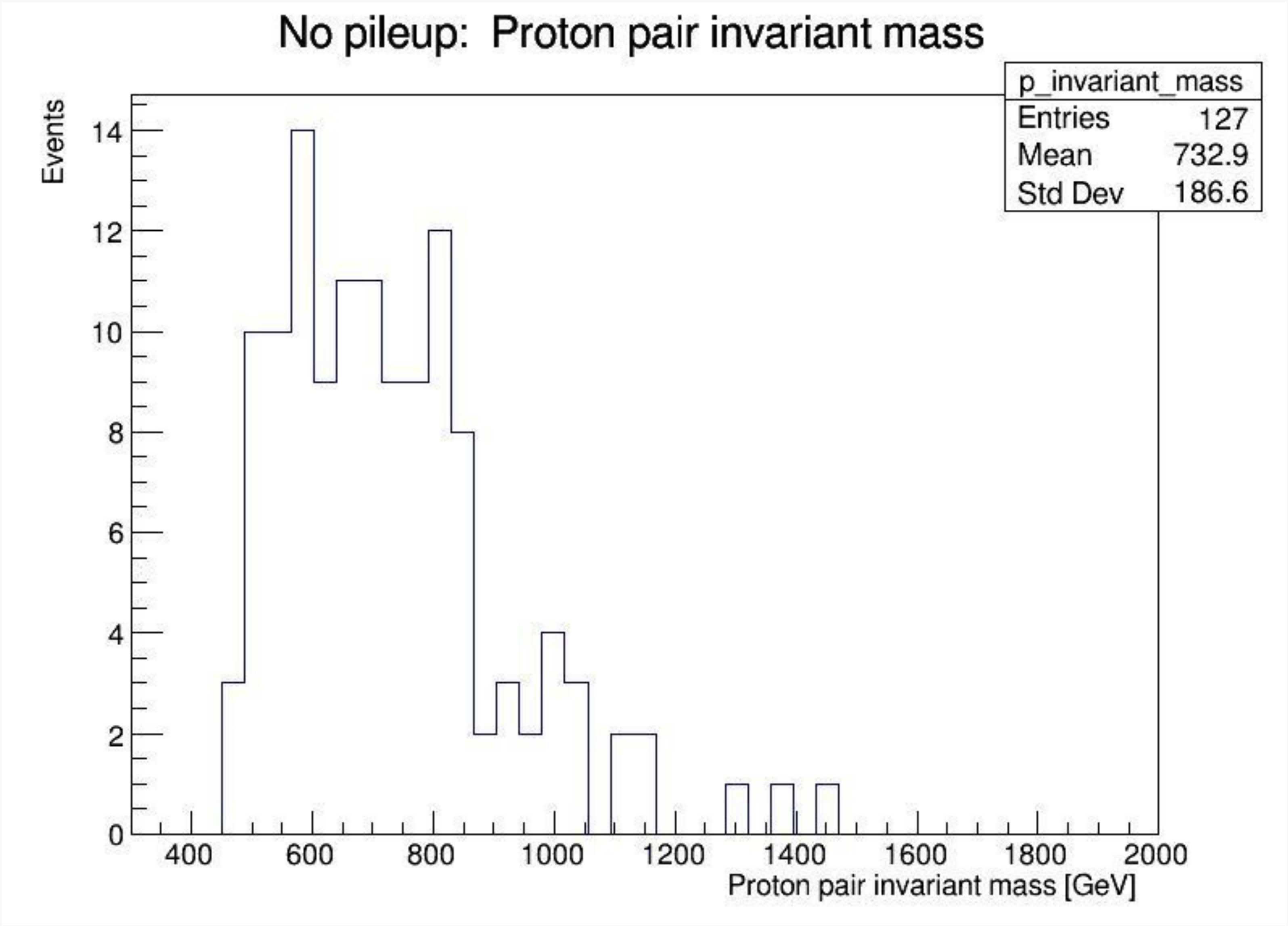
$$Y_X = \frac{1}{2}(\frac{E+p_z}{E-p_z})$$

$$Y_X = \frac{1}{2}log(\xi_1/\xi_2)$$

- E is total energy of tau pair
- P_z is total longitudinal moment of tau pair system

Plots



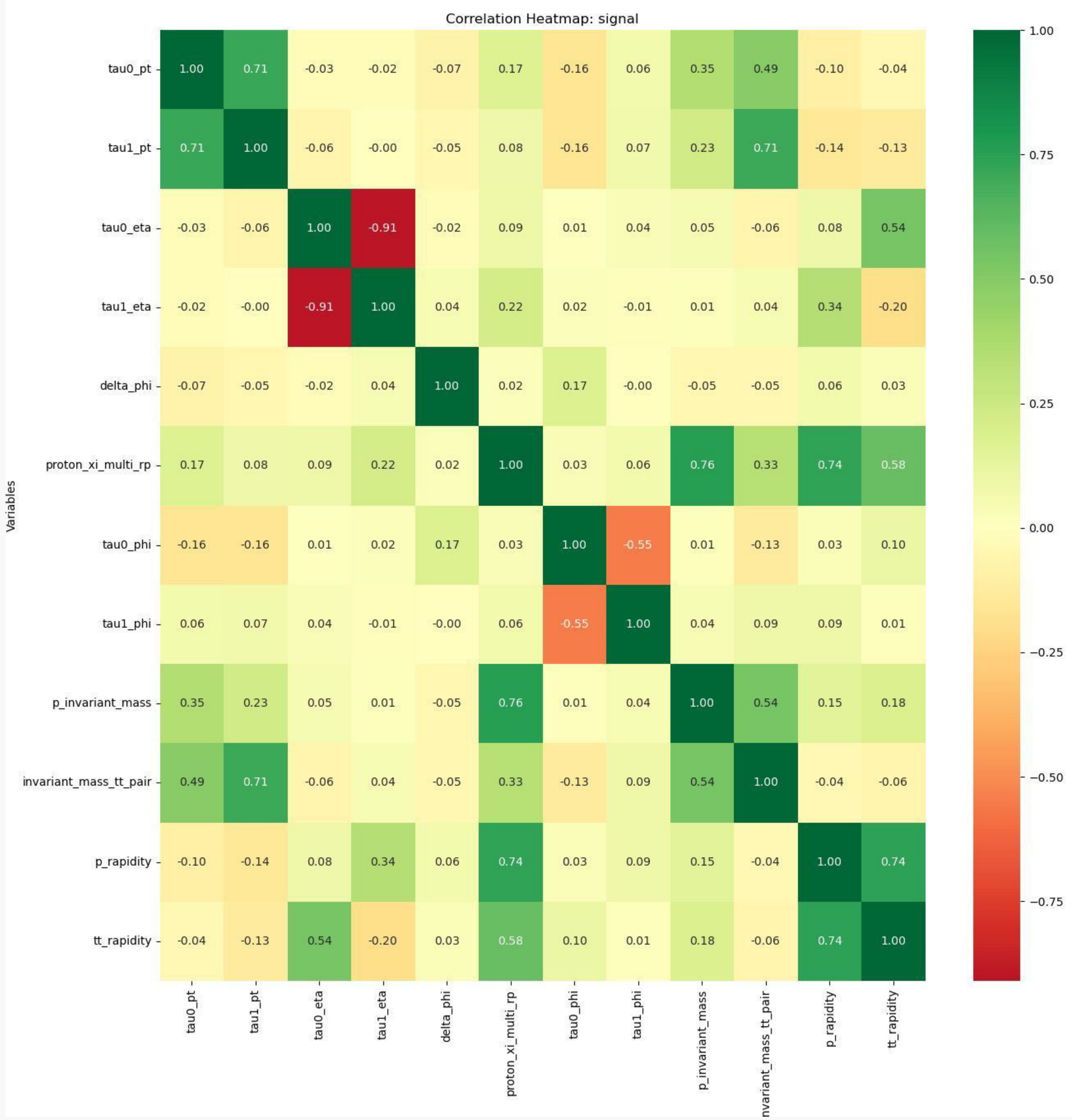


Used: $M^2 = 2pT_1pT_2(cosh(\Delta\eta) - cos(\Delta\phi))$

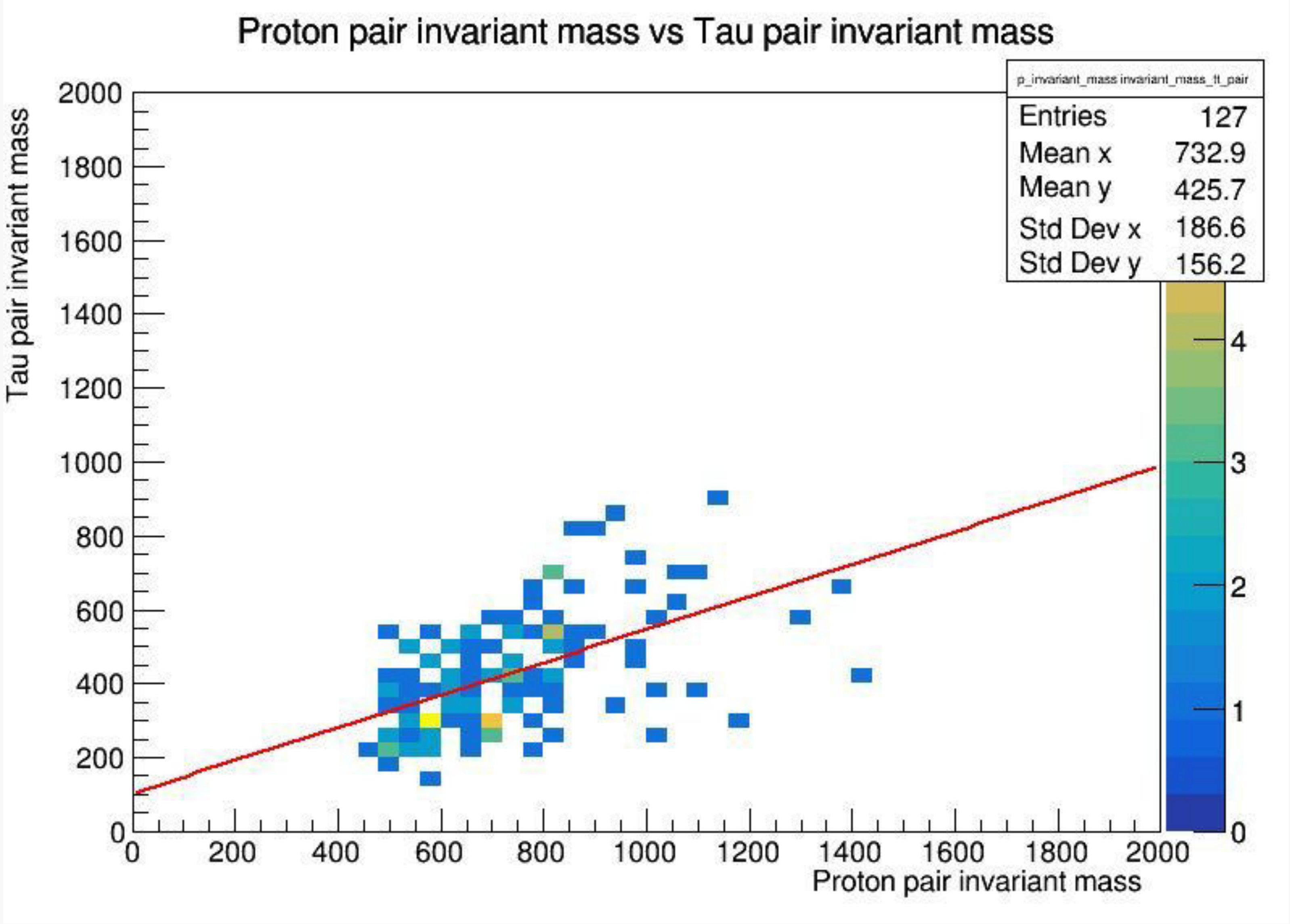
$M_X^2 = s\xi_1\xi_2$

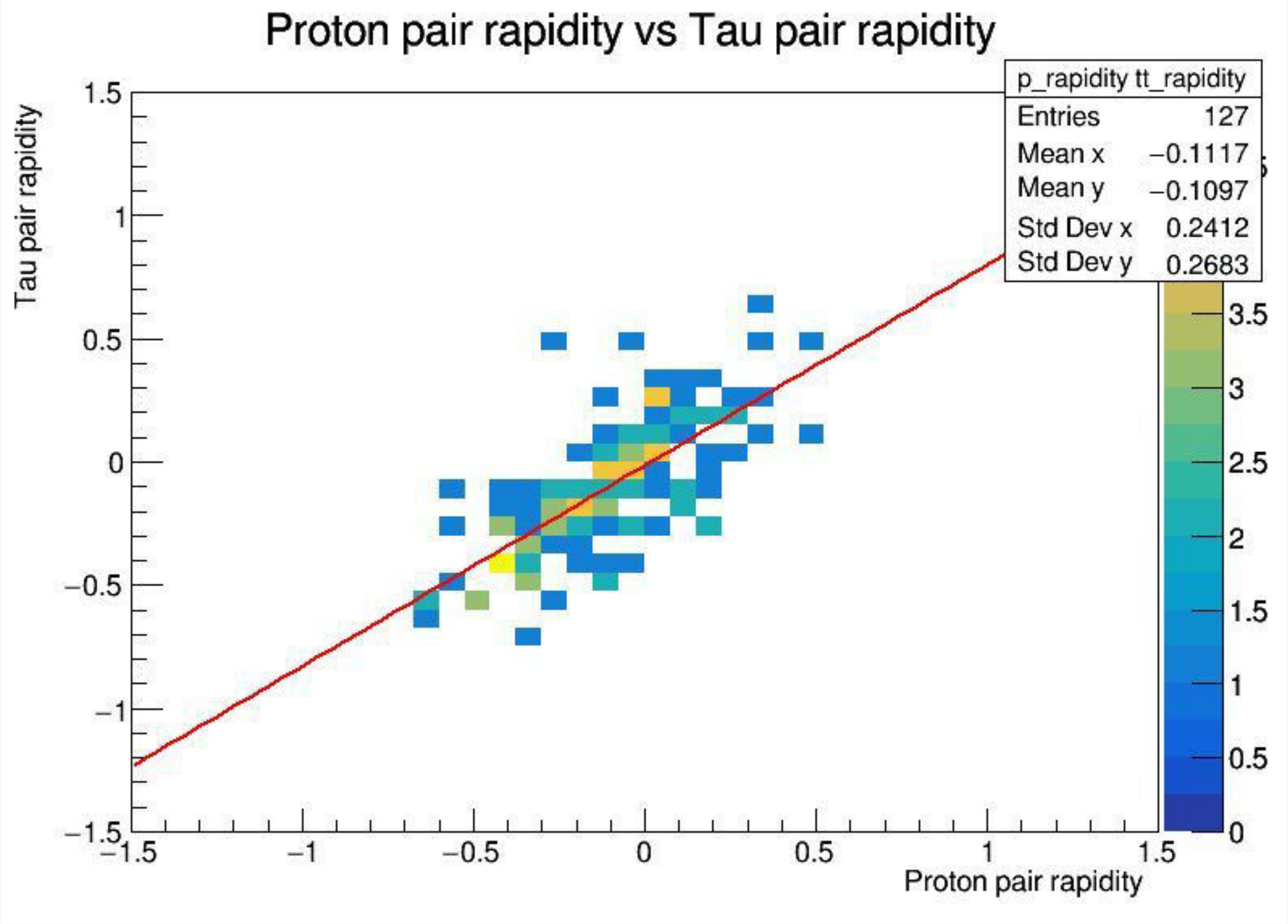
Change to formula that uses `TLorentzVector`

Variable Correlations (signal)



Variable Correlations (signal)





Variable Correlations (signal)

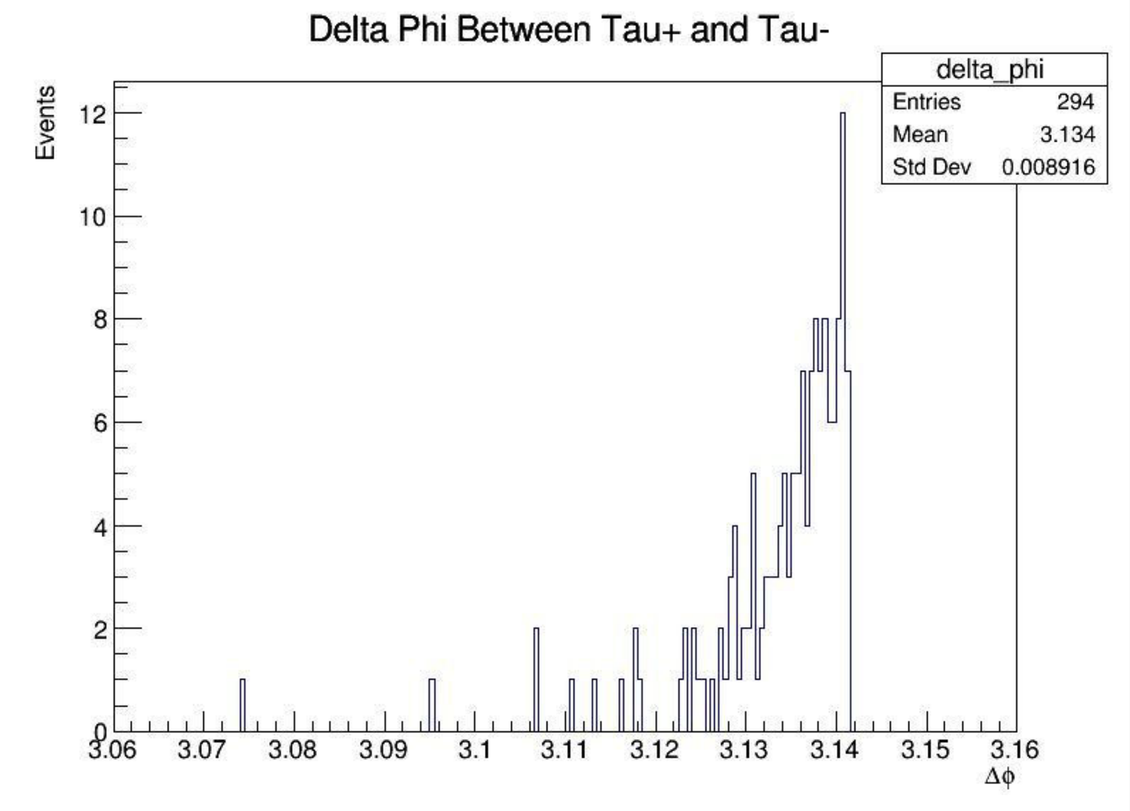
Next Steps

- **Repeat for sample with more events**
- **Add sample with QCD background**

01

Variables and distributions

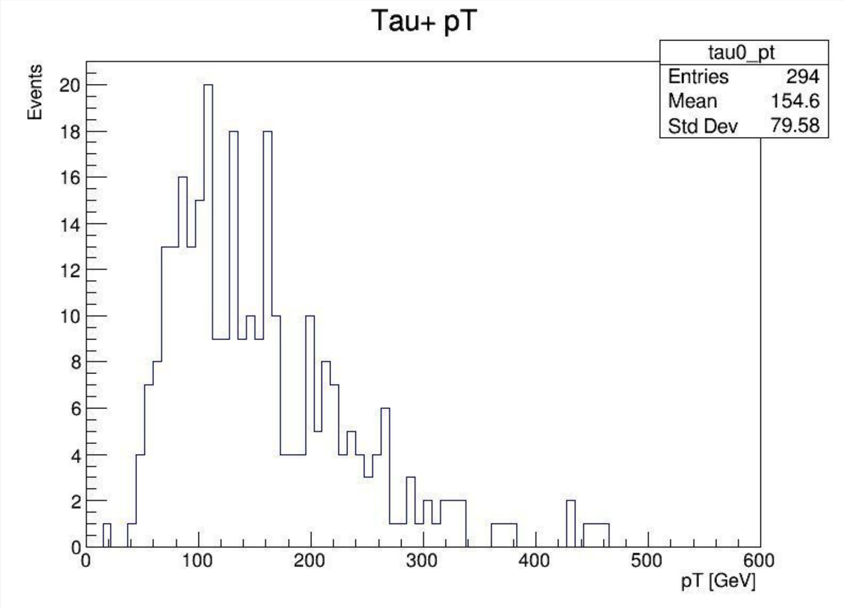
$\Delta\phi$

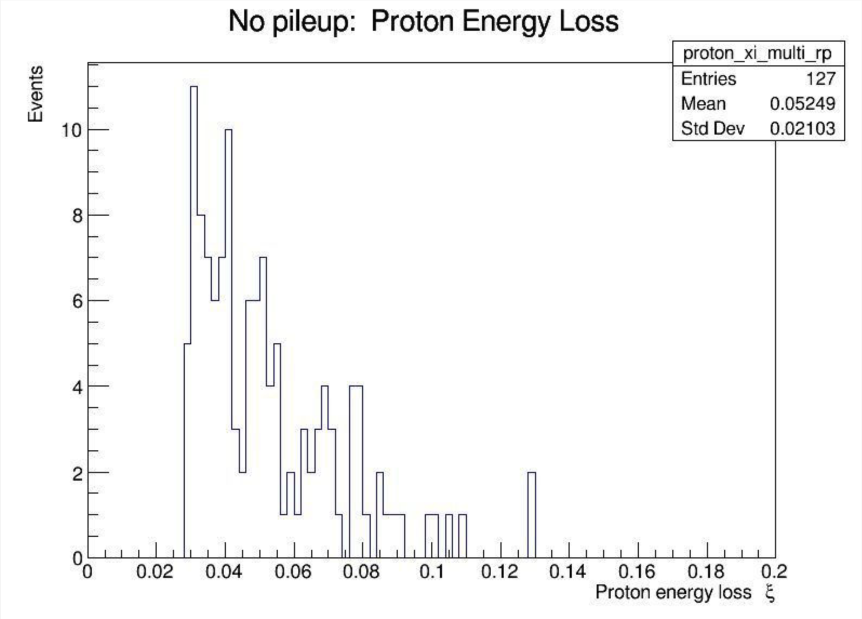


Delta Phi Between Tau+ and Tau-
delta_phi
Entries 294
Mean 3.134
Std Dev 0.008916
Events
0
2
4
6
8
10
12
3.06
3.07
3.08
3.09
3.1
3.11
3.12
3.13
3.14
3.15
3.16
Δϕ

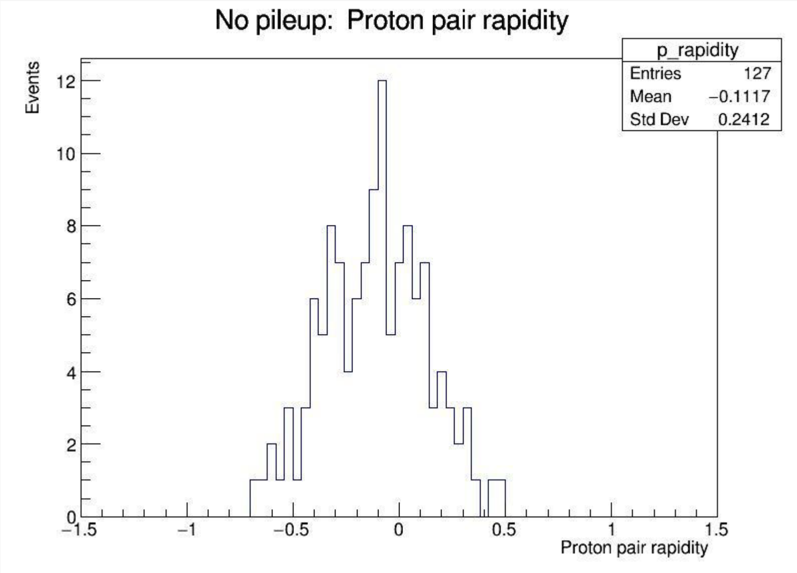
Pseudorapidity, η

Proton energy Loss, ξ

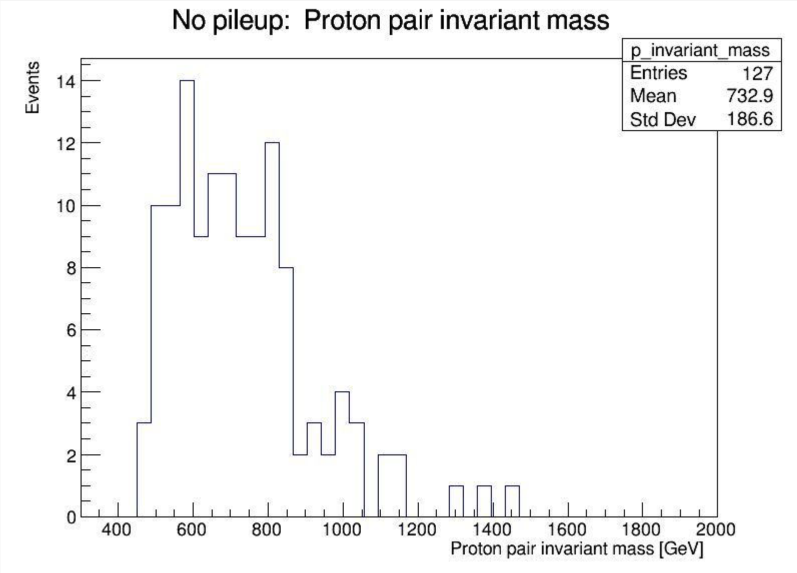




Proton pair rapidity

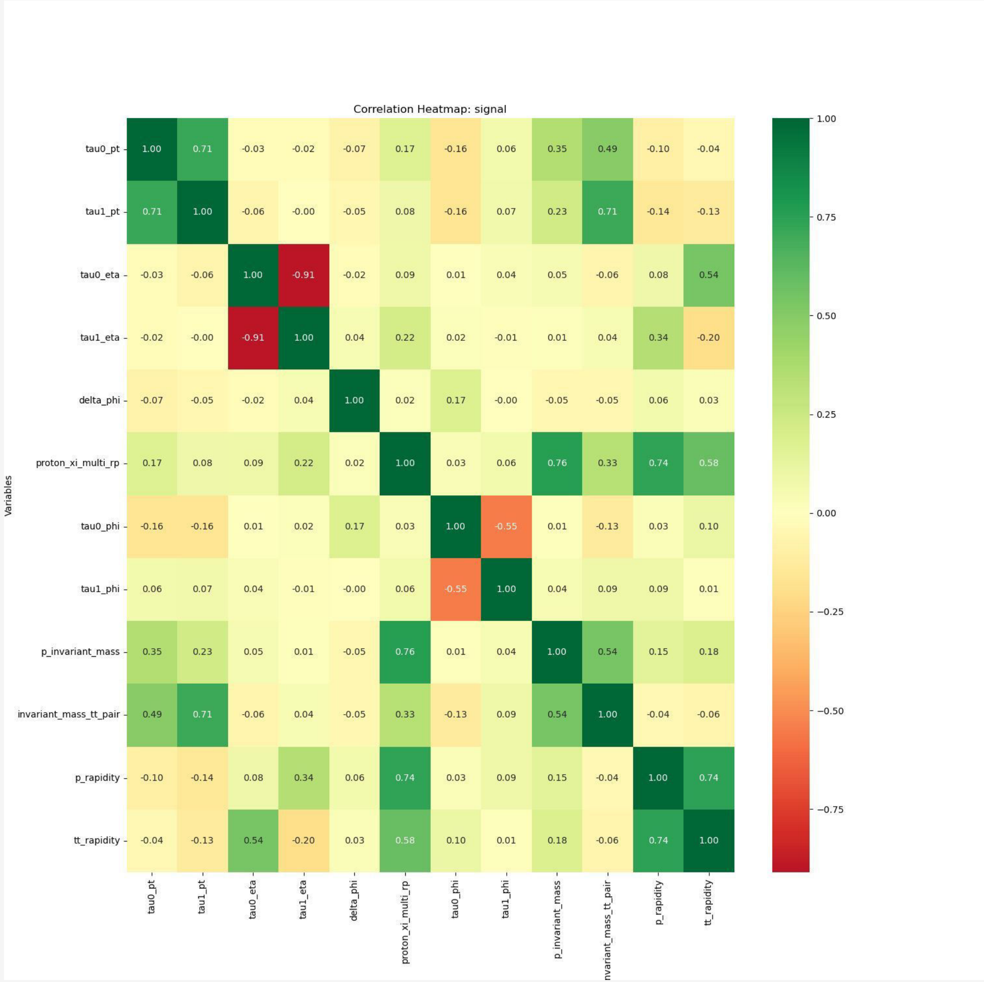


Proton pair invariant mass



02

Correlation between variables

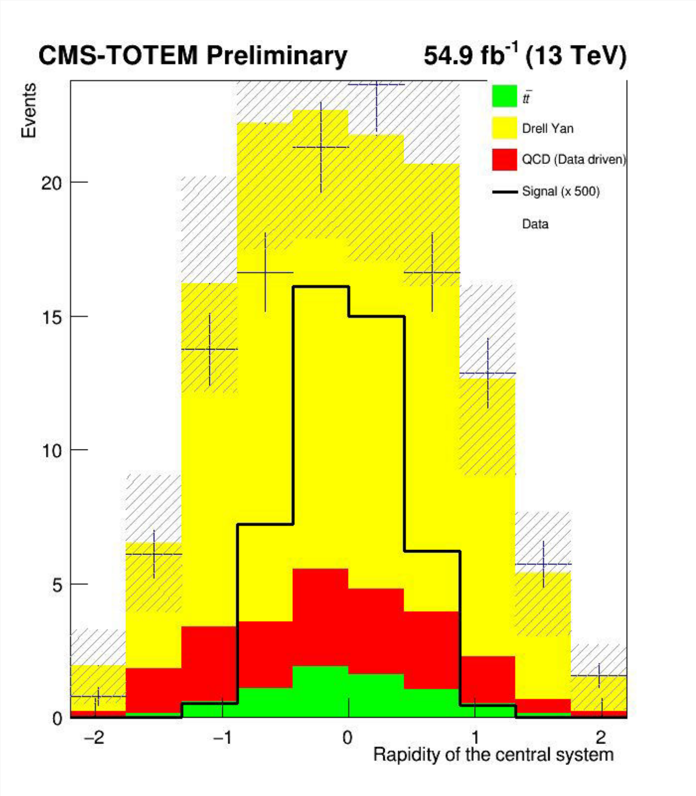


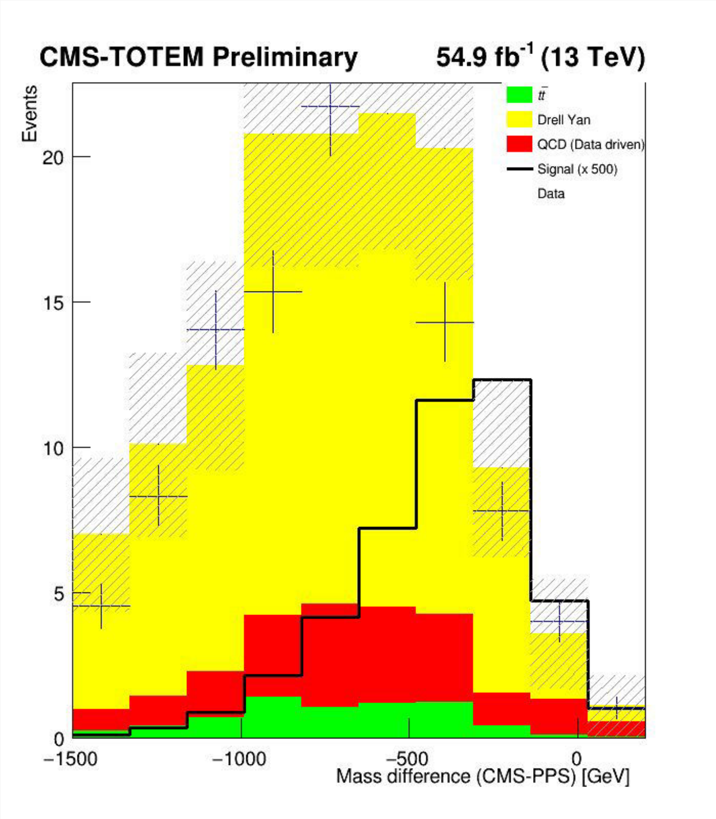
Expected:

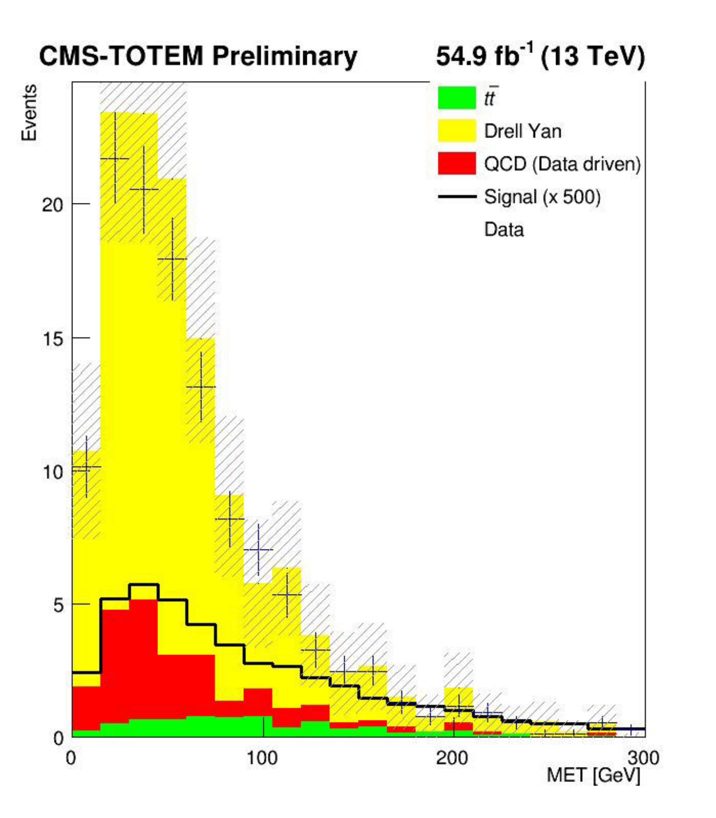
- Strong correlation between tau0 and tau1 phi
- Negative correlation between tau0 and tau1 eta
- Strong correlation between invariant mass of tau pair and tau0 and tau1 pt
- Some correlation between tau pair rapidity and tau0 eta

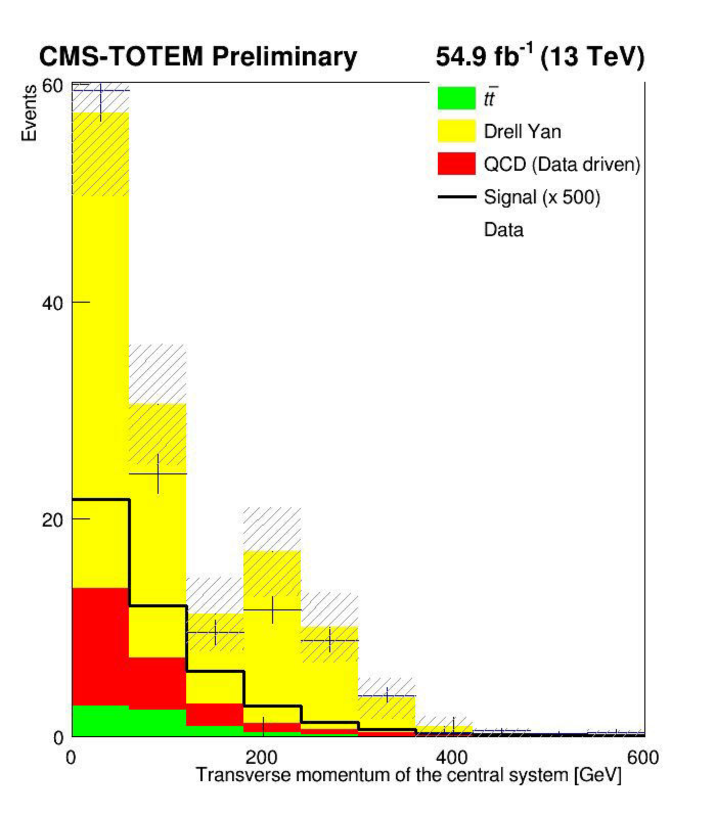
Input distributions to the BDT discriminator

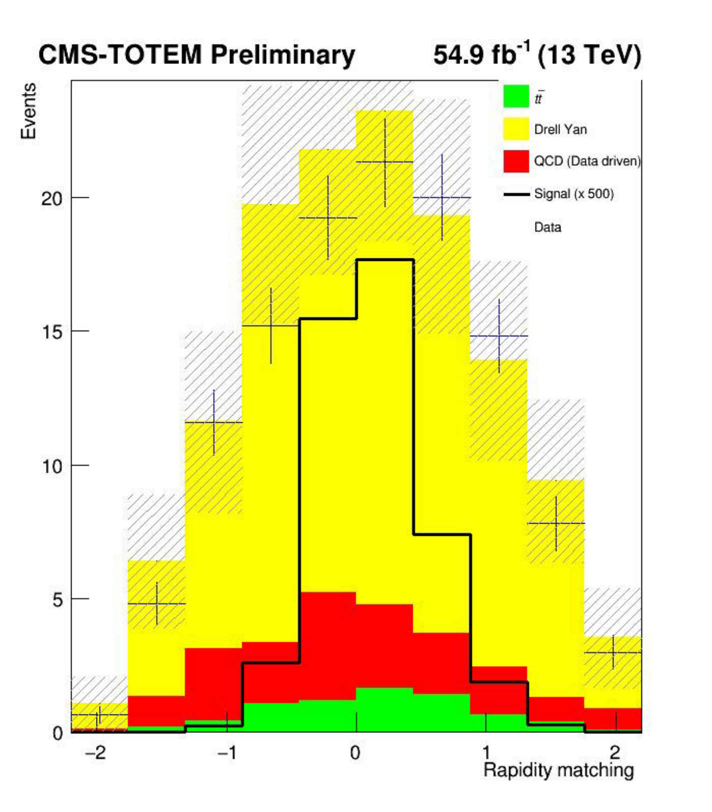
- Invariant mass of the central system
  → The total mass of all particles in the central detector, calculated from their 4-momenta.
- Rapidity of the central system
  → Describes how forward or backward the central system is moving along the beam axis.
- Mass difference
  → Difference between the central system mass and the mass reconstructed using the forward protons (CMS - PPS).
- MET (Missing Transverse Energy)
  → Represents undetected particles.
- Transverse momentum of the central system
  → Momentum of the central system perpendicular to the beam axis.
- Rapidity matching
  → Comparison of rapidity measured centrally..
- Transverse momentum of hadronic tau
  → Momentum perpendicular to the beam of the hadronically decaying tau lepton.
- Acoplanarity of central system
  → Measures how "non-back-to-back" the two leading tau candidates are; zero means perfect balance.

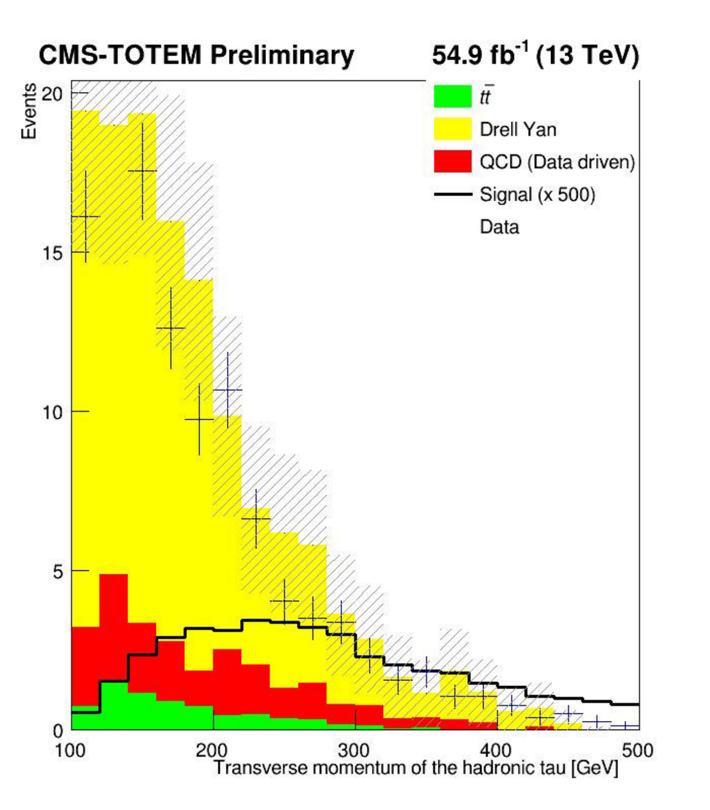

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Rapidity of the central system

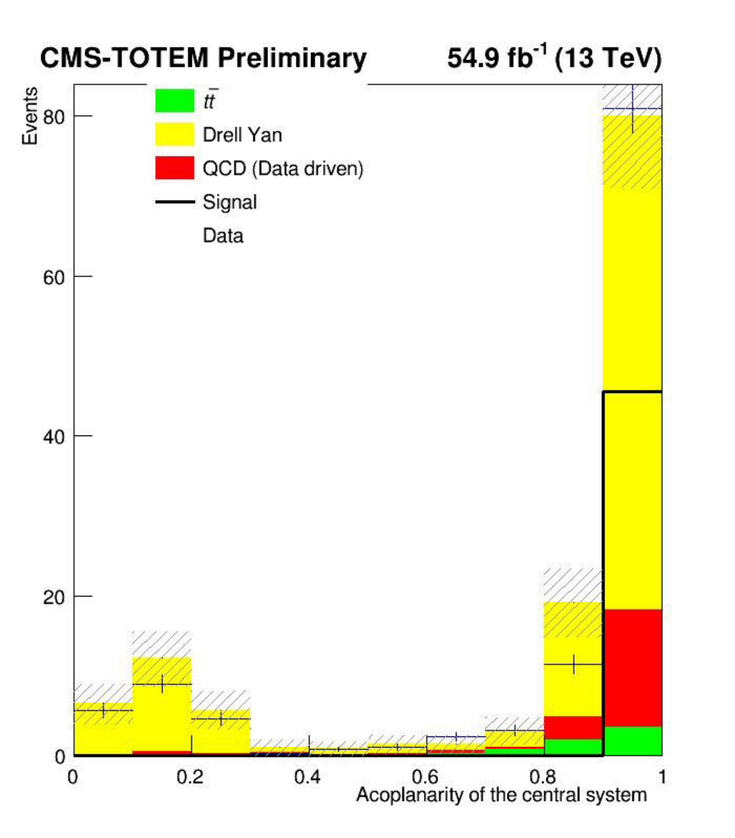

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Mass difference (CMS-PPS) [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Events
MET [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Events
Transverse momentum of the central system [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Rapidity matching


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Transverse momentum of the hadronic tau [GeV]


CMS-TOTEM Preliminary
54.9 fb-1 (13 TeV)
Events
tt̄
Drell Yan
QCD (Data driven)
Signal
Data
Acoplanarity of the central system
0
20
40
60
80
0
0.2
0.4
0.6
0.8
1

Weights applied in the previous plots

In 2018, due to an improved PPS detector configuration, silicon pixels are used. They allow the reconstruction of multiple protons and the scenario is different. Therefore, the technique used is the following:

- We estimate the probability $p_{n,m}$ to get n protons in arm 0 and m protons in arm 1;
- For the background samples, we observed that the probability to have at least one PU proton per arm is just 13%. Therefore, the weigth of each background event is rescaled of a factor 0.13;
- Subsequently, each background event is enriched with n protons in arm 0 and m protons in arm 1, according to the probability $p_{n,m}$ calculated earlier. If we get more than one proton per arm, the pair with the largest ξ is chosen;
- For the signal samples, they are directly enriched with the number of protons calculated according to the probability $p_{n,m}$.

```
//double w_data=1.;
double w_qcd=1.*0.8*0.13;
double w_ttjets=0.15*0.8*0.13;
double w_dy=1.81*0.8*0.13;
double w_sinal;
double w_data=1*0.13;
```

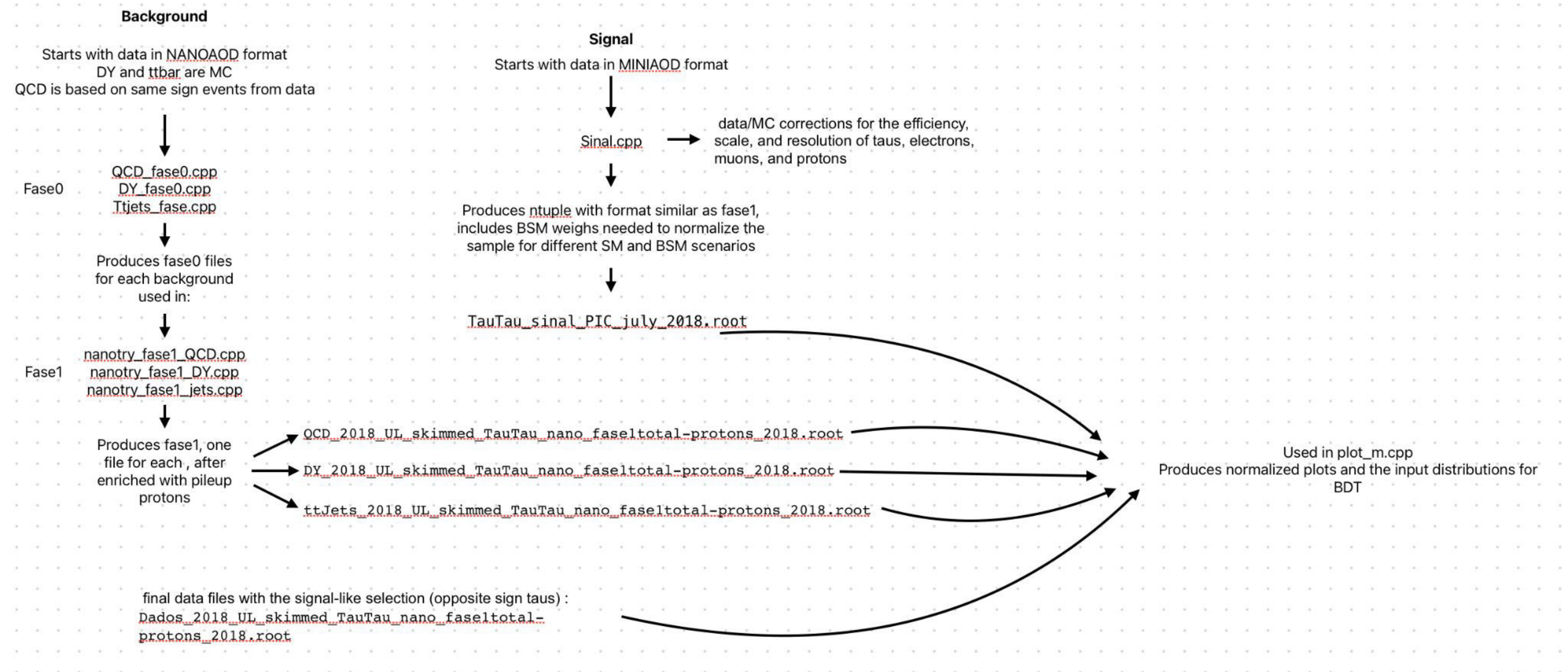
```
double weight = .15; //w= Numero monte carlo / Numero de 2018 , Numero de 2018 = luminosidade de 2018 * secção de 2018
```

0.13 is due to pps reconstruction technique -> probability of having at least one proton per arm

Still no explanation for 0.8 factor on the background only -> maybe some arbitrary normalization factor

03
Code specifications

Dataflow


Background
Starts with data in NANOAOD format
DY and ttbar are MC
QCD is based on same sign events from data
Fase0
QCD_fase0.cpp
DY_fase0.cpp
Ttjets_fase.cpp
Produces fase0 files for each background used in:
Fase1
nanotry_fase1_QCD.cpp
nanotry_fase1_DY.cpp
nanotry_fase1_jets.cpp
Produces fase1, one file for each , after enriched with pileup protons
QCD_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
DY_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
ttJets_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
final data files with the signal-like selection (opposite sign taus) :
Dados_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root
Signal
Starts with data in MINIAOD format
Sinal.cpp
data/MC corrections for the efficiency, scale, and resolution of taus, electrons, muons, and protons
Produces ntuple with format similar as fase1, includes BSM weighs needed to normalize the sample for different SM and BSM scenarios
TauTau_sinal_PIC_july_2018.root
Used in plot_m.cpp
Produces normalized plots and the input distributions for BDT

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
  System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
  Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1)

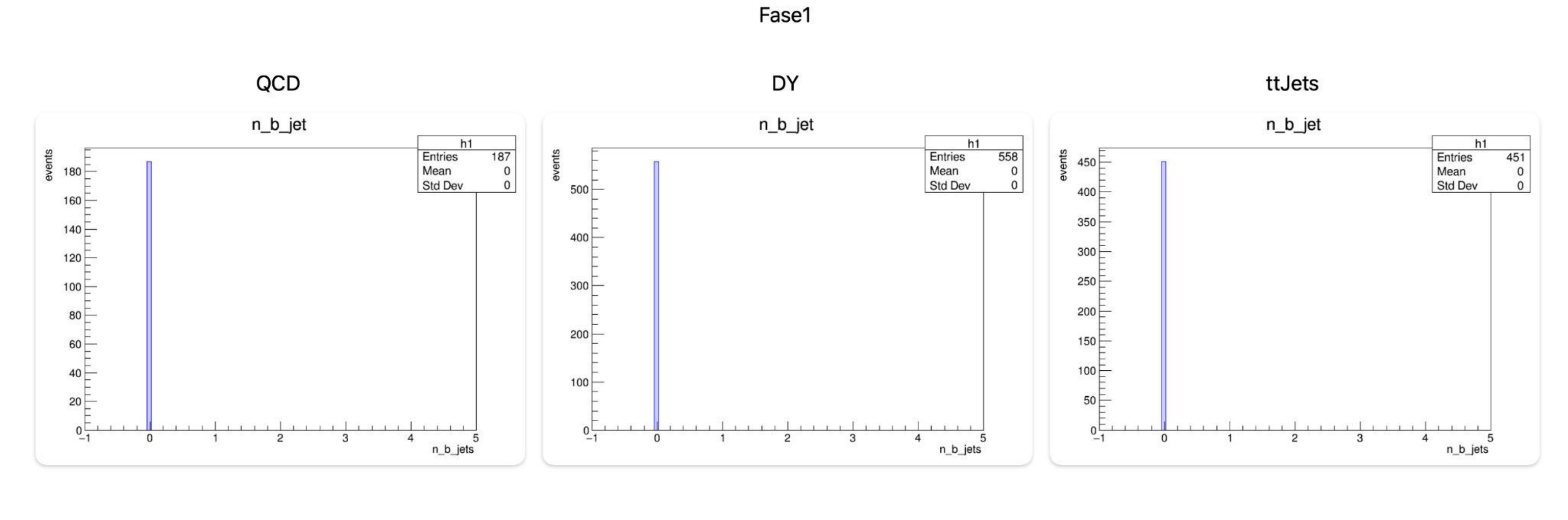
Code organization
Background

Goal: only one file for signal and one file for background

(with parallelization using rootEnableImplicit)

File name	reads		Output
DY_fase0	`DY_2018_UL.txt`		`DY_2018_UL_skimmed_TauTau_nano`
QCD_fase0	`QCD_2018_UL.txt` `dadosluminosidade.txt`	`Checks if luminosity in event matches with the data from file`	`QCD_2018_UL_skimmed_TauTau_nano_`
ttJets_fase0	`ttJets_2018_UL.txt`		`ttJets_2018_UL_skimmed_TauTau_nano_`
nanotry_fase1_DY	`DY_2018.txt` `Name of fase0 output files`		
nanotry_fase1_QCD	`QCD_fase1_PIC.txt` `Name of fase0 output files`		
nanotry_fase1_ttJets	`ttJets_fase1_PIC.txt` `Name of fase0 output files`		

Number of b jets: 0 for all samples

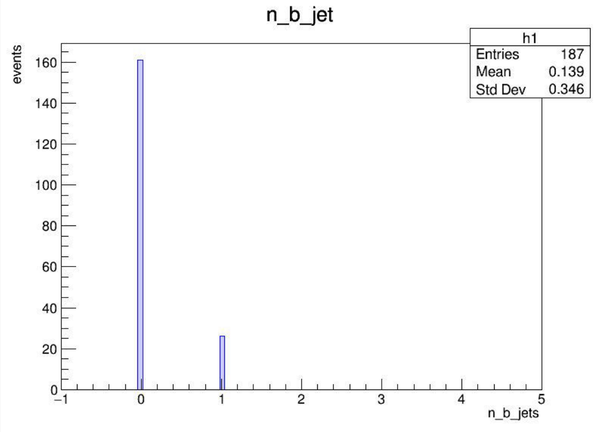


Cause: mismatch in the declaration of variable type in the proton enrichment code

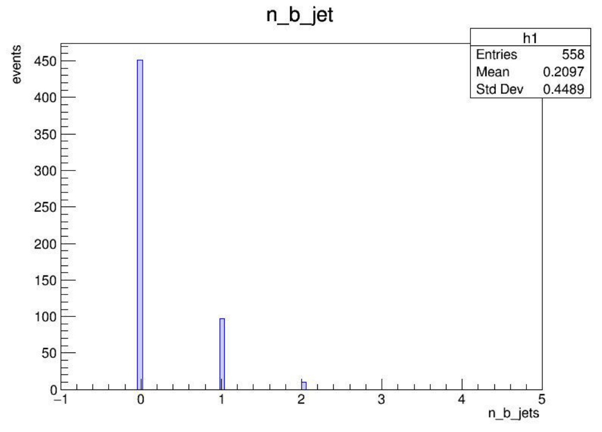
- Mismatch in the declaration of variable in the proton enrichment code (to introduce pileup protons);
    - N_b_jet was declared as *double* when it originally was *int*
- Handling of branch addresses

-> Fixed by passing the merged samples from phase1 through the fixed proton enrichment code

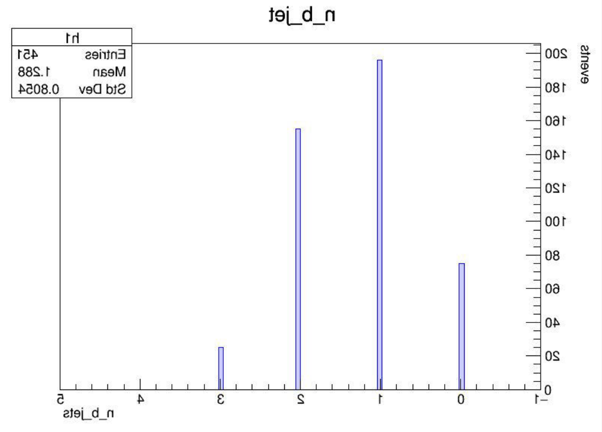
After fixing n_b_jet: Fase 1 samples



qcd



dy



ttjets

Control region observations

- worse data/simulation ratio for higher values on most variables;
- In (original) QCD CR the range of the plot of invariant mass of the central system [400,1000] does not add up with the limits of the control region [100,300] -> this was changed in the new plots;
- Also in QCD CR there are some discrepancies between the shape of the invariant mass of the central system and transverse momentum of central system plots;
- In TTJETS CR the cutoff of the transverse momentum also doesn't match the CR cut (in the original the plot goes to around 240 GeV, but the cut is pt<125 GeV;
- In the DY CR plots, the shapes look mostly similar, except for the invariant mass of the central system.

Exploring new physics with proton interactions at the LHC

Process: $pp \to pX(\to \tau\tau)p$;

Protons are tagged by PPS, while the decay products of the X system are detected by the CMS central detector.

Decay mode in study:

- Fully hadronic decay mode; Both τs decay hadronically

Index

Weights applied in the previous plots

In original code

```
weight_sample = 54900.*(1.)/(4000.*1000.);
```

- Weight_sample is normalization factor
- weight_sample=0.013725

$$\text{weight\_sample} = \frac{\mathcal{L} \cdot \sigma}{N_{\text{gen}}}$$

What worked to make the plots the same:

Is original weight_sample/1000 -> UNITS

```
weight_sample= 0.000013725;
```

Code organization
Background-> fase0 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	$t\bar{t}$ MC	Data-driven QCD
Luminosity block filtering			✓
Extra gen matching	✓	✓	
Generator weight	1.81	0.15	
HLT version	40_Trk1_eta2p1	40_Trk1_eta2p1	35_Trk1_eta2p1

Code organization
Background-> fase1 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	tt̄ MC	Data-driven QCD
Scale Factors	Tau ID + energy scale + sys	Tau ID + energy scale	
Systematic Variations	full		
Charge Selection	Opposite Sign (<0)	Opposite Sign (<0)	Same Sign (>0)
Tau Selection Cuts	pt > 100, η < 2.3	pt > 100, η < 2.4	pt > 100, η < 2.4
Gen-Match Handling	yes		

BDT training: specifications

Variables used:

Variable	Description	Units
`sist_acop`	Central system acoplanarity	dimensionless
`sist_pt`	Total momentum	GeV
`sist_mass`	Invariant mass	GeV
`sist_rap`	System rapidity	dimensionless
`met_pt`	Missing energy	GeV
Mass matching	`sist_mass − sqrt(13000²×ξ₁×ξ₂)`	GeV
Rapidity matching	`sist_rap − 0.5×log(ξ₁/ξ₂)`	dimensionless

TMVAClassification	Train BDT/ML models	Background file Signal file	Output_TMVA_tautau_2018.root Dataset weights
TMVAClassificationApplication	Apply trained models to data	Trained models Data samples: dy, ttjets,qcd and signal	On file for each type containing BDT scores for every event
Bdt_output	Process BDT results	TMVApp_*.root Event weights, sample information	Analysis histograms Bdt score distributions signal/background separation cuts Cut optimization results
TMVAOutPut_Distributions	Create TMVA evaluation plots	Output_TMVA_tautau_2018.root Training results Test results Method performance data	ROC curves Variable importance plots Correlation matrices

04
Control regions

In red: not applied

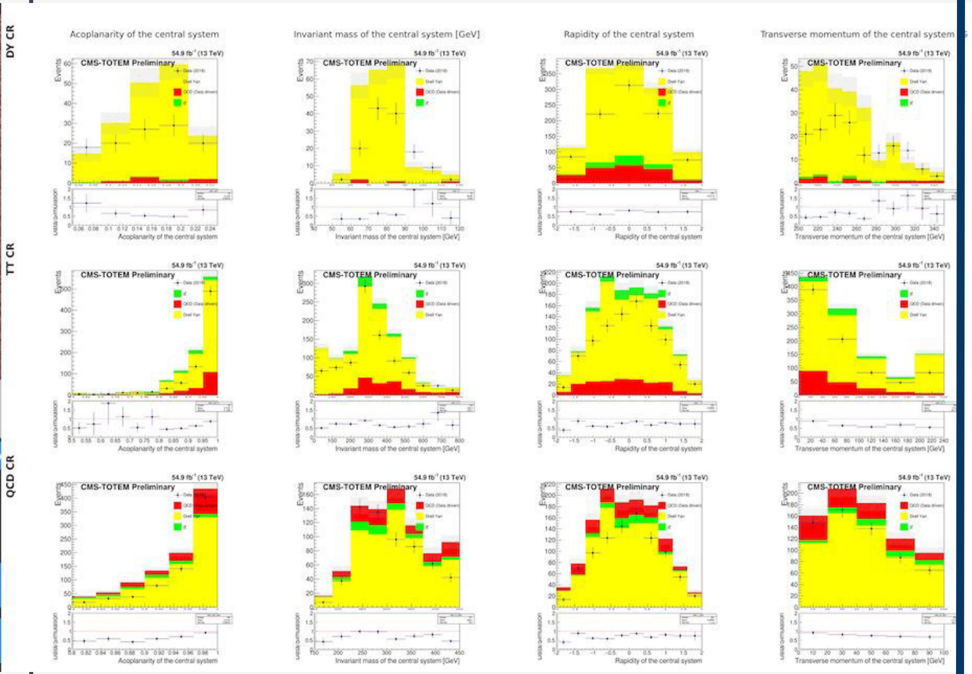
In green: applied in phase 0 or 1

Control regions: τhτh

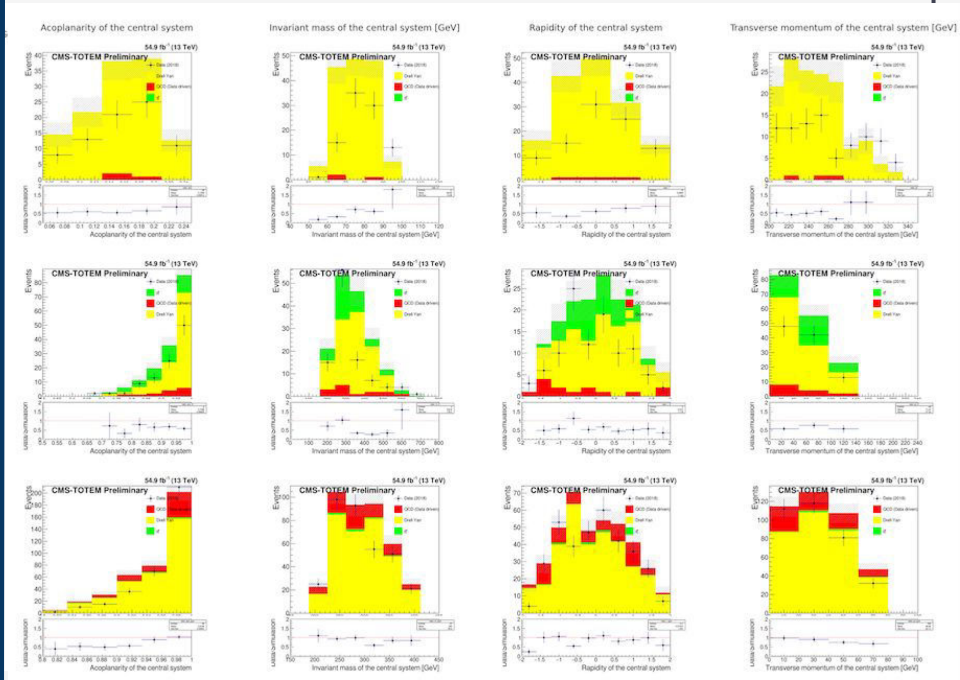
	Invariant mass (GeV)	acoplanarity	η	2 isolated ts	N of b-jets	p_t
DY	[40,100]	<0.3	<2.4	Opposite sign	0	
QCD	[100,300]	>0.8	<2.4	Same sign	0	<75
ttjets	[200,650]	>0.5	<2.4	Opposite sign	1 or more	<125

On top: original from analysis note
Bottom: new plots

Before cuts



After cuts



samples used and information

Background	Dataset	Cross section (pb)
Drell Yan (DY)	/DYJetsToLL_M*_TuneCP5_13TeV-amcatnloFXFX-pythia8/RunIISummer20UL18NanoAODv9-106X*/NANOAODSIM	6404
inclusive $t\bar{t}$	TTJets_TuneCP5_13TeV-amcatnloFXFX-pythia8/NANOAODSIM/106X_upgrade2018_realistic_v16_L1v1-v1	832
QCD, $e\tau_h$ channel	Run2018*/EGamma/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v1	data-driven – luminosity 54.9 fb^{-1}
QCD, $\mu\tau_h$ channel	Run2018*/SingleMuon/NANOAOD/UL2018_MiniAODv2_NanoAODv9-v2	data-driven – luminosity 54.9 fb^{-1}

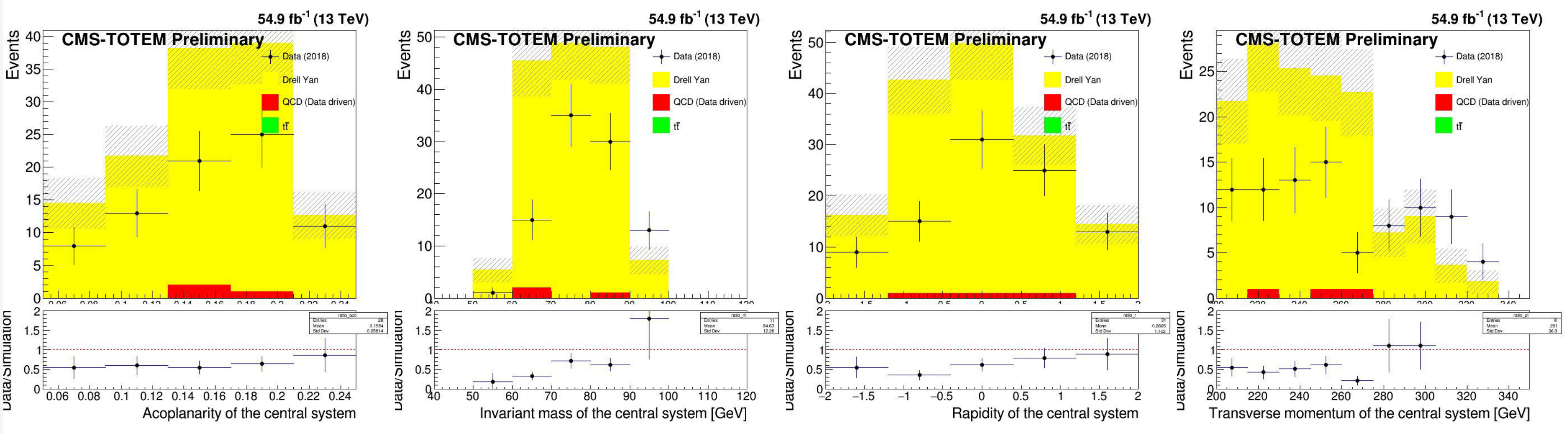
Table 3: List of the background considered. As for the Drell-Yan (DY), only events with M_X in the range [50,3000] GeV are considered (according to PPS acceptance [6]). For QCD, a data-driven approach is considered and data are used. This table refers to the 2018 data-taking period.

Questions

- The samples I am using (the ones mentioned in the twiki) come from the same datasets (referenced in the previous slide)?
-

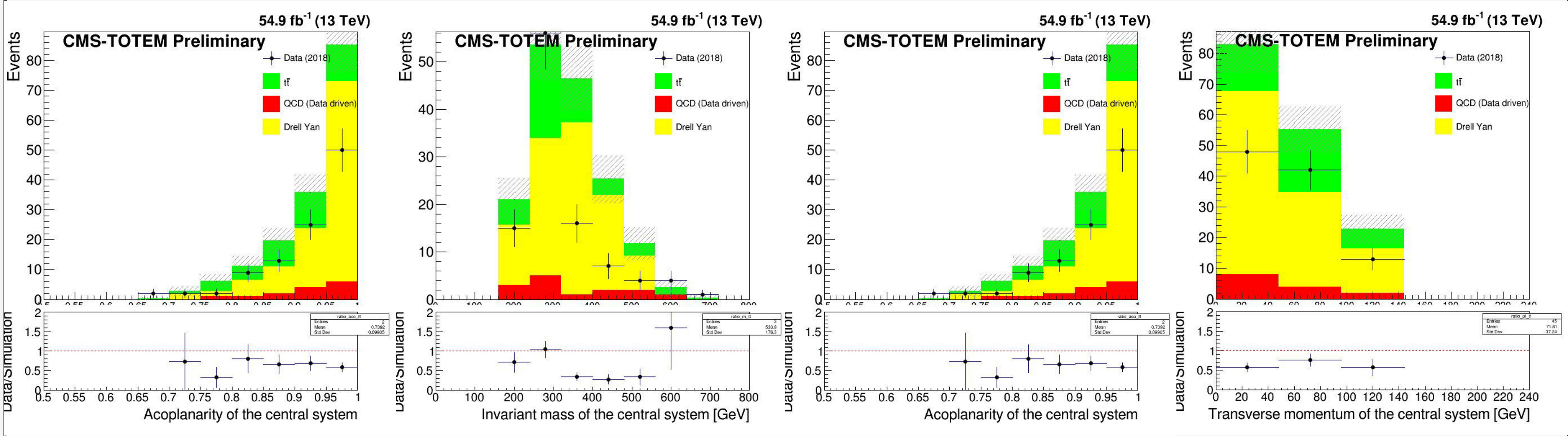
Control region plots - DY control region

tau mass [40,100]
system acoplanarity <0.3
nº of b jets =0



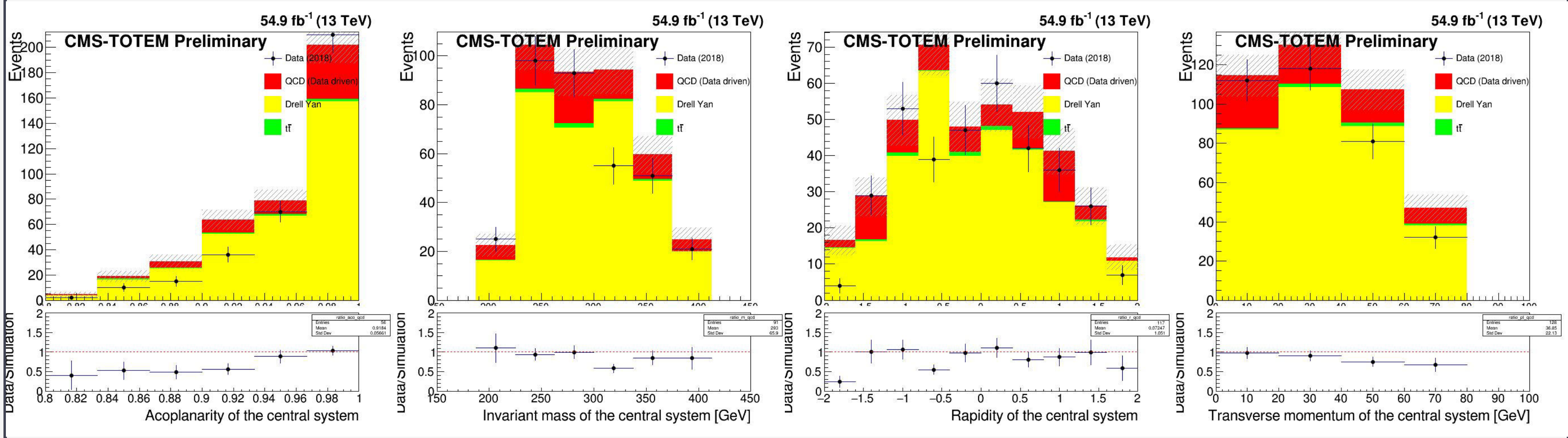
TT Control region

tau mass [200,650]
system acoplanarity >0.5
nº of b jets > 0
tau pair pt < 125



QCD control region

tau mass [100,400]
system acoplanarity >0.8
nº of b jets = 0
tau pair pt < 75



title

title

title

title

To dos:

In progress:

- Writing code to parallelize phase0 for all samples (using Rdataframe):
    - DY
    - QCD
    - TTjets ✓ (from 2-3 days to ~ 2 hours)
    - data
    - signal

Variable plots

- First sample: simulated signal 23k events → slide 3
  - Correlation between variables → slide 9

- Recovered data and code files → start in slide 20

Process: pp→pX(→ττ)p;

Protons are tagged by PPS, while the decay products of the X system are detected by the CMS central detector.

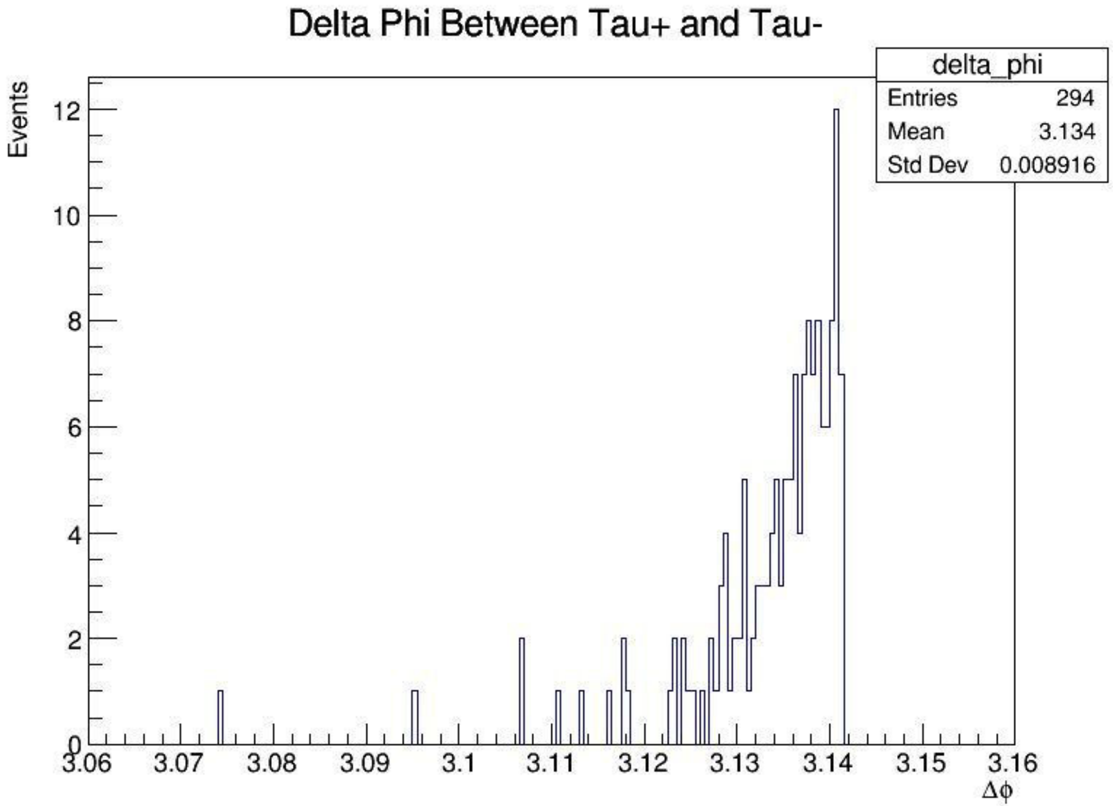
Decay mode in study:

- Fully hadronic decay mode; Both τs decay hadronically

Fase0 code 23k -> 294 events
127 for no pileup protons

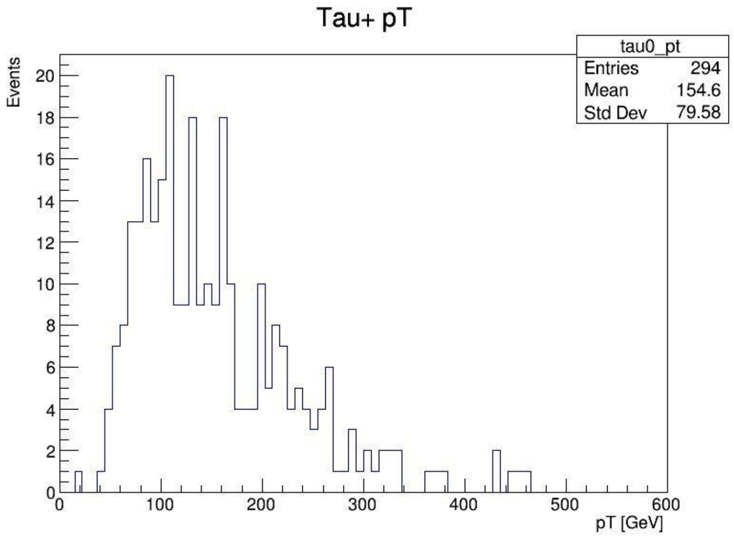
- Monte Carlo generated signal (23k events before processing);
- Events selected : Where trigger `HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg` is applied
- Protons are selected by checking if there is one on each arm. If there is more than one, the one with largest ξ is selected.
- proton selection was fixed
- Inv mass calculation for tau pair was changed

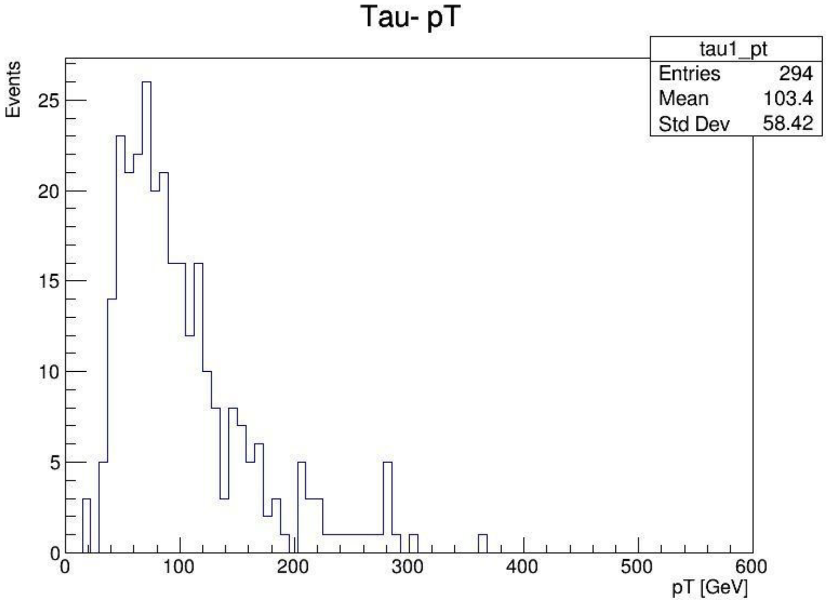
Δϕ



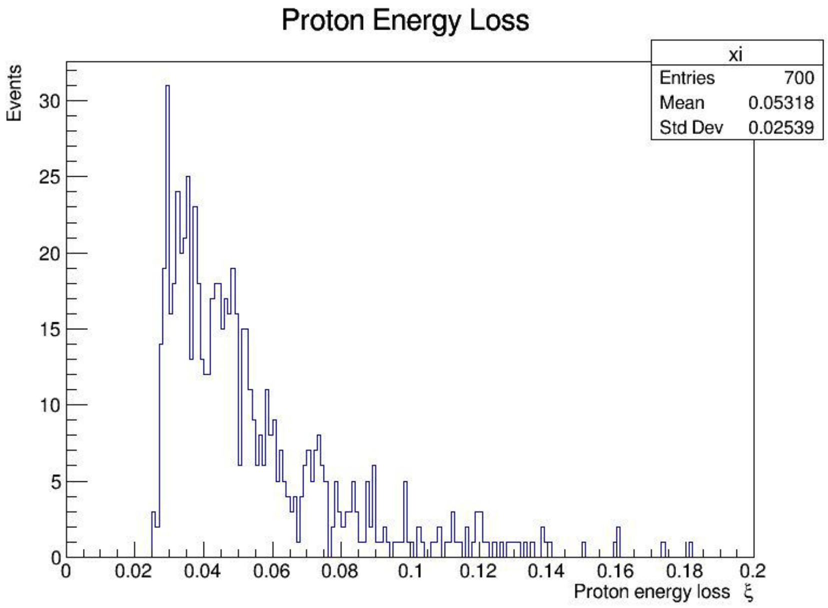
Delta Phi Between Tau+ and Tau-
delta_phi
Entries 294
Mean 3.134
Std Dev 0.008916
Events
0
2
4
6
8
10
12
3.06
3.07
3.08
3.09
3.1
3.11
3.12
3.13
3.14
3.15
3.16
Δϕ

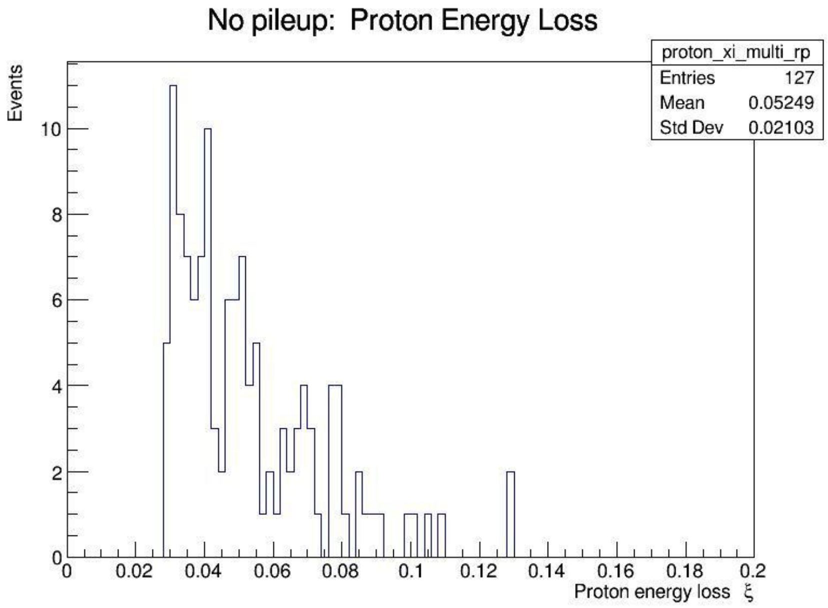
Pseudorapidity, η



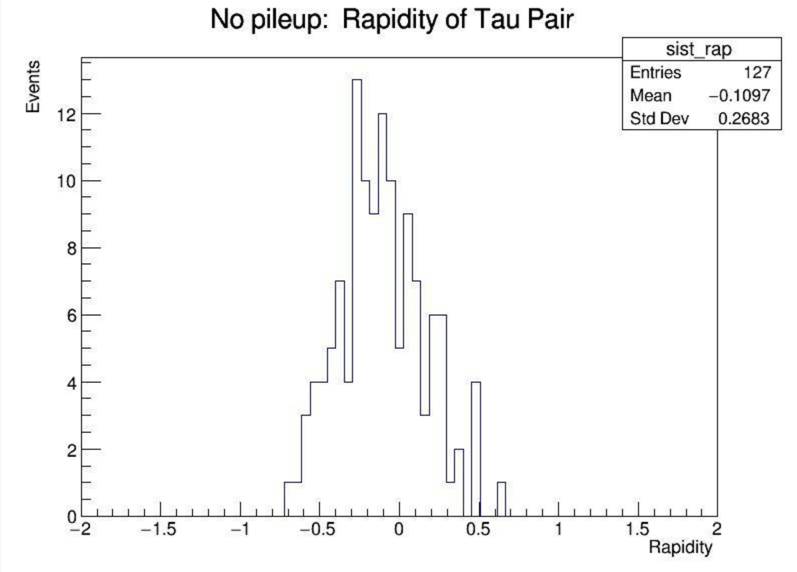


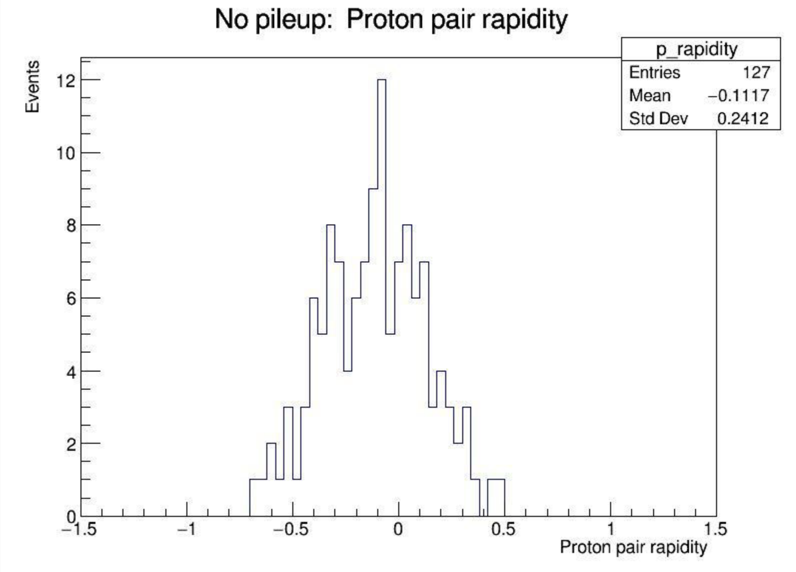
Proton energy Loss, ξ





Plots



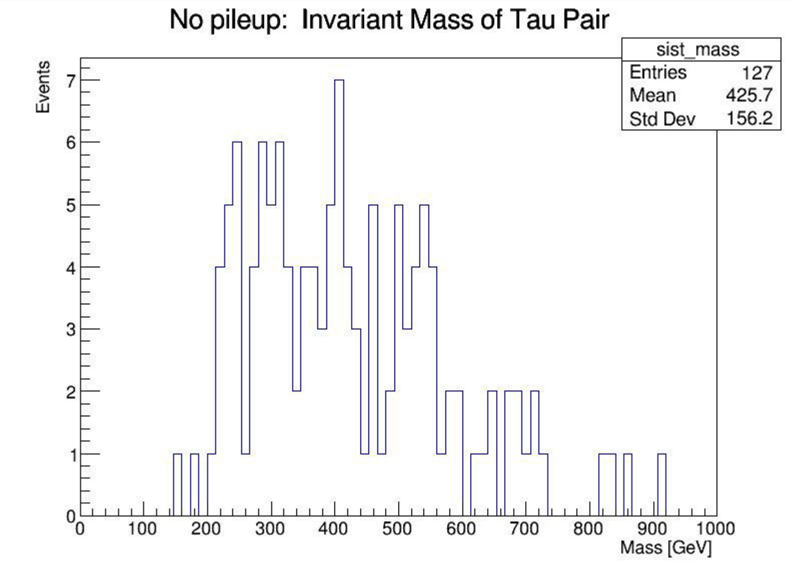


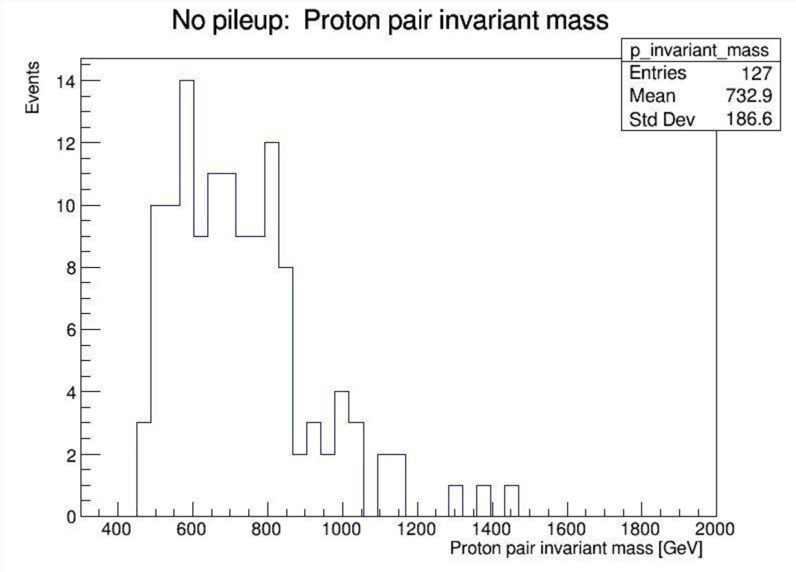
$$Y_X = \frac{1}{2}\left(\frac{E+p_z}{E-p_z}\right)$$

$$Y_X = \frac{1}{2} log(\xi_1/\xi_2)$$

- E is total energy of tau pair
- P_z is total longitudinal moment of tau pair system

Plots





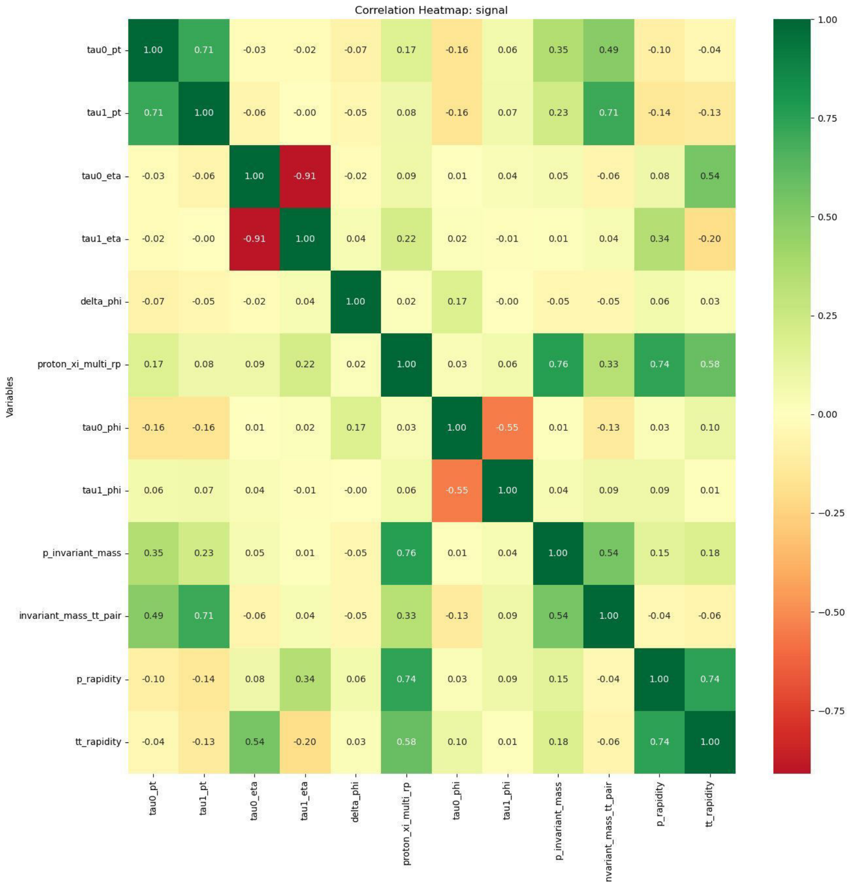
Changed to formula that uses

`TLorentzVector`

$$M_X^2 = s\xi_1\xi_2$$

Correlation between variables

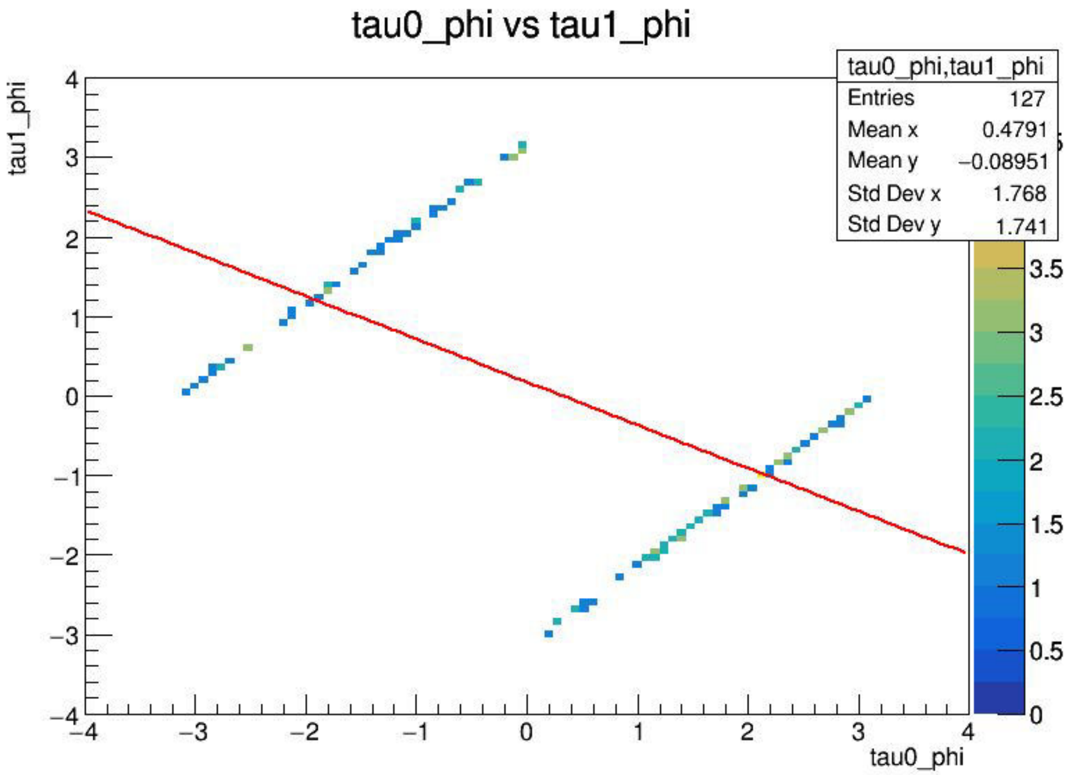
- tau0_phi,tau1_phi
- tau0_eta,tau1_eta
- tau0_pt,tau1_pt
- sist_mass,tau0_pt
- sist_mass,tau1_pt
- Sist_rap,tau0_eta
- Tt_rapidity,tau0_eta
- p_invariant_mass,invariant_mass_tt_pair
- p_rapidity,tt_rapidity



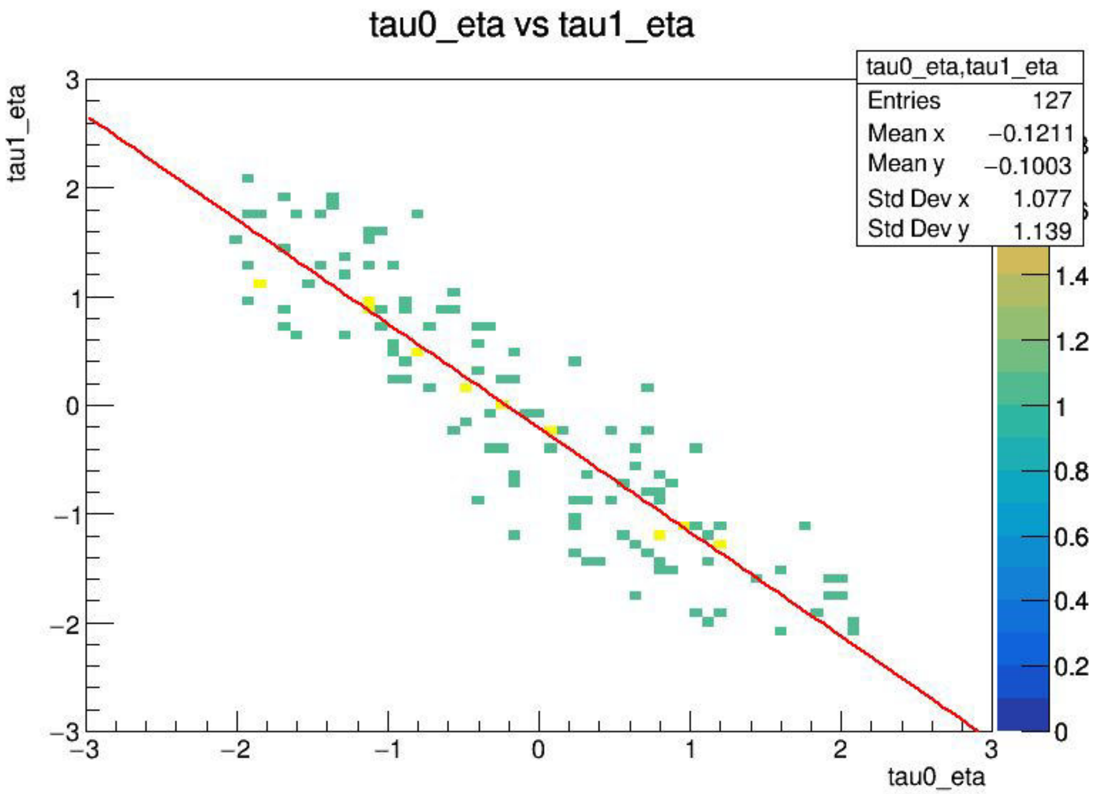
Expected:

- Strong correlation between tau0 and tau1 phi
- Negative correlation between tau0 and tau1 eta
- Strong correlation between invariant mass of tau pair and tau0 and tau1 pt
- Some correlation between tau pair rapidity and tau0 eta

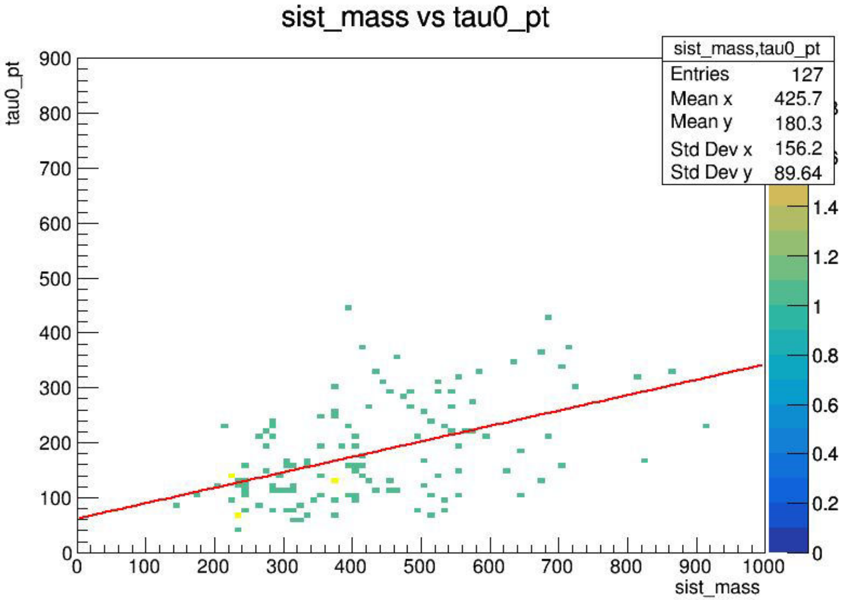
Tau0 and Tau1 phi

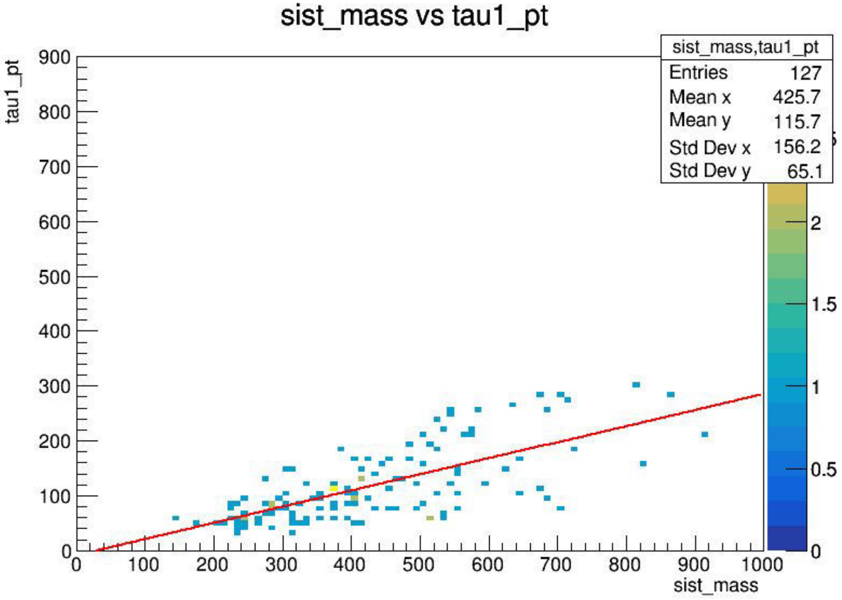


Tau0 and Tau1 eta

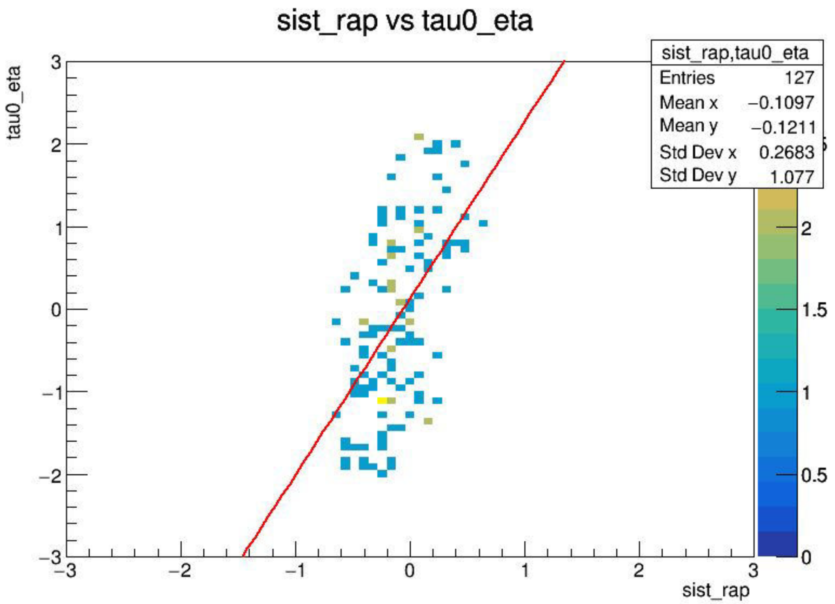


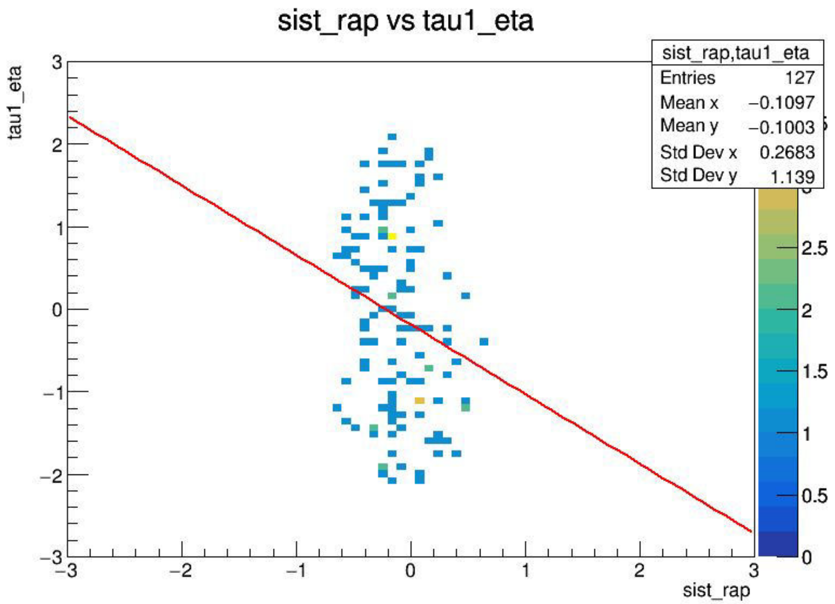
System mass, tau0 pT and tau1 pT



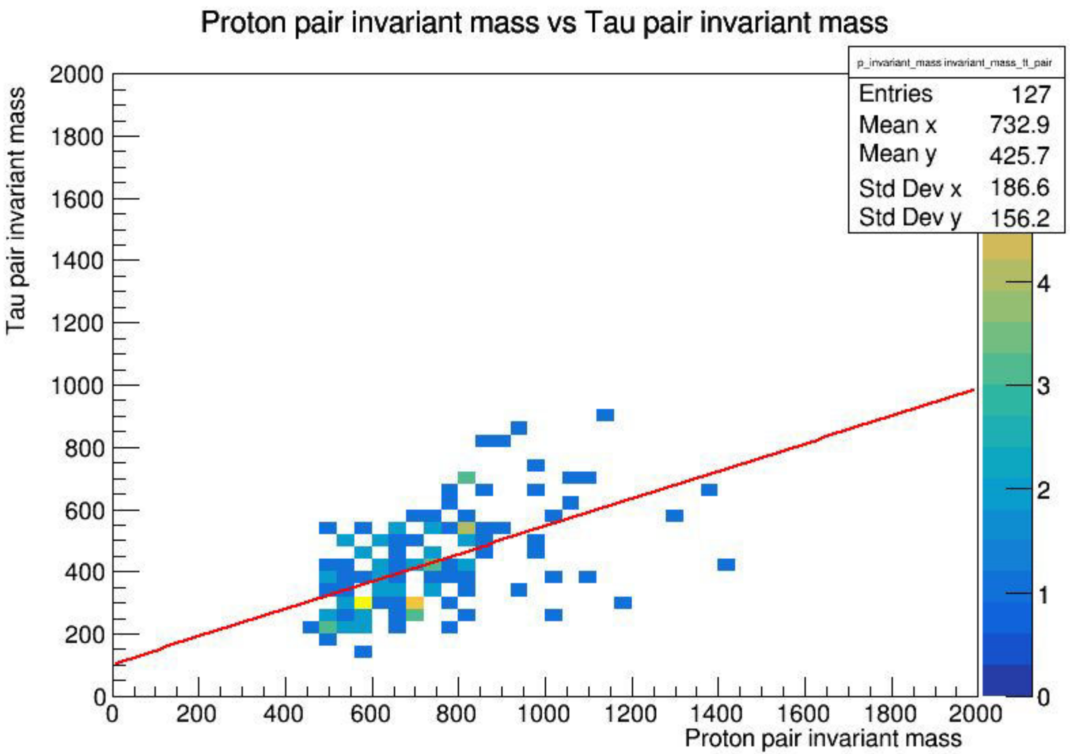


System rapidity, tau0 eta

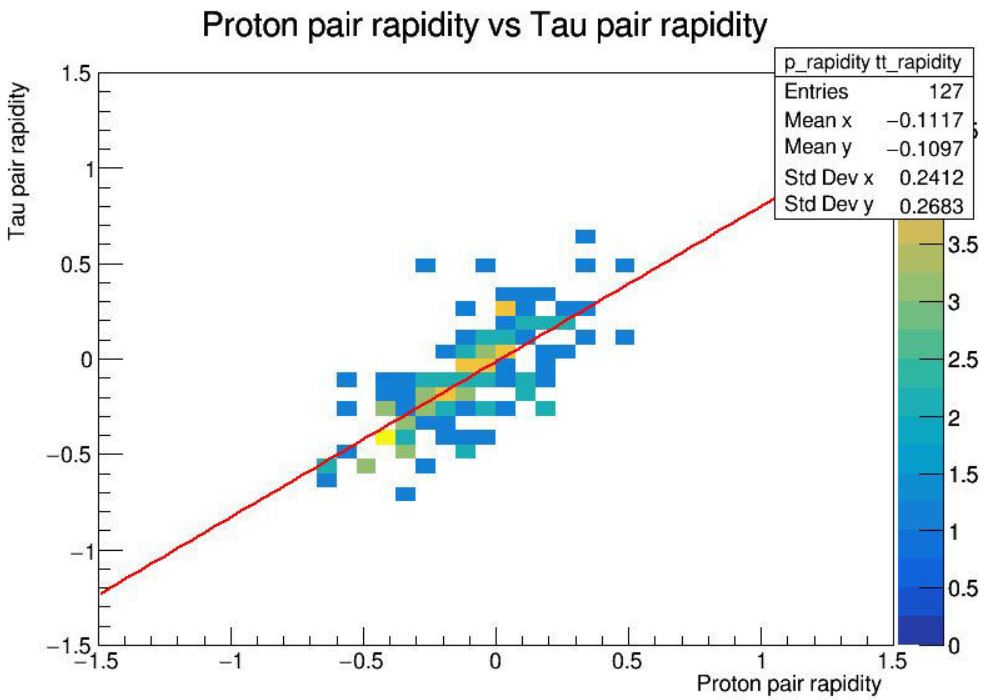




Invariant mas of tau pair, protons



Proton rapidity, tau pair rapidity

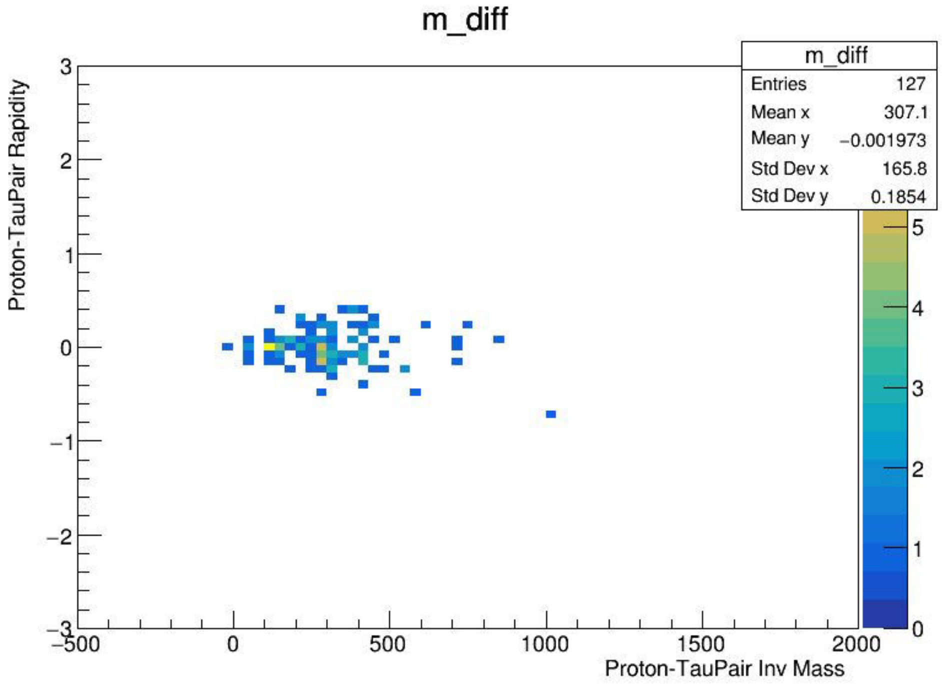


Proton-Tau pair invariant mass, Rapidity

Proton inv mass - tau pair inv mass

Vs

Proton rapidity - tau rapidity



		fase0		fase0		fase1	
Column 1	condition applied	with pileups	% of original data	# no pileups	% of original data	with pile	no pileu
	no cuts	~~23059~~	100	16837	100		
using HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1	trigger only	910	3.946398369	664	3.943695433		
	trigger and 2 or more taus	700	3.035691053	508	3.017164578		
	trigger and 1 proton on each arm	381	1.652283273	168	0.9978024589		
if(tree->GetLeaf("Tau_pt")->GetLen() < 2) continue;	2 taus or more	3963	17.18634806	2889	17.15863871		
	2 taus and one proton on each arm	1602	6.947395811	609	3.617033913		
if (arm == 0 && xi > proton_xi1) { // } else if (arm == 1 && xi > proton_xi2) {	1 proton on each arm	8874	38.48388915	3113	18.48904199		
	all 3	294	1.274990242	127	0.7542911445	55	31

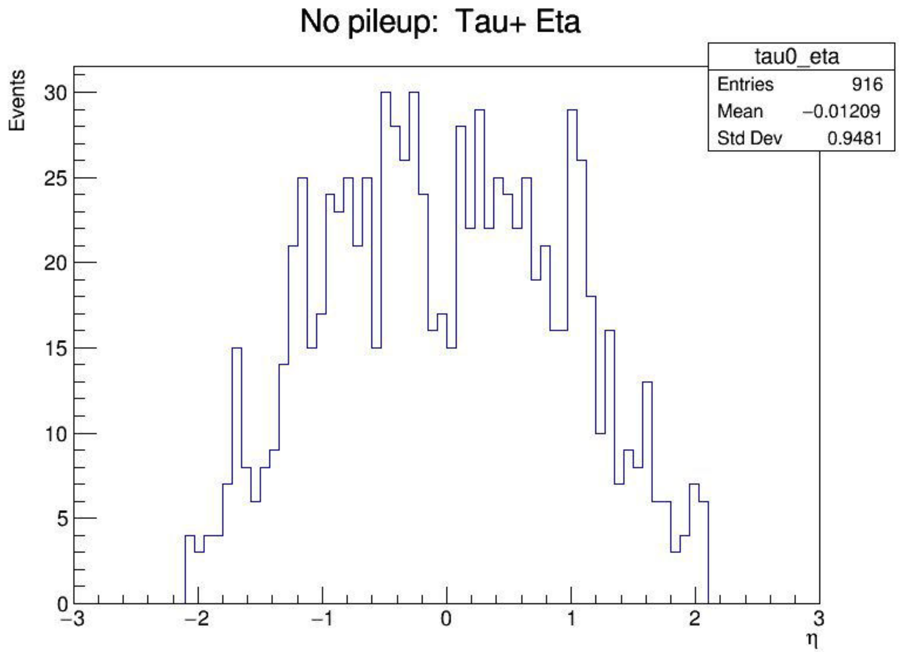
Tau cuts

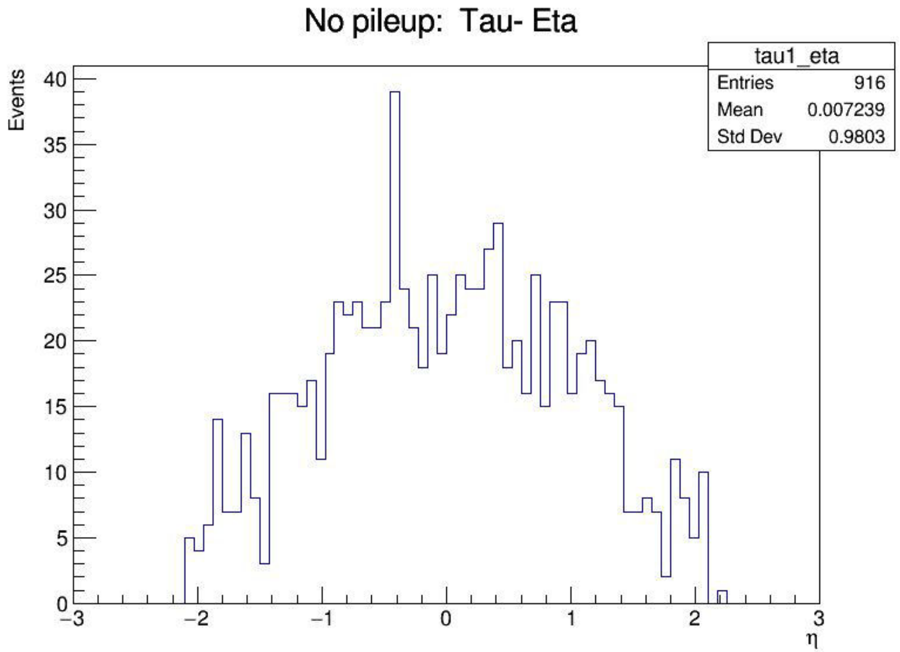
`Tau_idDeepTau2017v2p1VSjet`	6 → Tight
`Tau_idDeepTau2017v2p1VSe`	3 → VLoose
`Tau_idDeepTau2017v2p1VSmu`	1 → VVVLoose
`DeltaR`	>0.4
`tau0.Eta and tau1.Eta`	<2.4
`Tau0_pt and tau1_pt`	>100
`Opposite charge`	✔

New data and code

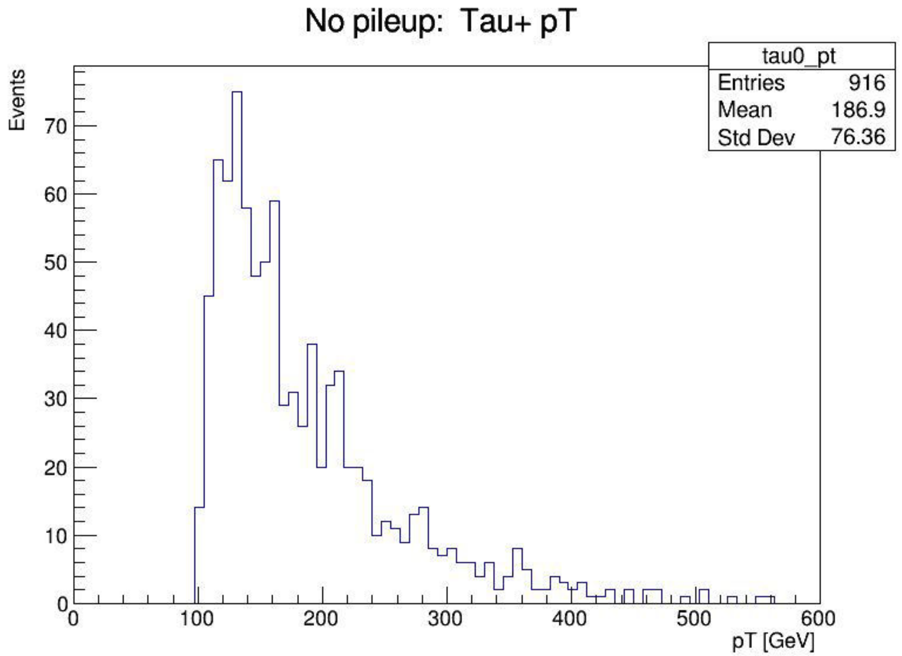
Variable name	Description

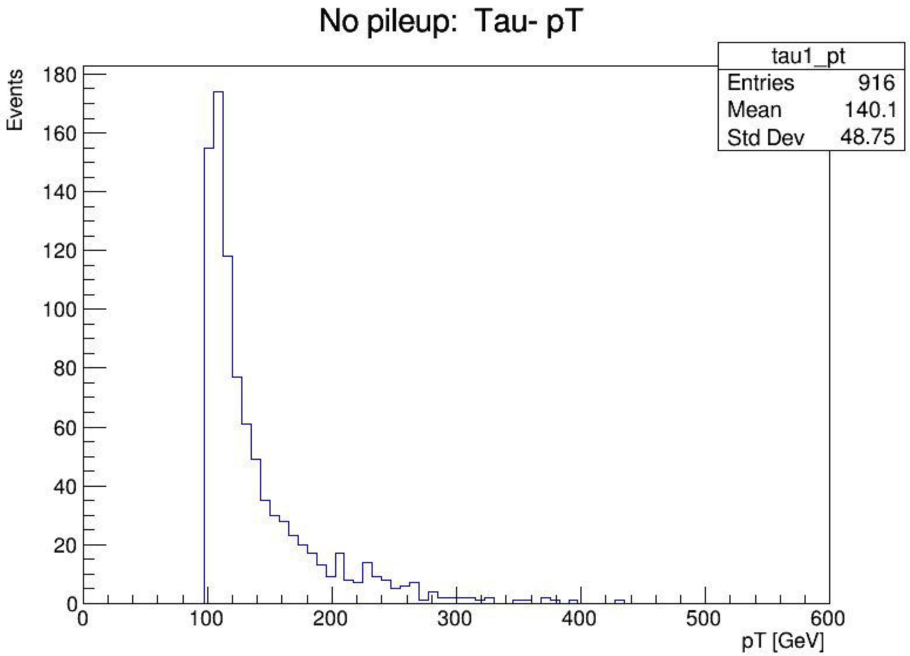
Plots full background with signal-like selection (opposite sign taus)



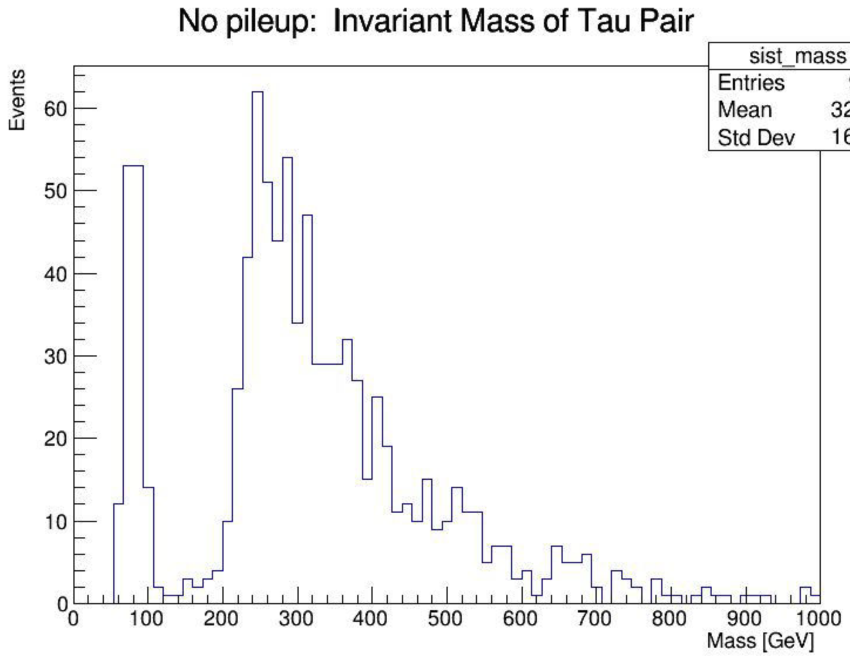


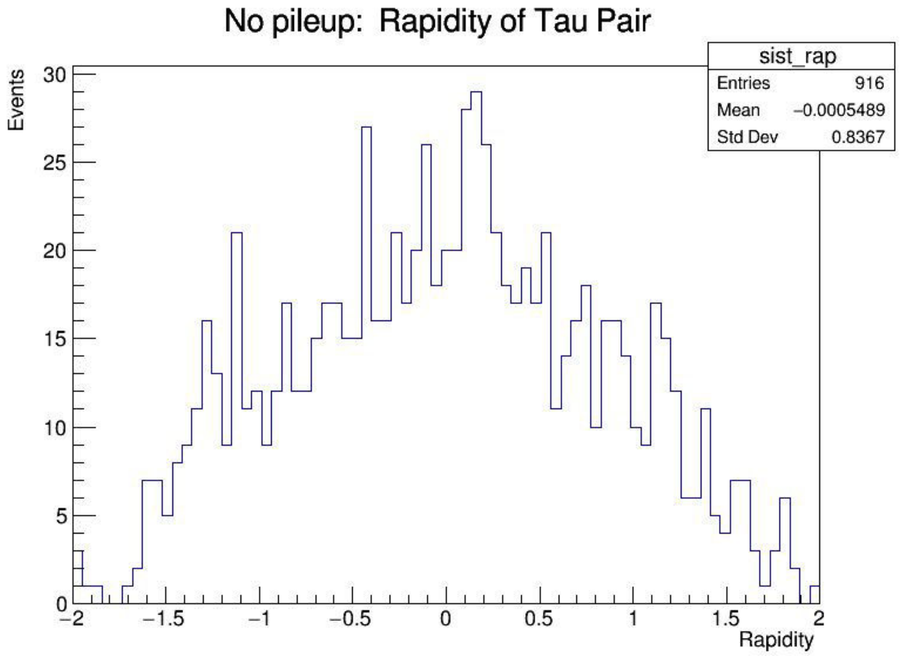
Plots full background with signal-like selection (opposite sign taus)



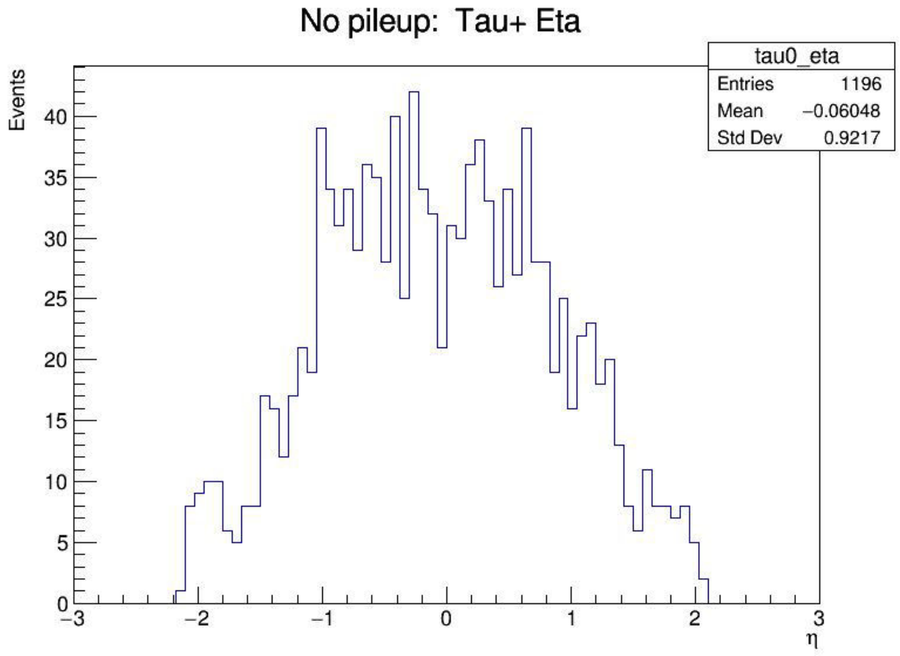


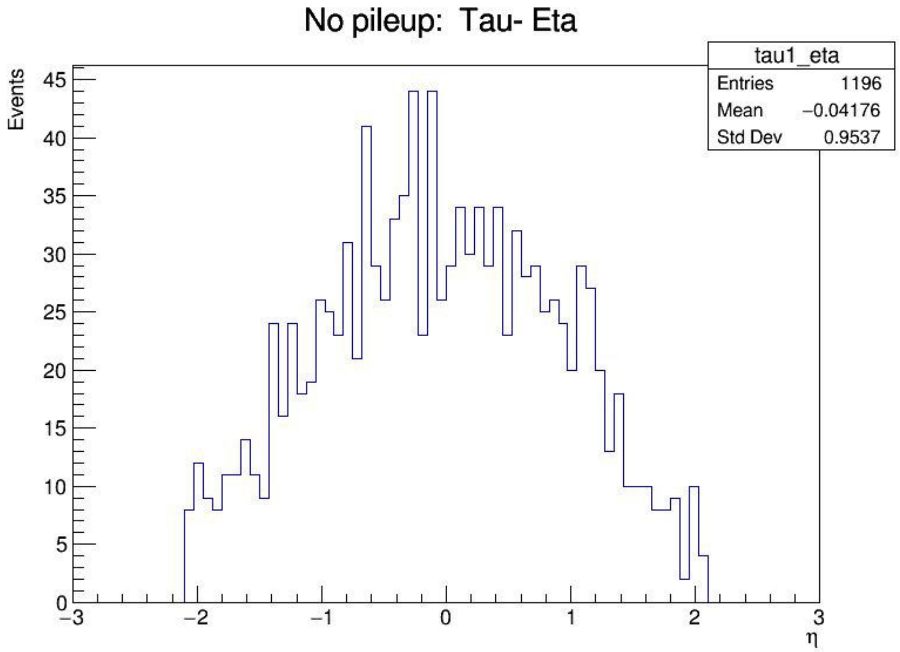
Plots full background with signal-like selection (opposite sign taus)



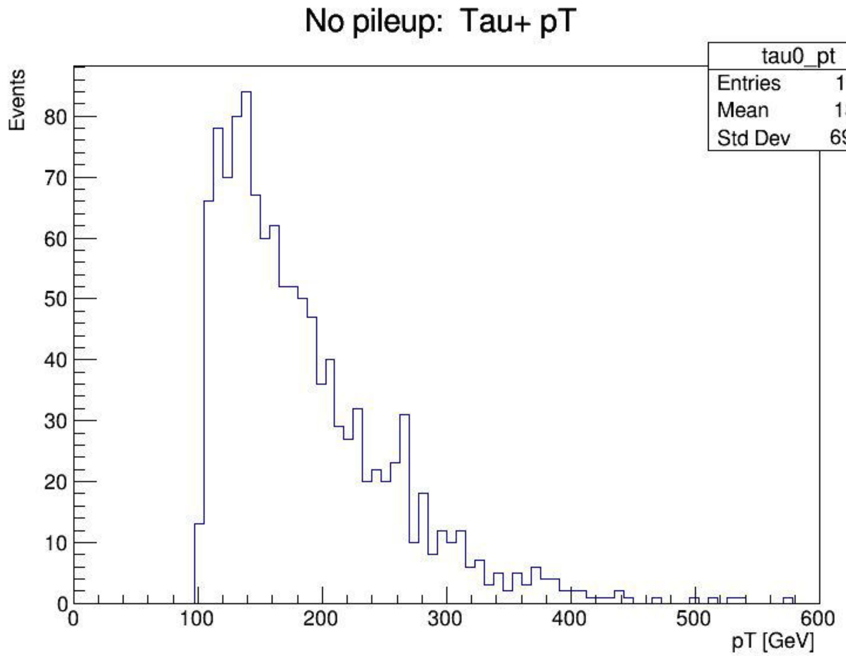


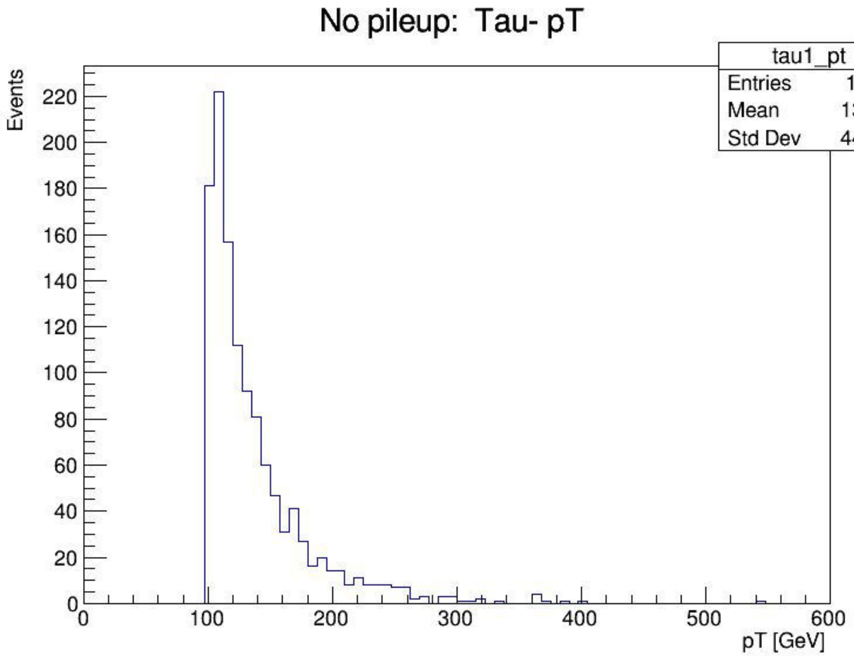
Plots fase1 files for the sum of all backgrounds, including pileup protons



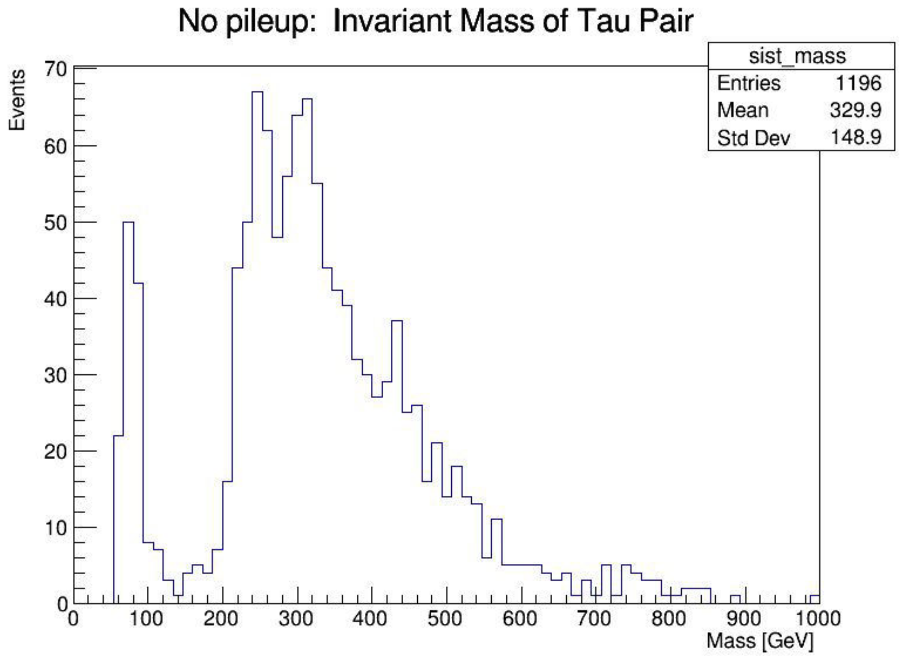


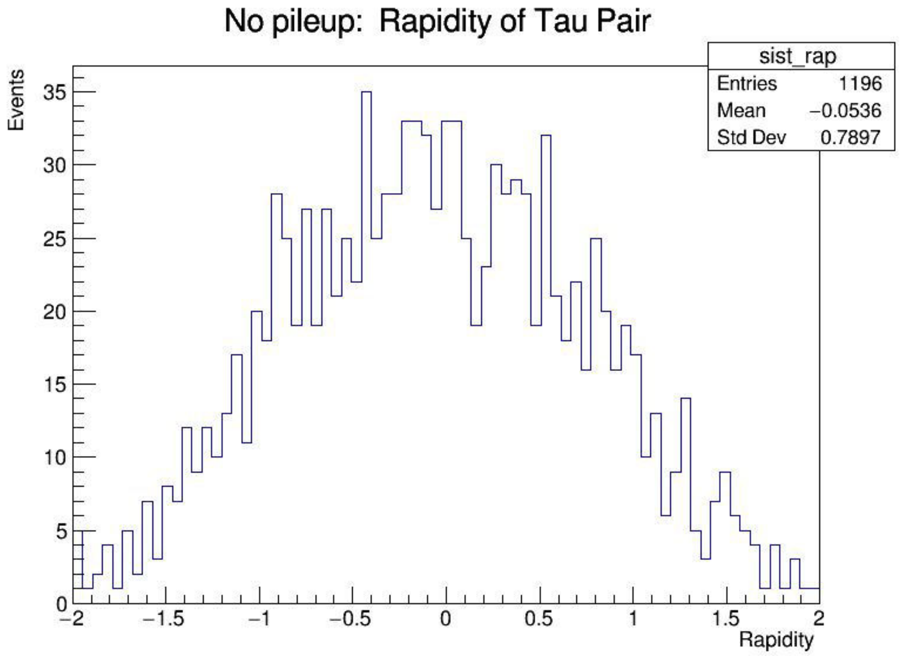
Plots fase1 files for the sum of all backgrounds, including pileup protons





Plots fase1 files for the sum of all backgrounds, including pileup protons



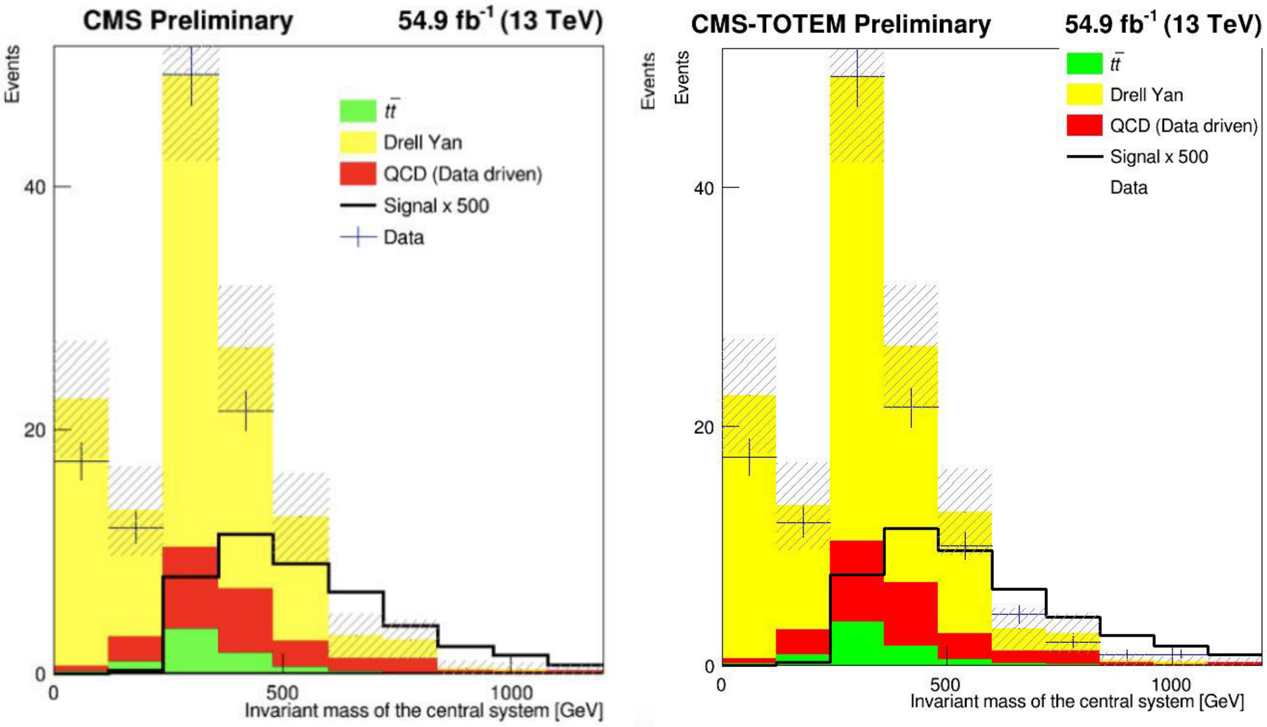


Input distributions to the BDT discriminator

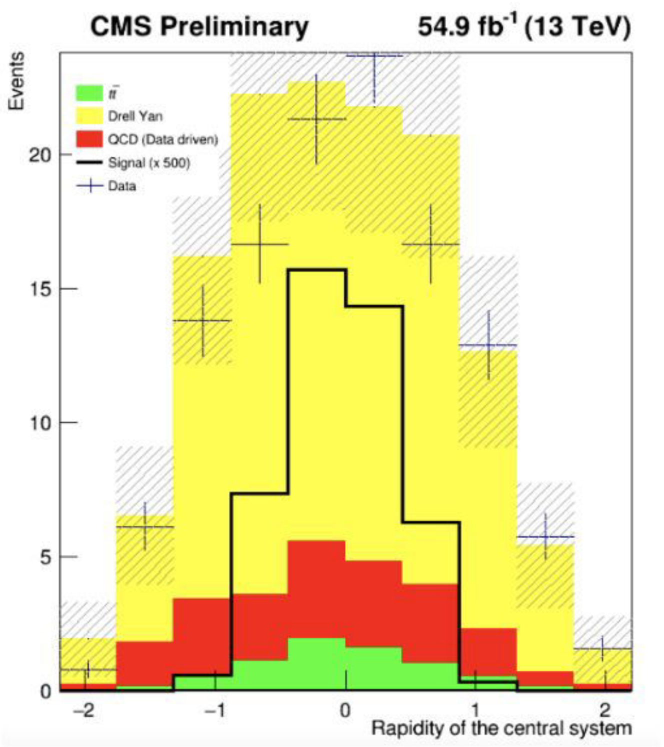
- Invariant mass of the central system
  → The total mass of all particles in the central detector, calculated from their 4-momenta.
- Rapidity of the central system
  → Describes how forward or backward the central system is moving along the beam axis.
- Mass difference
  → Difference between the central system mass and the mass reconstructed using the forward protons (CMS - PPS).
- MET (Missing Transverse Energy)
  → Represents undetected particles.
- Transverse momentum of the central system
  → Momentum of the central system perpendicular to the beam axis.
- Rapidity matching
  → Comparison of rapidity measured centrally..
- Transverse momentum of hadronic tau
  → Momentum perpendicular to the beam of the hadronically decaying tau lepton.
- Acoplanarity of central system
  → Measures how "non-back-to-back" the two leading tau candidates are; zero means perfect balance.

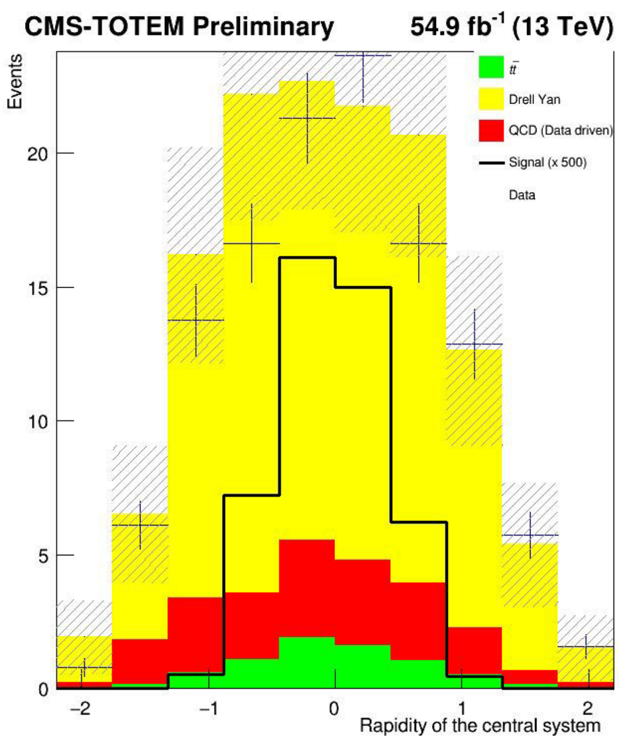
Invariant mass of the central system

Original - left vs New - right

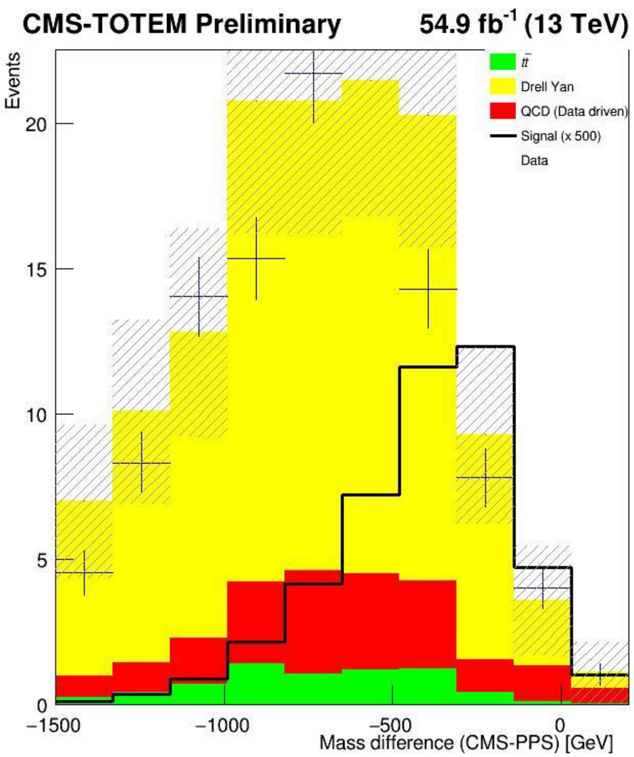


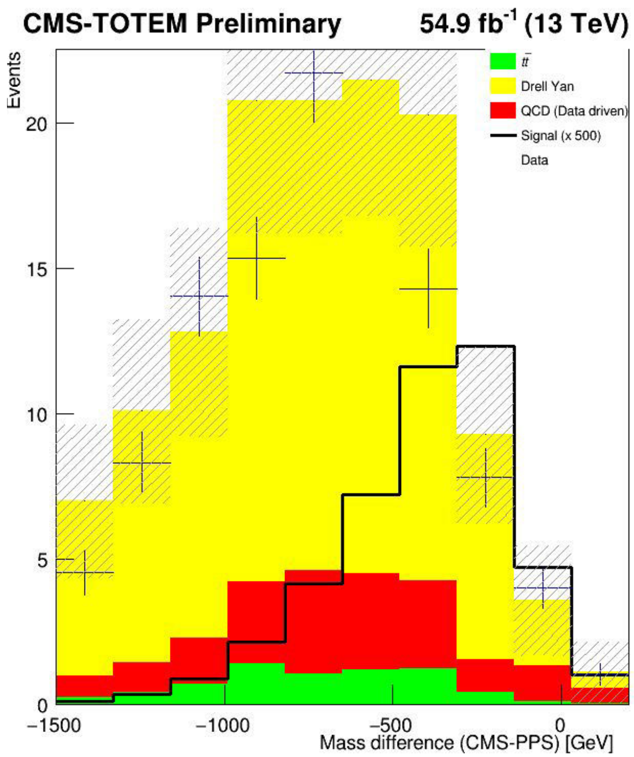
Rapidity of the central system
Original - left vs New - right



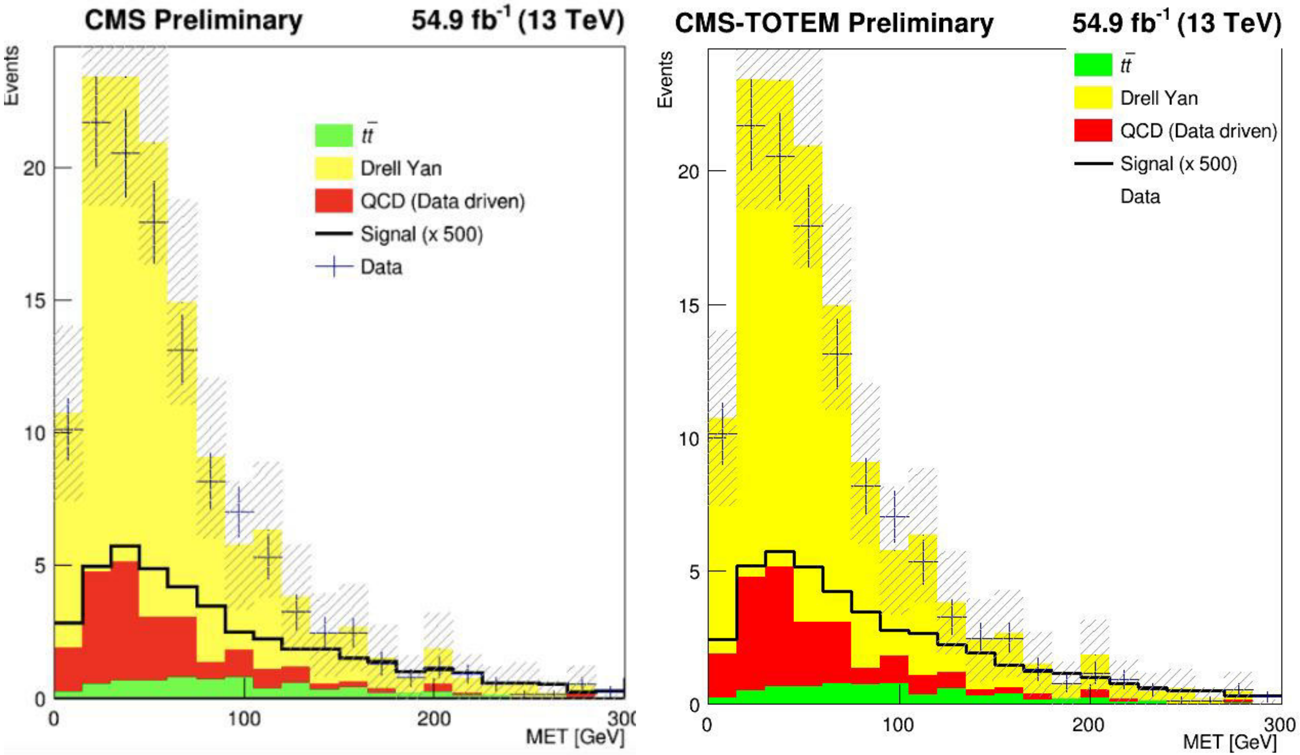


Mass Difference
Original - left vs New - right



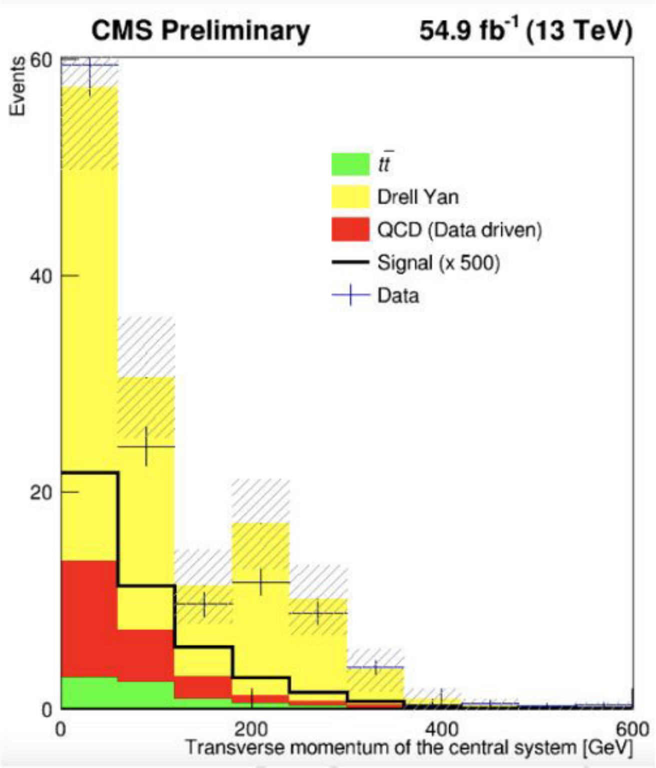


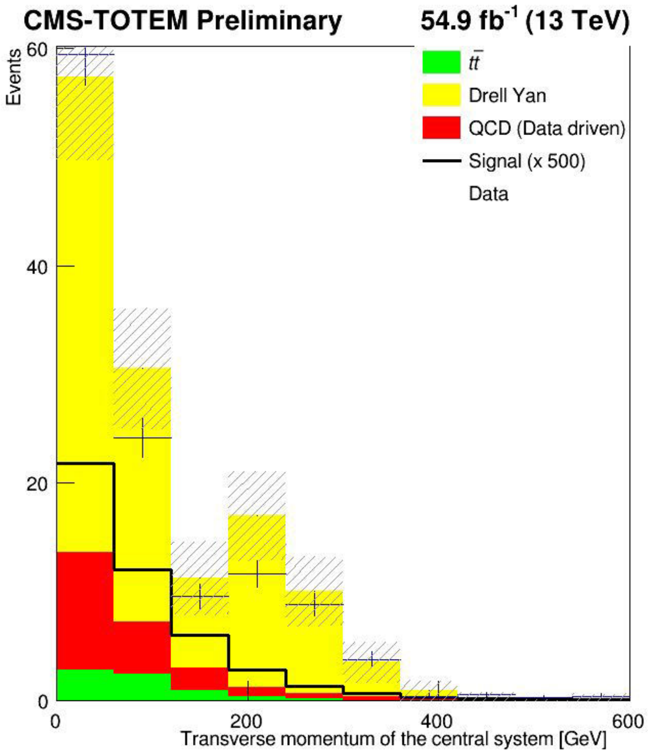
MET
Original - left vs New - right



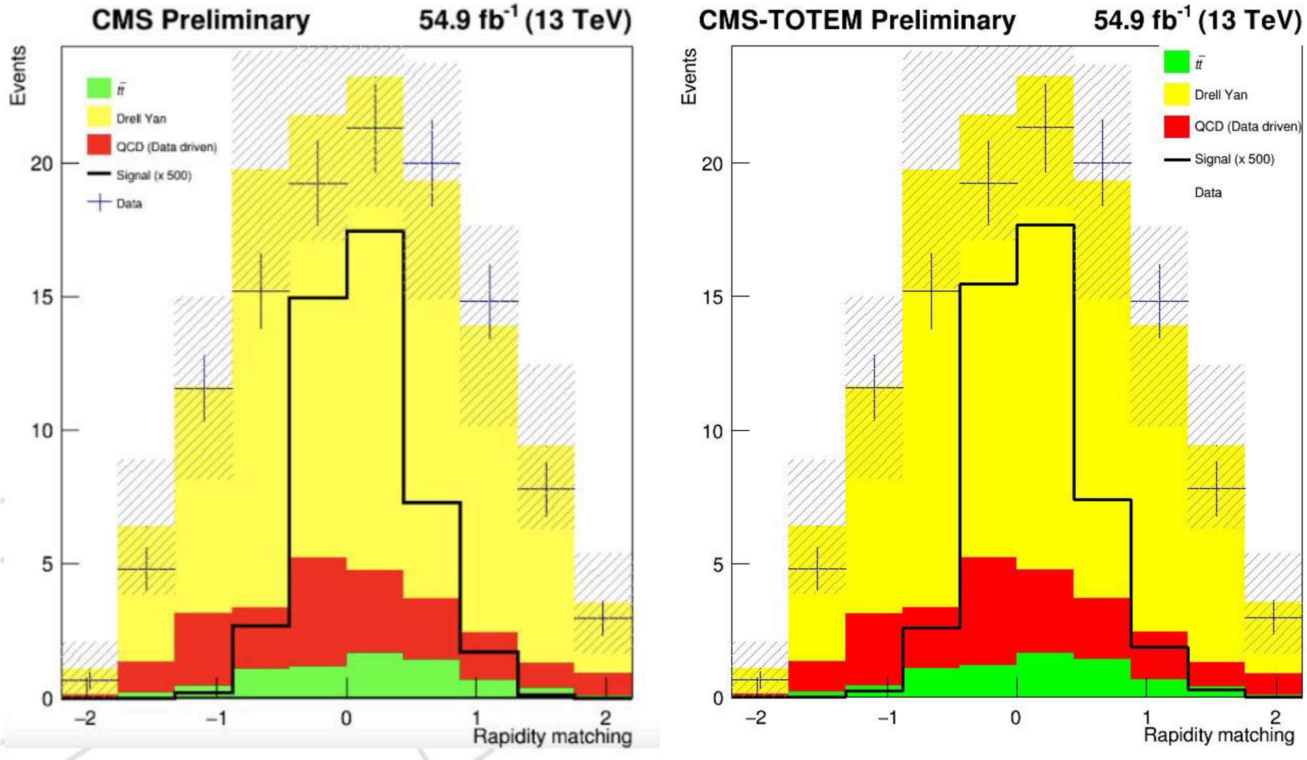
Transverse momentum of central system

Original - left vs New - right



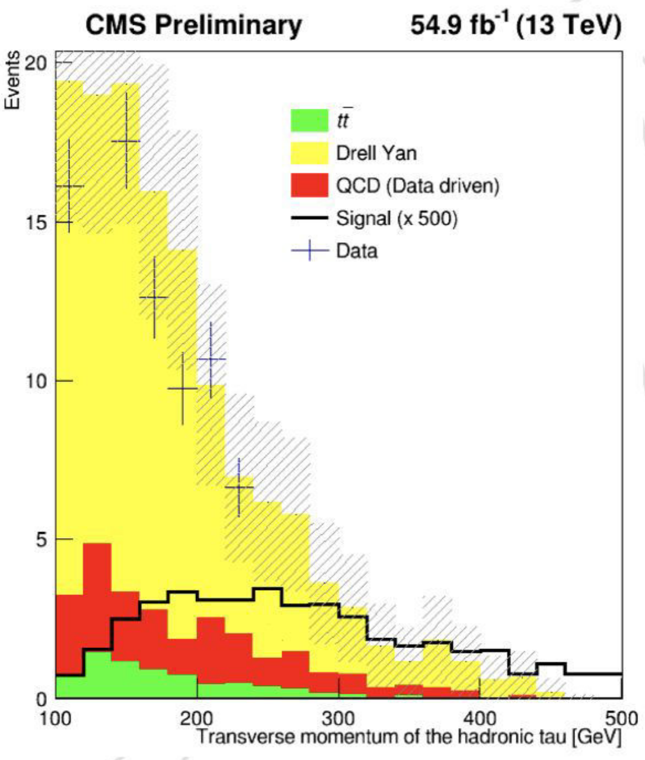


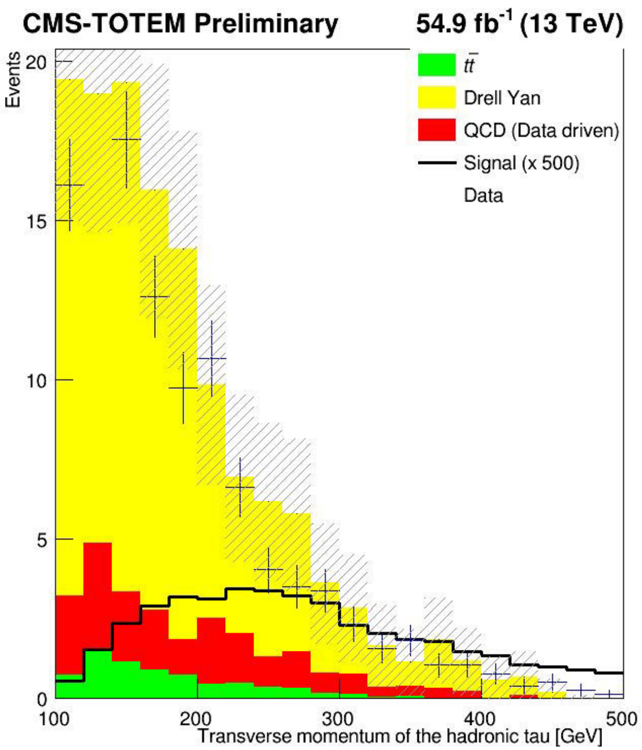
Rapidity matching
Original - left vs New - right



Transverse momentum of hadronic tau

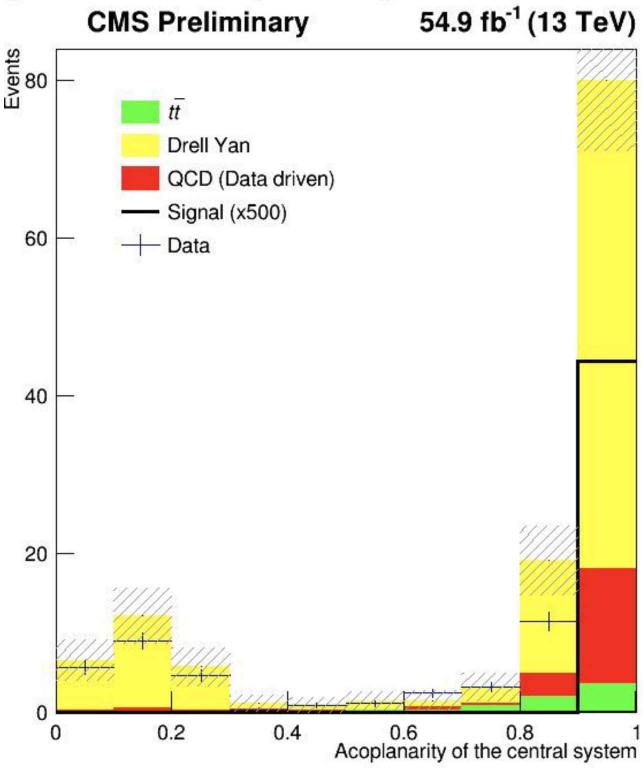
Original - left vs New - right

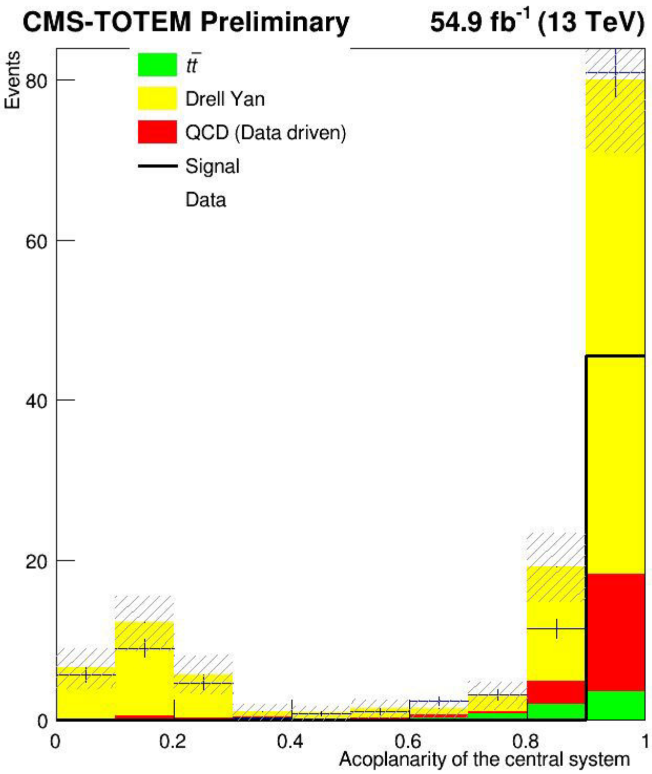




Acoplanarity of central system

Original - left vs New - right





Mystery weights

In original code

```
weight_sample = 54900.*(1.)/(4000.*1000.);
```

- Weight_sample is normalization factor
- weight_sample=0.013725

$$\text{weight\_sample} = \frac{\mathcal{L} \cdot \sigma}{N_{\text{gen}}}$$

What worked to make the plots the same:

```
weight_sample= 0.000013725;
```

Is original weight_sample/1000 -> UNITS

Mystery weights

```
//double w_data=1.;
double w_qcd=1.*0.8*0.13;
double w_ttjets=0.15*0.8*0.13;
double w_dy=1.81*0.8*0.13;
double w_sinal;
double w_data=1*0.13;
```

In 2018, due to an improved PPS detector configuration, silicon pixels are used. They allow the reconstruction of multiple protons and the scenario is different. Therefore, the technique used is the following:

- We estimate the probability $p_{n,m}$ to get n protons in arm 0 and m protons in arm 1;
- For the background samples, we observed that the probability to have at least one PU proton per arm is just 13%. Therefore, the weigth of each background event is rescaled of a factor 0.13;
- Subsequently, each background event is enriched with n protons in arm 0 and m protons in arm 1, according to the probability $p_{n,m}$ calculated earlier. If we get more than one proton per arm, the pair with the largest ξ is chosen;
- For the signal samples, they are directly enriched with the number of protons calculated according to the probability $p_{n,m}$.

```
double weight = .15; //w= Numero monte carlo / Numero de 2018 , Numero de 2018 = luminosidade de 2018 *
secção de 2018
```

0.13 is due to pps reconstruction technique -> probability of having at least one proton per arm

Still no explanation for 0.8 factor on the background only

Background

Starts with data in NANOAOD format
DY and ttbar are MC
QCD is based on same sign events from data

Fase0

QCD_fase0.cpp
DY_fase0.cpp
Ttjets_fase.cpp

Produces fase0 files
for each background
used in:

Fase1

nanotry_fase1_QCD.cpp
nanotry_fase1_DY.cpp
nanotry_fase1_jets.cpp

Produces fase1, one
file for each , after
enriched with pileup
protons

`QCD_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

`DY_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

`ttJets_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

Signal

Starts with data in MINIAOD format

Sinal.cpp

data/MC corrections for the efficiency,
scale, and resolution of taus, electrons,
muons, and protons

Produces ntuple with format similar as fase1,
includes BSM weighs needed to normalize the
sample for different SM and BSM scenarios

`TauTau_sinal_PIC_july_2018.root`

Used in plot_m.cpp
Produces normalized plots and the input distributions for
BDT

final data files with the signal-like selection (opposite sign taus) :
`Dados_2018_UL_skimmed_TauTau_nano_fase1total-protons_2018.root`

Fase0: general description

- Tau ID variables (DeepTau),
- Tau momenta (pt, eta, phi, mass),
  System variables (mass, pt, acoplanarity, rapidity),
- Jet info (pt, b-tagging),
- MET (missing transverse energy),
  Generator-level info.

Calculates:

- Invariant mass of the tau pair,
- Acoplanarity between taus,
- System pt and rapidity,
- Number of b-tagged jets (DeepB > 0.4506),

Applies trigger filter (HLT_DoubleMediumChargedIsoPFTauHPS40_Trk1_eta2p1_Reg == 1)

Code organization Background

Goal: only one file for signal and one file for background

(with parallelization using rootEnableImplicit)

File name	reads		Output
DY_fase0	`DY_2018_UL.txt`		`DY_2018_UL_skimmed_TauTau_nano_`
QCD_fase0	`QCD_2018_UL.txt` `dadosluminosidade.txt`	`Checks if luminosity in event matches with the data from file`	`QCD_2018_UL_skimmed_TauTau_nano_`
ttJets_fase0	`ttJets_2018_UL.txt`		`ttJets_2018_UL_skimmed_TauTau_nano_`
nanotry_fase1_DY	`DY_2018.txt` `Name of fase0 output files`		
nanotry_fase1_QCD	`QCD_fase1_PIC.txt` `Name of fase0 output files`		
nanotry_fase1_ttJets	`ttJets_fase1_PIC.txt` `Name of fase0 output files`		

Code organization

Background-> fase0 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	$t\bar{t}$ MC	Data-driven QCD
Luminosity block filtering			✓
Extra gen matching	✓	✓	
Generator weight	1.81	0.15	
HLT version	40_Trk1_eta2p1	40_Trk1_eta2p1	35_Trk1_eta2p1

Code organization
Background-> fase1 specifics

Feature	DY	ttjets	QCD
Sample Type	Drell-Yan MC	$t\bar{t}$ MC	Data-driven QCD
Scale Factors	Tau ID + energy scale + sys	Tau ID + energy scale	
Systematic Variations	full		
Charge Selection	Opposite Sign (`<0`)	Opposite Sign (`<0`)	Same Sign (`>0`)
Tau Selection Cuts	`pt > 100`, `η < 2.3`	`pt > 100`, `η < 2.4`	`pt > 100`, `η < 2.4`
Gen-Match Handling	yes		

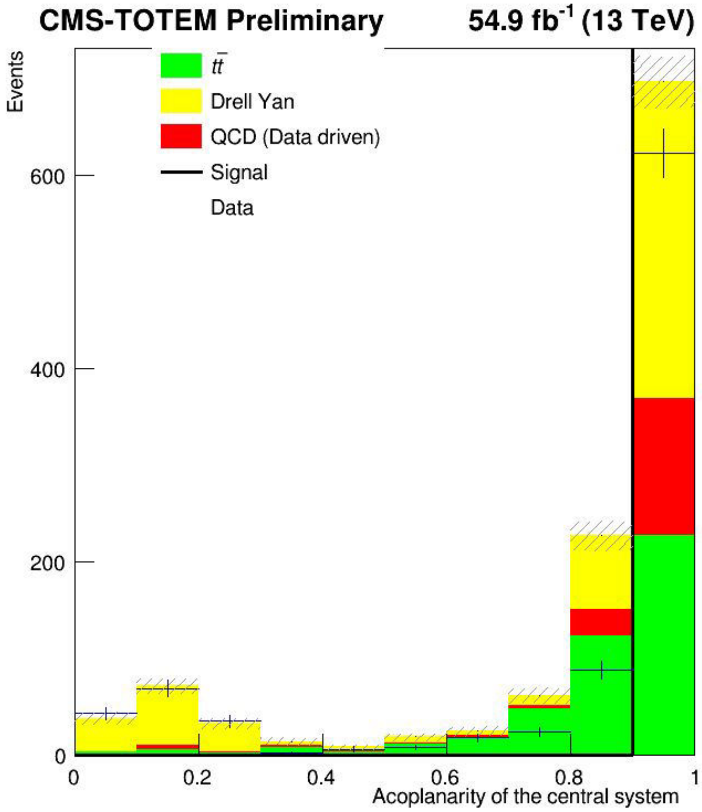
BDT: Code files information

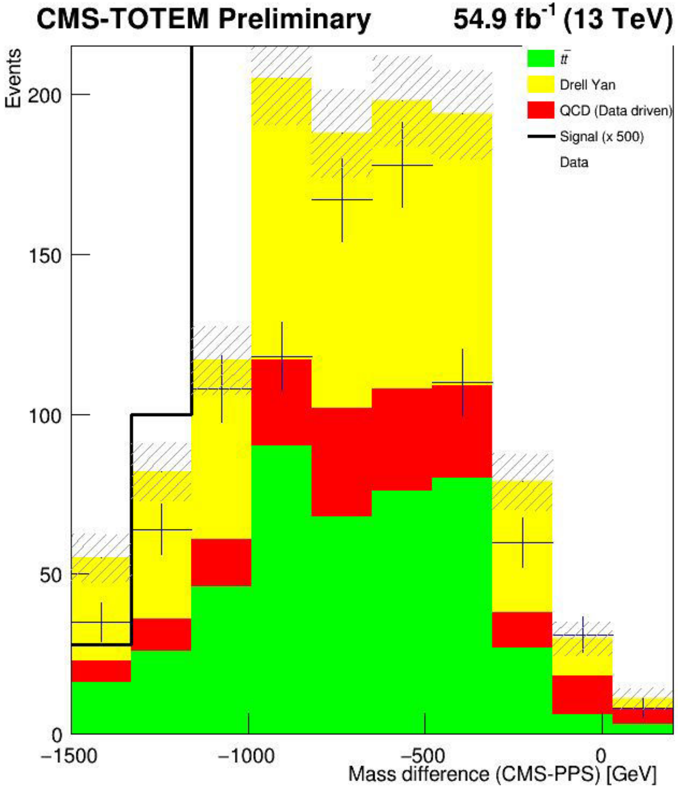
Variable	Description	Units
`sist_acop`	Central system acoplanarity	dimensionless
`sist_pt`	Total momentum	GeV
`sist_mass`	Invariant mass	GeV
`sist_rap`	System rapidity	dimensionless
`met_pt`	Missing energy	GeV
Mass matching	`sist_mass − sqrt(13000²×ξ₁×ξ₂)`	GeV
Rapidity matching	`sist_rap − 0.5×log(ξ₁/ξ₂)`	dimensionless

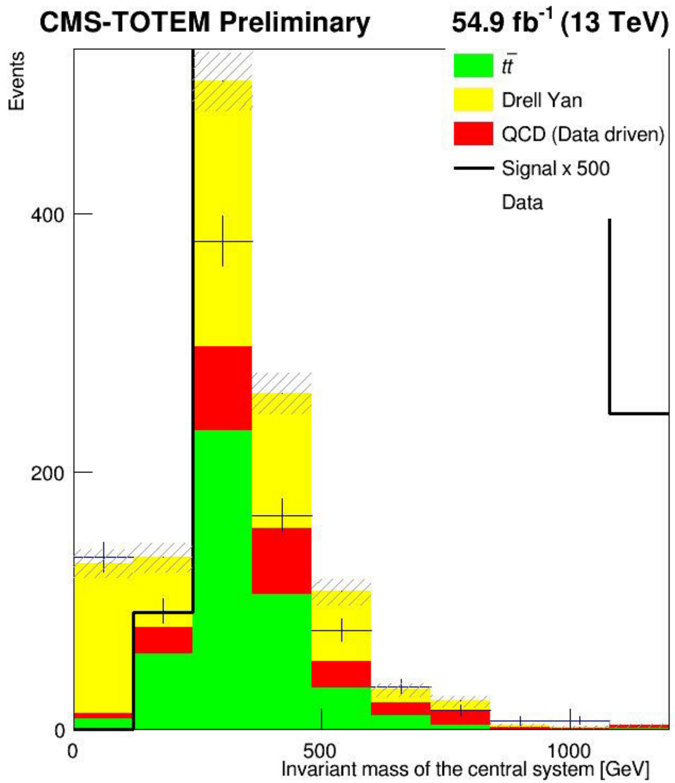
TMVAClassification	Train BDT/ML models	Background file Signal file	Output_TMVA_tautau_2018.root Dataset weights
TMVAClassificationApplication	Apply trained models to data	Trained models Data samples: dy, ttjets,qcd and signal	On file for each type containing BDT scores for every event
Bdt_output	Process BDT results	TMVApp_*.root Event weights, sample information	Analysis histograms Bdt score distributions signal/background separation cuts Cut optimization results
TMVAOutPut_Distributions	Create TMVA evaluation plots	Output_TMVA_tautau_2018.root Training results Test results Method performance data	ROC curves Variable importance plots Correlation matrices

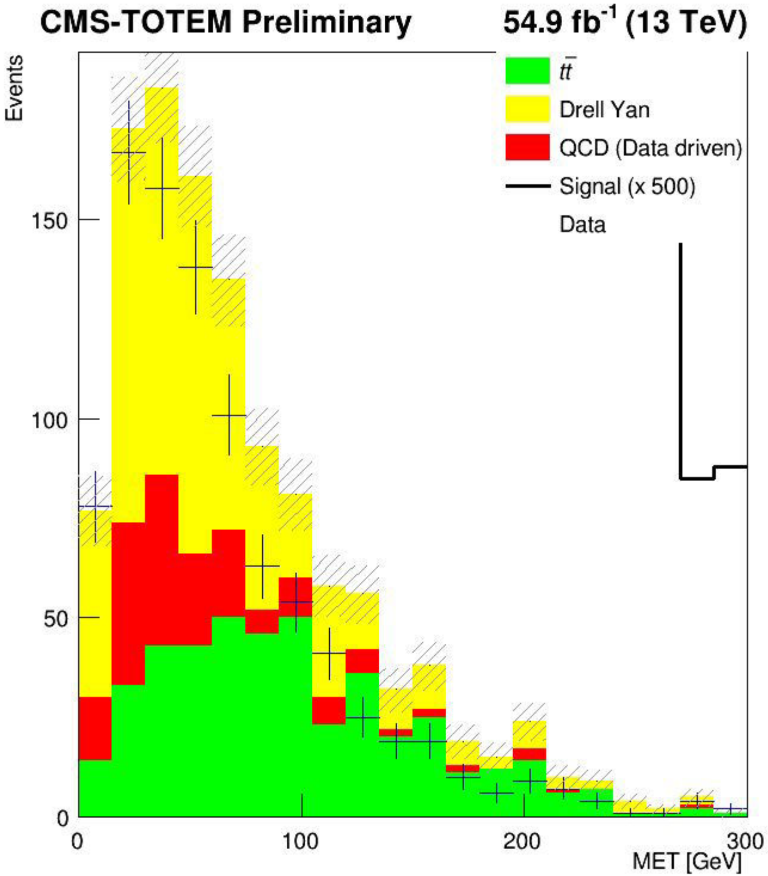
Plots of data without simulation weights applied

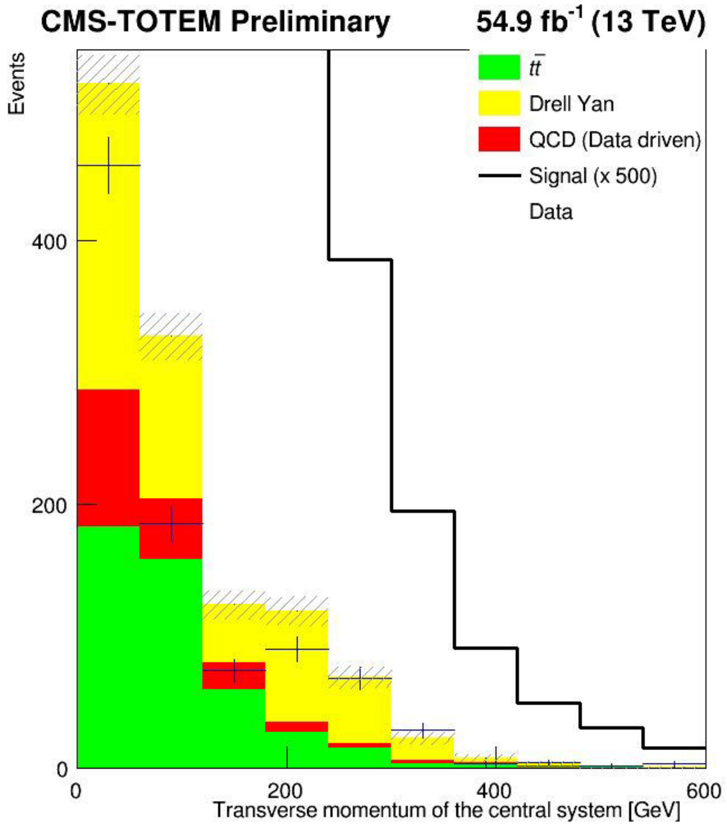
The following plots were obtained by setting the weight of each sample to 1

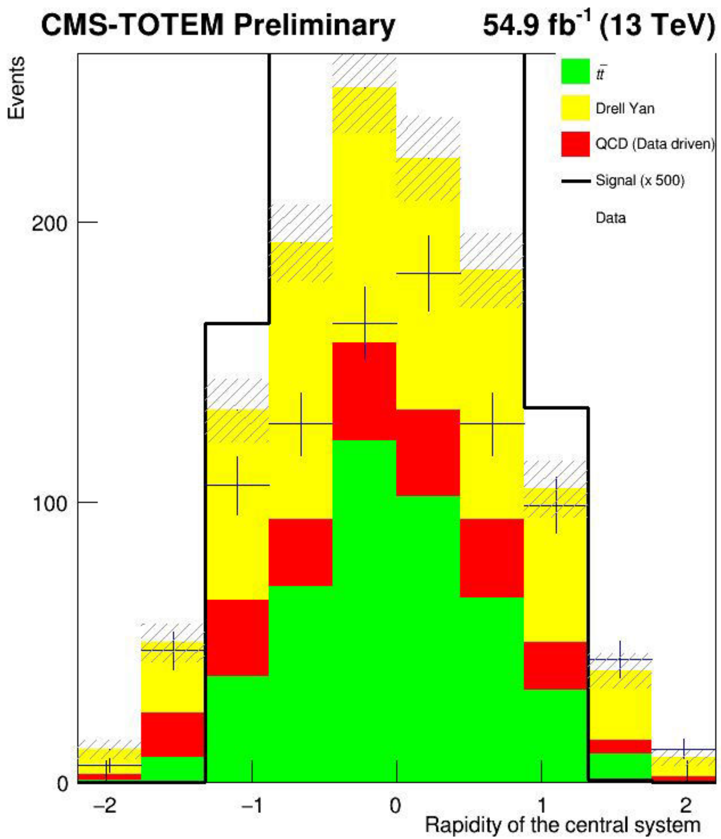

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal
Data
Acoplanarity of the central system

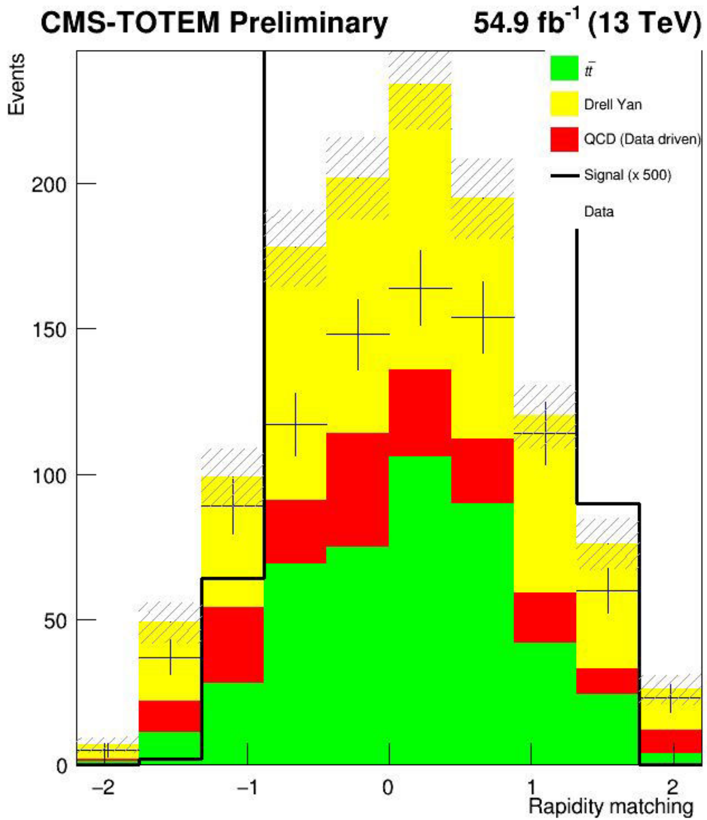

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Mass difference (CMS-PPS) [GeV]

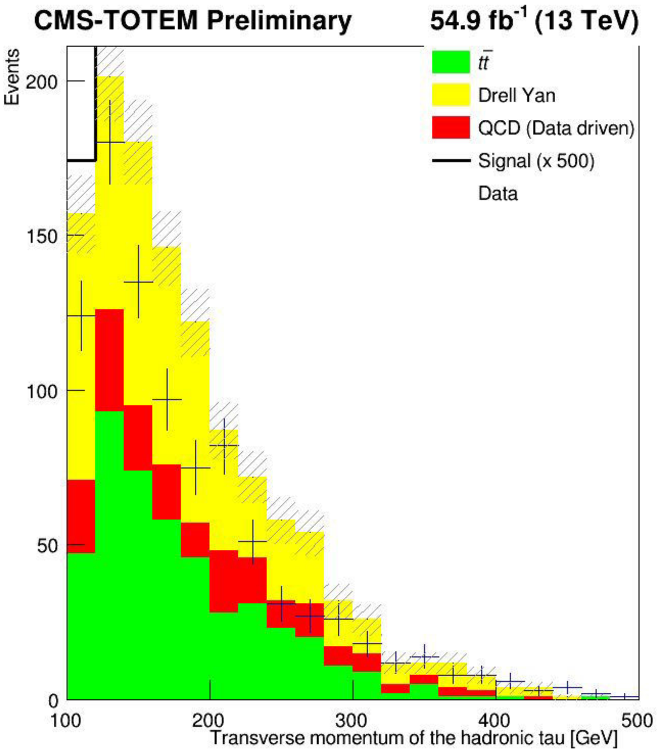

CMS-TOTEM Preliminary
54.9 fb⁻¹ (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal x 500
Data
400
200
0
0
500
1000
Invariant mass of the central system [GeV]


CMS-TOTEM Preliminary
54.9 fb⁻¹ (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
150
100
50
0
0
100
200
300
MET [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
0
200
400
600
Transverse momentum of the central system [GeV]


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
0
100
200
−2
−1
1
2
Rapidity of the central system

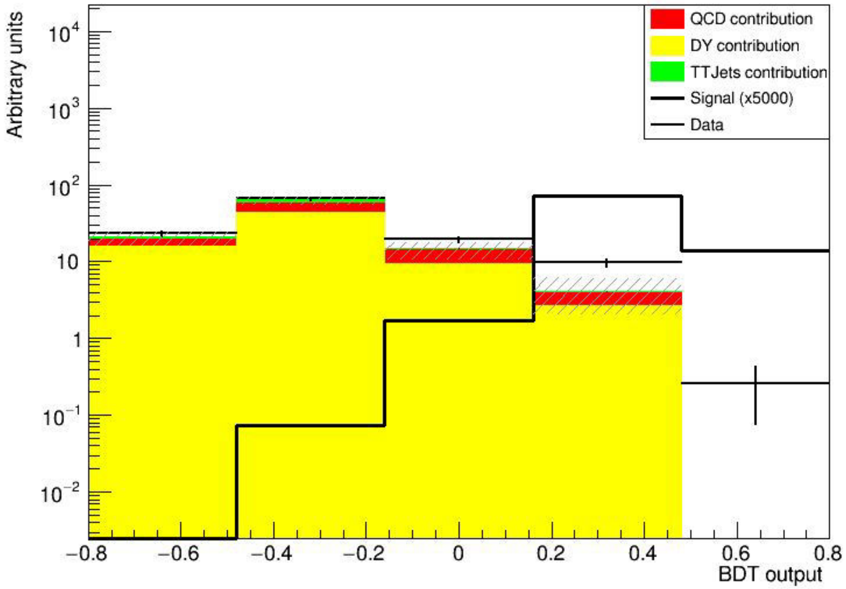

CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Rapidity matching


CMS-TOTEM Preliminary
54.9 fb^{-1} (13 TeV)
Events
$t\bar{t}$
Drell Yan
QCD (Data driven)
Signal (x 500)
Data
Transverse momentum of the hadronic tau [GeV]

BDT output scores for training with previous histograms

Same shape as before changing the weights

-> weights only change the size of the histograms, not the shape.



In red: not applied

In green: applied

Control regions: $\tau_h\tau_h$

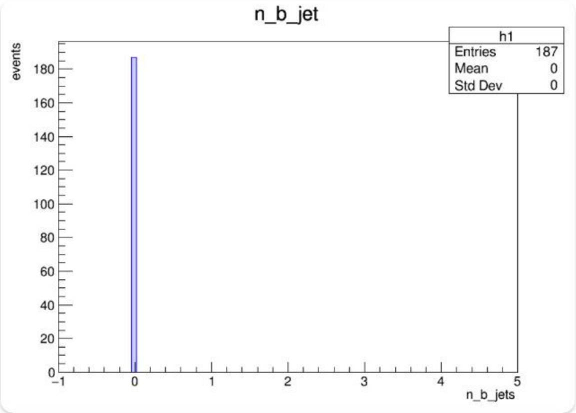
	Invariant mass (GeV)	acoplanarity	η	2 isolated ts	N of b-jets	p_t
DY	[40,100]	<0.3	<2.4	Opposite sign	0	
QCD	[100,300]	>0.8	<2.4	Same sign	0	<75
ttjets	[200,650]	>0.5	<2.4	Opposite sign	1 or more	<125

On top: original from analysis note
Bottom: new plots

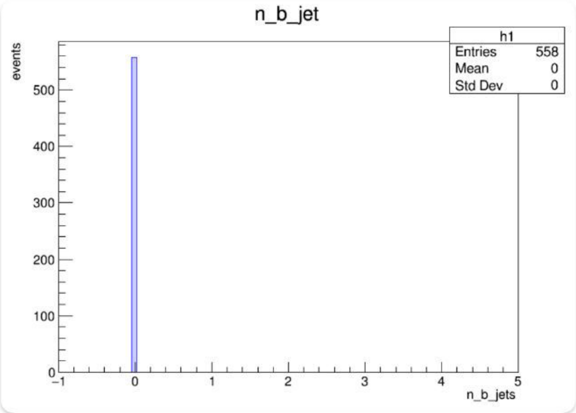
Number of b jets: 0 for all samples

Fase1

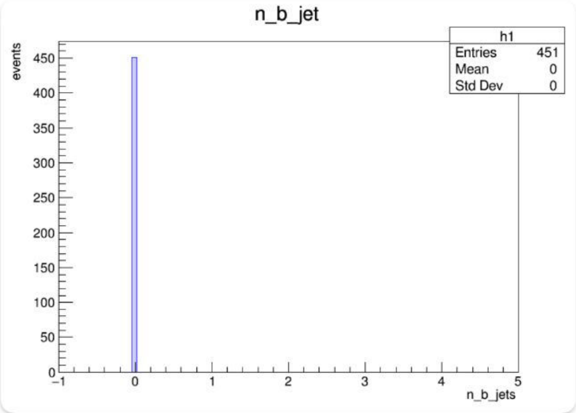
QCD



DY



ttJets

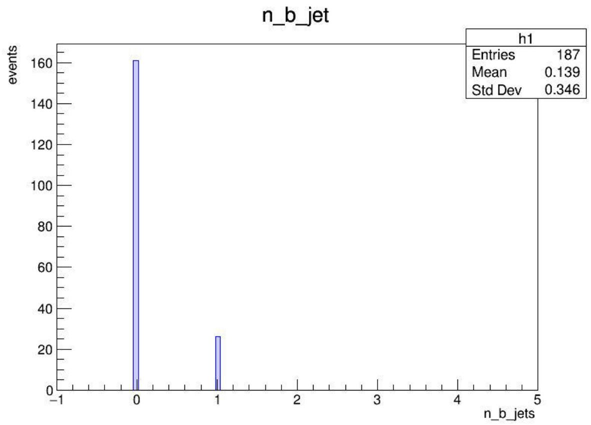


Cause: mismatch in the declaration of variable type in the proton enr

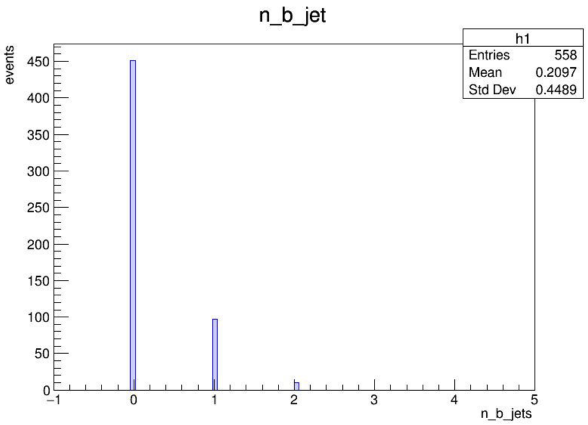
- Mismatch in the declaration of variable in the proton enrichment code (to introduce pileup protons);
    - N_b_jet was declared as *double* when it originally was *int*
- Handling of branch addresses

-> Fixed by passing the merged samples from phase1 through the proton enrichment code

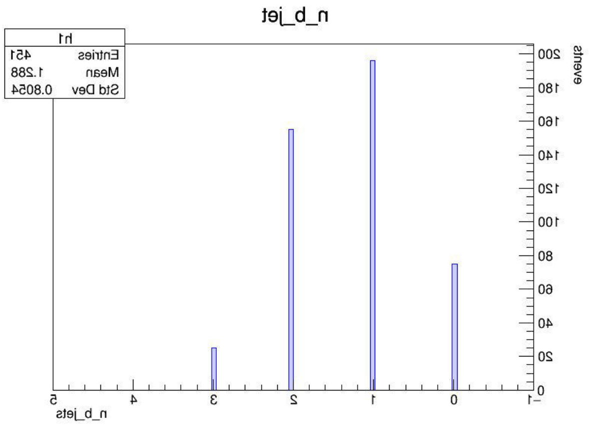
After fixing n_b_jet: Fase 1 samples



qcd

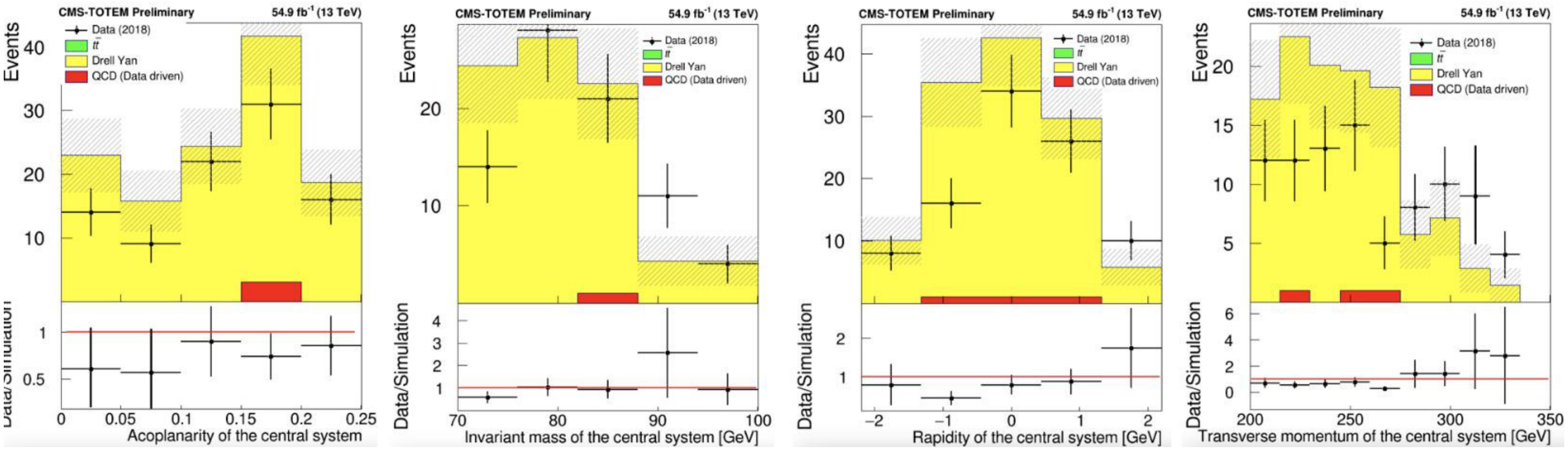


dy



ttjets

DY control region: inv_mass [40,100]; acoplanarity < 0.3; n_b_jets=0 ;



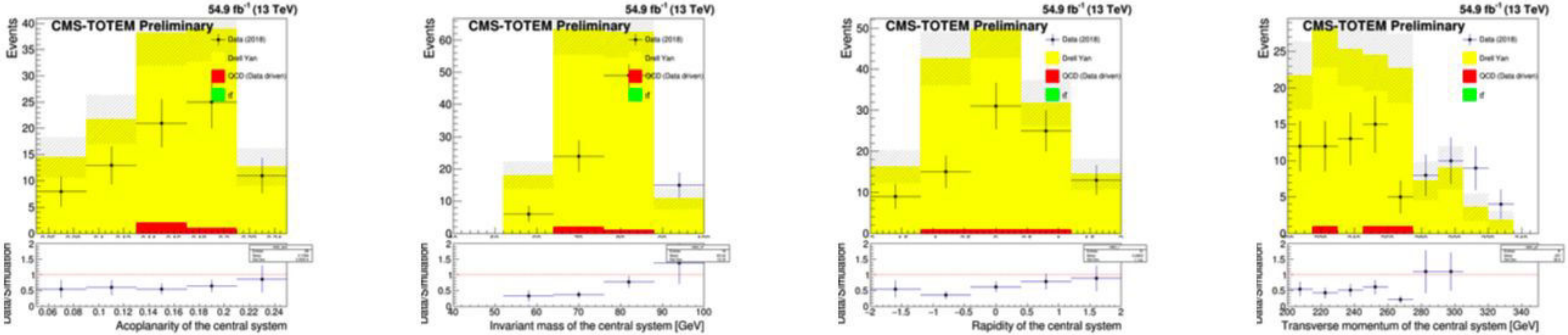
DY CR

Acoplanarity of the central system

Invariant mass of the central system [GeV]

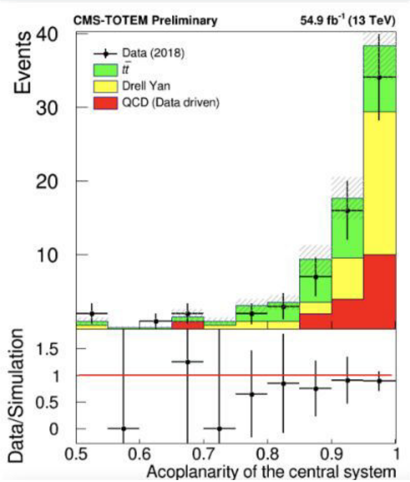
Rapidity of the central system

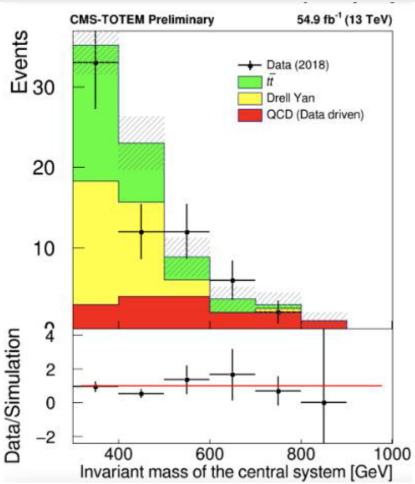
Transverse momentum of the central system [GeV]

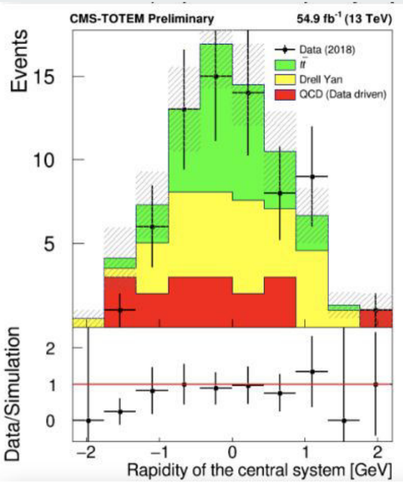


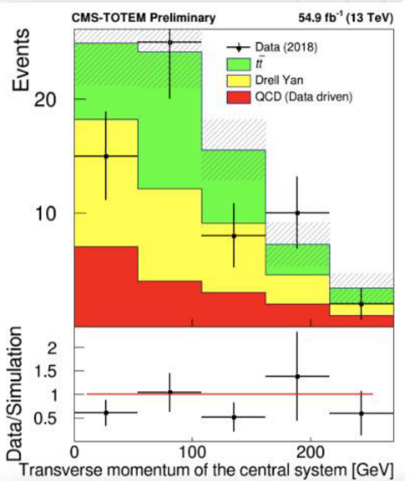
CR

TTJETS control region: inv_mass [200,650]; acoplanarity > 0.5; n_b_jets>=1 ; pt<125;

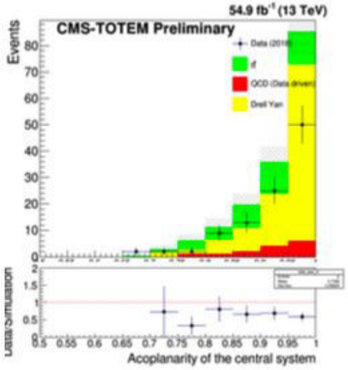


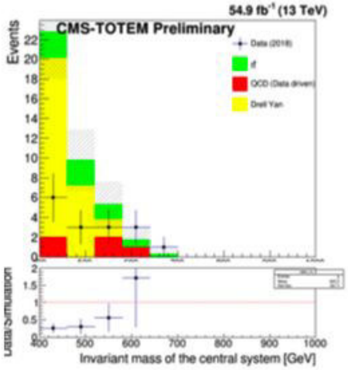


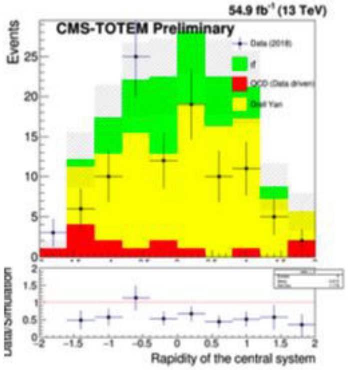


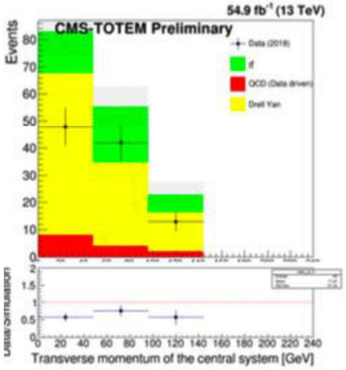


TT CR



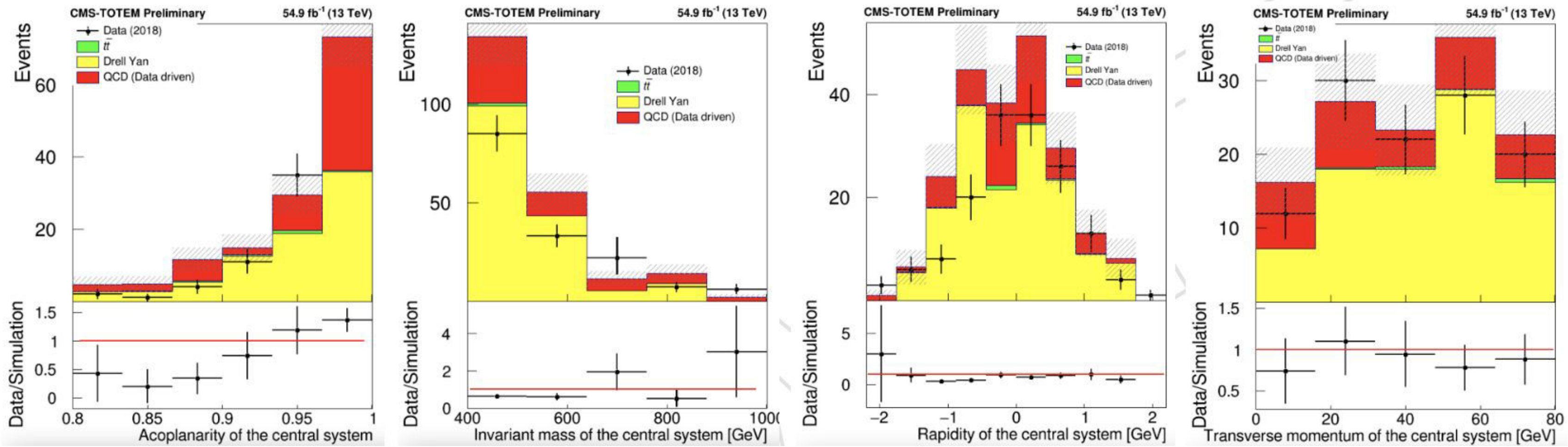






R

QCD control region: inv_mass [100,300]; acoplanarity < 0.3; n_b_jets<=0 ; pt<75;



QCD CR